



PiggyBank Protocol

Independent Public-Source Risk Review

Public-source review through 15 June 2026

by KallsFi

<https://x.com/KallsFi>

Subject: app.piggybank.fi and related public disclosures

For research and due-diligence purposes only. Not investment, financial, or legal advice. Third-party allegations are described as allegations unless independently corroborated.

Executive Summary

This report assesses PiggyBank Protocol as it appeared in the public record reviewed through 15 June 2026, rather than as a promotional product summary. It examines the protocol's vault design, pb-token accounting, off-chain NAV process, collateral routing, funding-rate strategy, external-venue dependencies, LAB incident, and transparency profile. The purpose is to identify what can be verified from public information and what remains dependent on operator disclosure.

This Executive Summary synthesizes findings that are developed and sourced in the body of the report. It is intended as a reader guide, not as a substitute for the underlying analysis; the main evidentiary discussions appear in Sections 2.1-2.3, 4.1-4.3, and 5.1-5.2.

Legal and reputational note: This report does not allege criminal conduct, fraud, market manipulation, or legal violation by PiggyBank or any other party. It evaluates publicly available information, protocol disclosures, third-party reporting, and identifiable transparency gaps for research and due-diligence purposes. Where this report discusses ZachXBT posts, media coverage, or LAB-related claims, those materials are treated as third-party allegations or public commentary unless the relevant point is independently corroborated by primary materials or on-chain evidence.

Central finding: PiggyBank should be viewed as an actively managed leveraged yield strategy. It is not a fully transparent, fully trustless on-chain protocol where all positions, valuations, and risks can be independently verified from the blockchain alone.

The on-chain vault layer is only one part of the risk surface. The economically material risks also sit in strategy selection, collateral borrowing, external lending venues, perpetual venues, off-chain valuation, administrative control, and the degree to which live positions can be independently verified.

Key findings

- **Vault and NAV model:** pb-tokens represent claims on vault value, but that value ultimately depends on operator-reported NAV. Public information provides partial vault-level visibility: DefiLlama's adapter identifies three PiggyBank vault state accounts and reads on-chain vault accounting fields. That does not, by itself, disclose the complete treasury, venue-position, strategy-allocation, or NAV-attestation data needed to reconstruct the full backing of each vault independently.
- **Strategy risk:** The advertised funding-rate and delta-neutral strategy can generate yield in favorable market regimes, but it can also lose money through funding-rate reversals, negative carry, crowding, execution slippage, leverage stress, and poor liquidity. The LAB episode shows

that the strategy mandate has included assets and structures that go beyond a simple major-asset funding trade.

- **Collateral, integration, and composability risk:** The non-stablecoin vaults add a collateral-and-borrow layer through Solana lending venues. The report does not treat Kamino, Loopscale, Jupiter Lend, Backed/xStocks, or JitoSOL as weak counterparties merely because PiggyBank uses them; several are reputable, widely integrated parts of the Solana or tokenized-asset stack. The relevant risk is narrower: PiggyBank depositors inherit those external protocols' smart-contract, oracle, liquidation, market-pause, issuer/custody, redemption, and liquidity dependencies, while PiggyBank does not publish enough live-position and routing data to let outsiders measure the size or concentration of those dependencies.
- **Security assurance:** The Formal Land audit is useful but narrow. It addresses the on-chain program perimeter, not the full strategy book, off-chain valuation process, venue positions, admin-key management, or economic soundness of the deployed strategy.
- **LAB incident significance:** The LAB incident is the key stress case for the report because it tests the gap between conservative marketing language and observed strategy behavior. It also shows the importance of operator discretion in position selection, accounting treatment, and user-facing NAV marks.
- **Compensation and live-data update:** A later PiggyBank X post dated 10 June 2026 is included as an official protocol statement concerning compensation, and a 15 June 2026 live-data check adds DefiLlama's current PiggyBank protocol page to distinguish historical interface figures from current third-party TVL. The compensation statement is recorded as part of PiggyBank's public response, but it is not treated as independent payment-level verification because no externally verifiable distribution record, per-user schedule, or completed reimbursement proof is included in the public materials reviewed here.^{152, 153}
- **Transparency and governance:** Material information remains undisclosed or insufficiently verifiable: team identity, operating entity, a consolidated program/authority and treasury/signer register, live venue positions, routing weights, NAV methodology, post-incident control changes, and independent proof-of-reserves or NAV attestations. The existence of public vault state accounts narrows the address-disclosure gap, but it does not close the strategy-level verification gap.

Risk snapshot

The report assesses smart-contract risk and counterparty/composability risk as medium, strategy risk and team risk as high, and transparency risk as critical. Transparency risk is assessed as

critical because, although three vault state accounts can be observed through DefiLlama's adapter, the most important live-position, treasury, strategy-allocation, NAV-attestation, and operator-identity information needed for full independent verification remains unavailable.

Credibility improvements that would materially change the assessment

The most important improvements would be publication of a consolidated address register covering the program ID, upgrade authority, treasury, signer, vault, strategy, and venue-position accounts; publication of position limits, eligible-asset rules, LTV targets, routing logic, and NAV methodology; independent NAV or proof-of-reserves attestations; and a documented post-incident risk-control framework. Until then, the central unresolved issue is not whether the strategy can generate yield, but whether depositors can independently verify the risks taken to generate it.

Research Scope and Method

Public-source review date: the original source review was completed through 13 June 2026, with targeted follow-up updates added on 15 June 2026 for PiggyBank’s 10 June compensation statement, current DefiLlama TVL, reported post-incident TVL contraction, the Formal Land audit figure replacement, and publication-readiness corrections. Live website metrics, TVL figures, APY displays, vault availability, dashboard screenshots, and social-media materials remain date-bound and may have changed after the review date.

Conflict-of-interest disclosure: the author previously had funds in PiggyBank in the form of SPYx deposited on the platform. After the LAB incident and the resulting loss, the author withdrew the position and began preparing this report. This prior depositor exposure is a potential source of bias and is disclosed so readers can evaluate the report accordingly; the analysis therefore relies on a source-weighted method and separates personal experience from independently cited public evidence.

Right-of-reply note: PiggyBank was not contacted for comment before publication. The analysis is limited to public-source materials available at the review date.

The analysis applies a source-weighted public-source due-diligence method. It is not a forensic audit and should not be read as full forensic proof. Protocol statements are treated as claims unless corroborated; third-party reporting is used to reconstruct public timelines; and independently verifiable evidence is separated from information that depends on the protocol’s own representations.

- **Primary-source review:** protocol documentation, public disclosures, terms, audit materials, and official communications are treated as the baseline record.
- **Third-party corroboration:** news reporting, investigator commentary, venue documentation, and public ecosystem data are used where they clarify or challenge the baseline record.
- **Verification standard:** on-chain verifiability is assessed separately from what the protocol reports through its interface, marketing pages, or social channels.
- **Source hierarchy:** primary protocol disclosures, official documentation, audit materials, public code, on-chain data, and issuer documentation carry the highest weight. Third-party reporting is used for chronology and public reaction, but not as independent proof where it merely repeats the same original posts or protocol statements. Educational explainers and media summaries are used only for background or context.

- **Allegation-handling standard:** investigator posts, quoted social-media commentary, and media characterizations are treated as allegations or public claims unless this report independently verifies the underlying point. This standard is especially important for the LAB and ZachXBT sections, where claims about LAB market structure are relevant to asset-risk context but are not treated as findings about PiggyBank insiders or legality.

Key limitation: public information now provides partial vault-level visibility, including three PiggyBank vault state accounts listed in DefiLlama’s adapter. However, PiggyBank still does not publish a complete address set covering treasury, strategy, venue-position, and signer/authority accounts. This report therefore does not independently reconstruct all PiggyBank-controlled wallets, all venue positions, or all NAV marks from raw on-chain data. Several exposure-level and NAV-related claims can be assessed only from public disclosures, reported data, and partial vault-state reads, not from complete independent on-chain reconstruction.

TABLE OF CONTENTS

Executive Summary.....	2
Research Scope and Method.....	5
1. Introduction & Protocol Overview.....	11
1.1 Protocol Category, Positioning, and Network.....	11
1.2 Use Case and Value Proposition.....	12
1.3 Protocol Background, Team, and Milestones.....	13
2. Protocol Architecture and Strategy Mechanics.....	14
2.1 Vault Architecture, pb-token Design, and Epoch/NAV Accounting.....	14
2.1.1 On-Chain Vault Structure and Smart-Contract Logic.....	14
2.1.2 The pb-token: Issuance, Mathematical Meaning, Accrual, and Redemption.....	16
2.1.3 The Epoch System.....	17
2.1.4 NAV Accounting and Trust Assumptions.....	18
2.1.5 On-Chain Verifiability and Transparency Assessment.....	19
2.2 Collateral-and-Borrow Flow across Kamino, Loopscale, and Jupiter Lend.....	20
2.2.1 Which Vaults Enter the Collateral-and-Borrow Flow, and How Collateral Is Routed..	20
2.2.2 Borrow Mechanics, Loan-to-Value, and Position Management.....	22
2.2.3 Collateral-Health Management: Liquidation Thresholds, Margin Calls, Deleveraging, and Loss Handling.....	22
2.2.4 Trust and Composability Risk Introduced by Each Lending Venue.....	23
2.2.5 Known Incidents and Near-Misses Involving These Collateral Venues.....	23
2.3 Delta-Neutral Funding-Rate Strategy, Leverage Mechanics, and Rebalancing Logic.....	25
2.3.1 Delta-Neutrality in PiggyBank’s Specific Context: Position Construction and the Venue Set.....	25
2.3.2 The Funding-Rate Capture Mechanic: When It Pays and When It Bleeds.....	28
2.3.3 Leverage: Layered Application up to Approximately 5x, Amplification, and the Protocol’s Safety Claims.....	29
2.3.4 Rebalancing Logic: Triggers, Automation, and Latency.....	31

2.3.5 Critical Analysis: Breakdown Conditions, Generic Failure Modes, and Their Application to PiggyBank.....	32
3. Security, Disclosure, and Technical Risk Surface.....	36
3.1 The Formal Land Audit — Scope, Methodology, Findings, and Limitations.....	36
3.1.1 Formal Land: Background, Verification Methodology, and Track Record.....	37
3.1.2 Audit Scope: Repository, Commit, Code Perimeter, and Exclusions.....	37
3.1.3 Methodology: Manual Review, Not Formal Verification.....	38
3.1.4 Findings: Inventory, Severity, and Remediation Status.....	40
3.1.5 Critical Evaluation: What the Audit Guarantees, What It Does Not, and Sufficiency for a Leveraged-Strategy Protocol.....	42
3.2 Protocol-Disclosed Risks vs. Actual Observed Risk Exposure.....	45
3.2.1 The Disclosed Risk Surface: An Inventory of PiggyBank’s Risk Representations.....	45
3.2.2 Conspicuous Absences: Risks the Mechanics Imply but the Disclosures Omit.....	47
3.2.3 Disclosed-versus-Observed: Mapping the Representations onto the Record.....	49
3.2.4 The Delta-Neutral Narrative versus Observed Behavior: Was the LAB Hedge Delta-Neutral?.....	52
3.2.5 Admin-Key and Governance Risk: Who Controls the Strategy Layer?.....	54
3.3 Smart Contract Risk, Oracle Risk, and Counterparty/Composability Risk.....	56
3.3.1 Smart Contract Risk: Vault-Program Vulnerabilities, Upgrade Authority, and the Boundary of Audited Assurance.....	57
3.3.2 Oracle Risk: How PiggyBank Prices Its Vaults, the Oracle Dependency Chain, and NAV-Marking Failure Modes.....	59
3.3.3 Integrated-Lending Dependencies: Kamino, Loopscale, and Jupiter Lend.....	61
3.3.4 Perpetual-DEX Dependencies: Insolvency, Liquidity Fragmentation, and Smart-Contract Failure Across Approximately Ten Venues.....	62
3.3.5 Concentration and Correlated Risk: Shared Dependencies, Network-Outage Exposure, and Blast Radius.....	63
4. LAB Token Incident and Protocol Response.....	64
4.1 The LAB Token Hedge: Public-Source Reconstruction of Events.....	64

4.1.1 The LAB Token: Origin, Purpose, and Market Characteristics.....	64
4.1.2 Chronological Reconstruction of the Incident.....	66
4.1.3 Mechanics of the Loss: From Hedge to NAV Drawdown.....	69
4.1.4 Disclosure to Users: Timeline, Content, and Omissions.....	72
4.1.5 Consistency with the Protocol’s Stated Strategy.....	73
4.2 ZachXBT’s Public Critique, On-Chain Claims, and Community Response.....	75
4.2.1 ZachXBT: Profile, Methodology, and Standing.....	75
4.2.2 The Specific Claims Regarding PiggyBank.....	75
4.2.3 On-Chain Evidence and the Limits of Independent Verification.....	76
4.2.4 The Community Response.....	77
4.2.5 PiggyBank’s Official Response and Its Adequacy.....	78
4.2.6 Beyond the LAB Incident: Structural Risk-Management Concerns.....	78
4.3 Protocol Response, NAV Drawdown Analysis, and User Impact.....	79
4.3.1 The Official Response: Statements, Actions, Timeline, and Remediation.....	80
4.3.2 Quantitative NAV Drawdown Analysis.....	83
4.3.3 User Impact: Scope, Aggregate Loss, and Recourse.....	85
4.3.4 Post-Incident Changes and Their Verifiability.....	86
4.3.5 Assessment of the Response Against Industry Standards.....	87
5. Risk Posture and Transparency as of the Review Date.....	89
5.1 Ongoing Risk Management Concerns Post-LAB Incident.....	89
5.1.1 Persistence of the Structural Conditions That Enabled the Incident.....	89
5.1.2 The Current Risk-Management Framework: Controls and Their Verifiability.....	90
5.1.3 The Funding-Rate Environment and the Conditions for Sustained Loss.....	90
5.1.4 Leverage Risk Under Prevailing Market Conditions.....	92
5.1.5 Systemic Risk from Solana-Ecosystem Concentration.....	93
5.2 Transparency Gaps: Anonymous Team, Self-Reported Metrics, and Unverified Claims.....	95
5.2.1 Team Anonymity.....	95
5.2.2 TVL and User Metrics.....	97

5.2.3 Unverified Partnership and Integration Claims.....	99
5.2.4 NAV Reporting Integrity.....	100
5.2.5 The “First on Solana” and Similar Marketing Claims.....	101
6. Conclusion & Assessment.....	102
6.1 Protocol Summary.....	102
6.2 Risk Profile Verdict.....	103
6.3 The Core Tension.....	105
6.4 Open Questions.....	107
6.5 Closing Assessment.....	108
Source Table.....	110

1. Introduction & Protocol Overview

1.1 Protocol Category, Positioning, and Network

PiggyBank is a decentralized finance (DeFi) yield protocol that operates a system of asset-specific vaults on the Solana blockchain. Users deposit assets such as USDC, SPYx, and the underlying JitoSOL liquid-staking token into PiggyBank's vaults, which are then allocated to a range of strategies, including funding-rate arbitrage and liquidity provision. In exchange, a yield-bearing token is minted to the depositor representing their allocation in the corresponding vault (for example, pbUSDC for USDC), which can later be burnt to redeem the underlying token and the performance generated. The protocol thus sits within the broader category of automated yield-aggregation and "delta-neutral" vault protocols, with a specific specialization in tokenized real-world assets (RWAs).¹

In terms of positioning, the protocol describes itself as the first decentralized protocol bringing extra yield to xStocks and stablecoins on Solana, marketing the proposition with the tagline "RWAs on steroids." This "first-mover" framing is the protocol's own marketing claim and could not be independently verified through this research; it should be read as a positioning statement rather than an established fact.²

The protocol offers two categories of strategy — stablecoin yield, and extra yield on top of other asset exposure — both accessed through its own Solana vault infrastructure. As of the access date, three vaults are live — the USDC vault (stablecoin), the SPYx vault (tokenized S&P 500 exposure), and the JitoSOL vault (built on the underlying JitoSOL liquid-staking token) — with TSLAx and NVDAx vaults listed as "Coming Soon." For clarity, this report uses "JitoSOL vault" for PiggyBank's deposit vault, "pbJitoSOL" for the corresponding vault-share token, and "JitoSOL" or "underlying JitoSOL token" for the liquid-staking token itself. The protocol advertises a target net APY of approximately 15% for the USDC vault and approximately 7% for the xStocks vaults. These are stated targets; the protocol's own disclosures emphasize that yields are not guaranteed.^{1, 2}

PiggyBank operates exclusively on Solana, and its mechanics are tightly coupled to Solana-native infrastructure. The protocol's documentation states that deposited assets are used as collateral on Solana lending protocols such as Kamino or Loopscale, against which stablecoins are borrowed. Those stablecoins are then deployed into automated, delta-neutral strategies based on funding-rate arbitrage across approximately ten perpetual DEXs, taking long-short positions on crypto majors and tokenized stocks with up to approximately 5x leverage.²

1.2 Use Case and Value Proposition

The problem PiggyBank seeks to address is the friction and complexity involved in generating yield on tokenized equities and stablecoins. The underlying asset class — tokenized stocks (“xStocks”) — emerged only recently: on 30 June 2025, Backed Finance launched xStocks, bringing 60-plus fully collateralized U.S. equities such as AAPL, NVDA, and TSLA on-chain as SPL tokens on Solana, each backed 1:1 by shares held in custody under Swiss regulatory oversight. Because these tokens are composable, they can be used as collateral, lent, or pooled, unlocking on-chain composability and automated DeFi yield strategies that are not native to ordinary brokerage interfaces. PiggyBank positions itself within this nascent ecosystem as an abstraction layer over those strategies.^{3, 4}

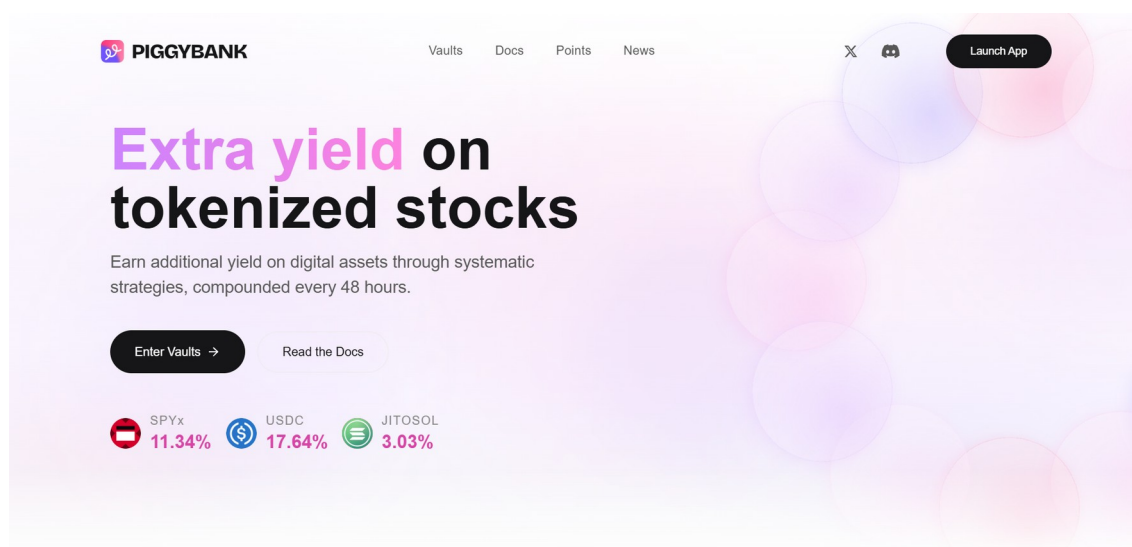


Figure 1. PiggyBank homepage marketing/interface positioning and yield proposition. Screenshot captured / source reviewed 13 June 2026. Source: #2.

The value proposition, as characterized both by the protocol and by third-party coverage, is the simplification of access to that yield. PiggyBank is described as letting users deposit tokenized stock assets and stablecoins to earn automated yield via delta-neutral strategies, with the core proposition being to simplify access to sustainable yield on real-world assets so that users deposit into vaults and the protocol handles the strategy layer rather than requiring them to navigate complex DeFi mechanics manually. Relative to manually executing a basis trade, the protocol’s stated differentiators include: liquid pb-tokens that can be held, transferred, or used across DeFi while the underlying deposits earn yield; automated margin rebalancing and deleveraging based on market indicators; and RWA-specific features such as support for dividend distributions, stock-split adjustments, and hook-based compliance integrations enforced at the vault-token level.^{5, 2}

The target user base is therefore holders of xStocks, stablecoins, and Solana liquid-staking tokens who seek passive yield without active position management. Notably, the protocol is

geographically restricted: its own disclosures state that PiggyBank is not available to residents of the United States or other restricted jurisdictions, a constraint inherited from the xStocks/Backed access framework. Operationally, the protocol runs on a 48-hour cycle: vaults operate on an epoch-based system in which net asset value is reported every 48 hours, and upon submitting a redemption request the underlying asset becomes claimable after two epochs have elapsed — in practice, withdrawals taking 48 to 96 hours.^{2, 1}

1.3 Protocol Background, Team, and Milestones

Launch. Multiple sources place the mainnet launch on 30 October 2025. Third-party tracking states that PiggyBank launched on mainnet on 30 October 2025 and is currently live with multiple active vaults. This is corroborated by the protocol’s own pre-launch page, which announced that the first vaults for xStocks would launch on PiggyBank on 30 October, with an xStock-holder snapshot on the 29th at 6 p.m. UTC granting two weeks of exclusive vault access and a first-month 10,000 USDC incentive.^{5, 6}

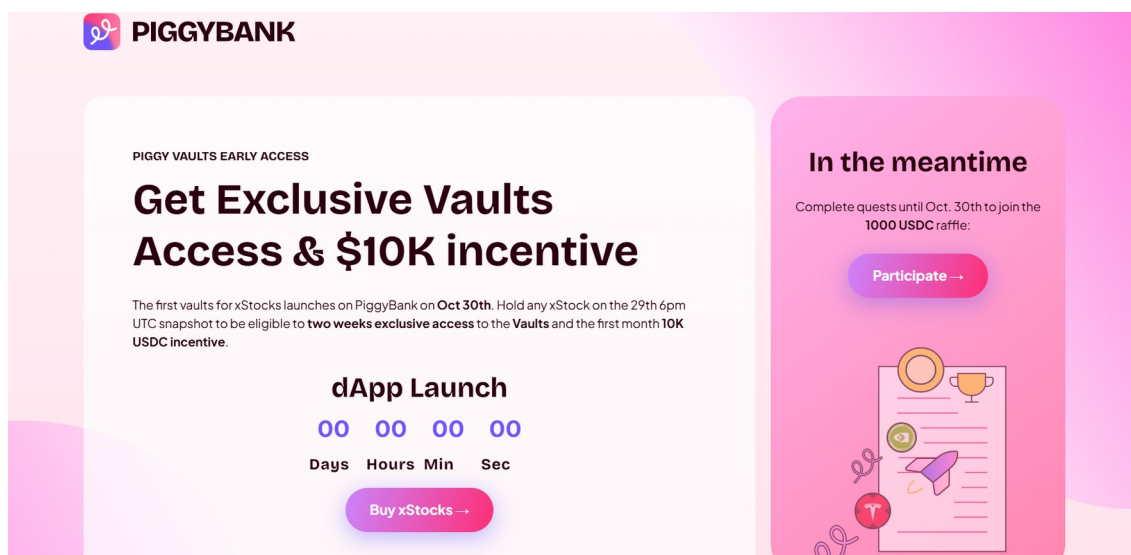


Figure 2. PiggyBank early-access page announcing vault launches and incentives. Screenshot captured / source reviewed 13 June 2026. Source: #6.

Founding team and funding. The founding team or operating entity behind PiggyBank is not publicly disclosed, and this could not be verified through available sources. The official site names no founders and lists only a generic contact address (hi@piggybank.fi) and a “© 2026 Piggybank” footer. The documentation describes the team only in general terms, stating that it has expertise in blockchain, financial engineering, and quantitative research. Independent coverage similarly notes that funding details and backers have not been publicly disclosed. Accordingly, the identity, operating jurisdiction, and financial backing of the team should be treated as unconfirmed at the time of writing.^{1, 5}

Audits and integrations. The protocol states it is audited by Formal Land, the team behind the formal-verification work for Tezos with Nomadic Labs; an audit document is linked from the site, though an independent review of that audit’s scope and findings is reserved for a later section. The site additionally displays integration or “trusted by” relationships with Solflare, xStocks, Jupiter Lend, Ink, Binance Wallet, and others, and the strategy layer references Kamino, Loopscale, and Jupiter Lend as lending venues. The depth and formality of these relationships (logo display versus contractual partnership) is not independently established here.²

Traction and TVL visibility. The protocol reports a completed incentive “Season 0” and a rewards mechanism it calls “Oinks.” Per the official site, Season 0 recorded 1,832 unique depositors and 69 contributor-program participants, and the protocol reports more than 2,000 total users. On total value locked, the site displays vault TVL figures of approximately \$3.06M (USDC), \$823K (SPYx), and \$639K (JitoSOL vault), implying an aggregate on the order of \$4.5M. These user figures remain self-reported, but the TVL picture is now partly externally checkable: DefiLlama’s adapter lists three vault state accounts associated with pbUSDC, pbSPYx, and pbJitoSOL and calculates underlying value from on-chain vault fields. This corroborates that specific vault accounts exist and expose partial vault-state data, but it does not verify the full strategy book, treasury positions, venue exposures, or user-count claims.^{2, 149}

2. Protocol Architecture and Strategy Mechanics

2.1 Vault Architecture, pb-token Design, and Epoch/NAV Accounting

2.1.1 On-Chain Vault Structure and Smart-Contract Logic

PiggyBank’s vaults are implemented not as a family of independent contracts but as a set of accounts managed by a single Solana program. The protocol’s security audit, conducted by Formal Land at commit 6c930b34 of the repository github.com/zsociety-io/piggybank-program, records that the entire protocol is contained in one self-contained smart contract written in Rust against the Pinocchio library — a low-level Solana program framework used in place of the more common Anchor framework — and characterizes the design as one in which users are provided “managed funds by a trusted admin.” At the time of the audit (27 October 2025, three days before the documented mainnet launch) the repository was private, and as of the access date it does not appear among the public repositories of the zSociety GitHub organization, indicating that the program source remains closed. Some vault state accounts are publicly visible through DefiLlama’s adapter, which lists vault state accounts associated with pbUSDC, pbSPYx, and pbJitoSOL; however, a consolidated official register tying the program ID, upgrade authority,

vault accounts, treasury/signers, and strategy/venue-position addresses together was not located in the public materials reviewed here. This is the more precise transparency gap.^{9, 10, 149, 151}

The audit's file listing — which enumerates only those files that drew findings — exposes part of the program's instruction set, the on-chain logic that governs the vault lifecycle. Instructions evident from the report include `create_vault` (vault initialization), `claim` (redemption settlement), `roll_epoch` (epoch advancement and accounting update, treated in §2.1.3–§2.1.4), and `update_admin` (rotation of the administrative authority); dedicated deposit and redemption-request instructions are implied by the deposit and redemption accounts the audit references but are not individually enumerated. Two findings bear directly on the protocol's trust model. First, in reviewing `create_vault` the auditors asked whether the instruction should honor the program's paused state; the team replied that vault creation is intentionally exempt because the administrator can in any case unpause the program — confirming that an administrative key holds pause and unpause control over the entire program. Second, the redemption path and other user-facing vault instructions were flagged for missing validation of the supplied vault account, a check the team agreed to add, indicating that at the audited commit certain account-consistency safeguards were still being hardened.⁹

Each vault is asset-specific: the protocol operates, in its own words, a single vault per digital asset. As of the access date three vaults are live — the USDC vault, the SPYx vault, and the JitoSOL vault — with TSLAx and NVDAx listed as forthcoming. The vault infrastructure is described as compatible with both the legacy SPL Token standard and the Token-2022 standard, and as supporting transfer hooks, stock-split adjustments, and dividend distributions so that vault positions can track corporate-action events on the underlying real-world assets. The documentation does not, however, specify which token standard is used for each individual vault or pb-token, nor the address or logic of any transfer-hook program enforcing compliance rules at the token level (transparency gap).^{2, 1}

A critical architectural qualification concerns whether the asset-specific vaults are also risk-isolated. The protocol's own description of yield generation indicates they are not. USDC deposits are deployed directly into the funding-rate-arbitrage strategy, whereas deposits into the SPYx vault and the JitoSOL vault are posted as collateral on third-party Solana lending markets (Kamino or Loopscale) to borrow stablecoins, and those borrowed stablecoins are then deployed into the same market-neutral funding-rate strategies used by the USDC vault. Because all vaults ultimately fund a common strategy book, a loss originating in that book is not contained within a single vault. This was demonstrated empirically in June 2026, when a single losing position led the protocol to project simultaneous net-asset-value declines across all three vaults — approximately 15% (USDC

vault), 12% (SPYx vault), and 9% (JitoSOL vault). The uniform direction but differing magnitude of the impact is consistent with shared strategy exposure layered on top of differing per-vault leverage; the protocol does not publish a per-vault segregation or loss-attribution policy that would allow this to be characterized precisely (transparency gap). A full account of that episode is reserved for §3 and §4.^{2, 7, 11}

2.1.2 The pb-token: Issuance, Mathematical Meaning, Accrual, and Redemption

When a user deposits into a vault, the program mints a corresponding pb-token to the depositor — for example pbUSDC for the USDC vault, or pbNVDAX for the forthcoming NVDAX vault — with the token-minting and -burning logic handled inside the program’s token_helper module. The protocol’s illustrative example shows a one-to-one issuance at deposit: a deposit of 10 NVDAX yields 10 pbNVDAX. That 1:1 figure corresponds to a unit net-asset-value-per-share at vault inception. The documentation does not publish the general issuance formula for deposits made after a vault has accrued value, and in particular does not state explicitly that later deposits are minted in proportion to the prevailing NAV per share (transparency gap); on the standard vault-share model the design otherwise follows, the quantity minted would equal the deposited amount divided by the current NAV per share.^{1, 9, 2}

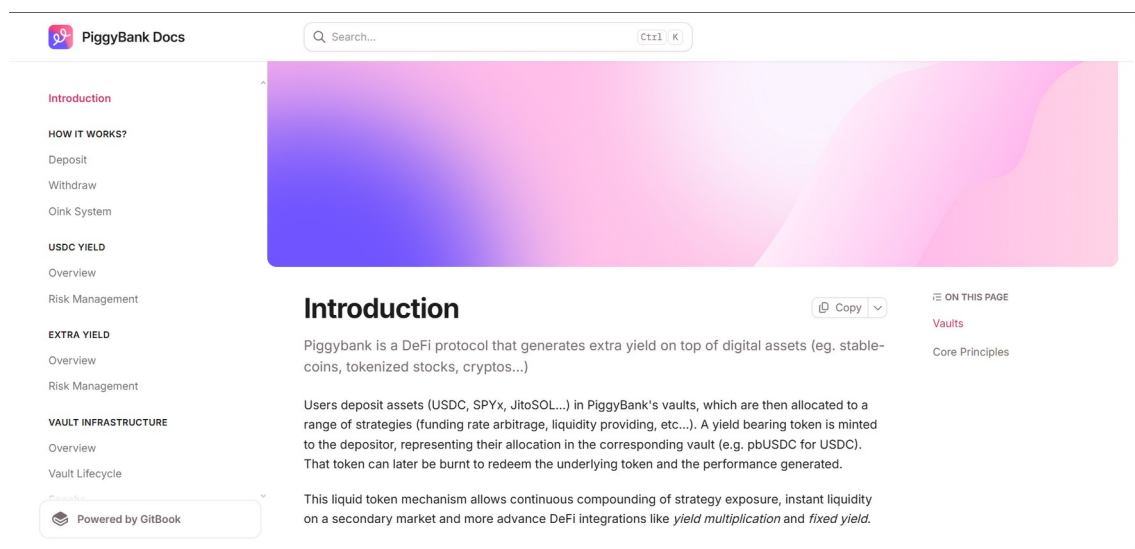


Figure 3. PiggyBank documentation page describing deposit-to-vault mechanics and pb-token issuance. Screenshot captured / source reviewed 13 June 2026. Source: #1.

Mathematically, a pb-token represents a pro-rata claim on the assets attributed to its vault: the redemption value of one pb-token equals the vault’s net asset value divided by the outstanding pb-token supply. Value accrues to holders through appreciation of this per-token claim rather than through an increase in token balances. The protocol states that the token continuously reflects the vault’s performance and that there is nothing to claim manually, and it illustrates accrual by way of redemption — for instance, redeeming a position originally deposited as 10 NVDAX for 11

NVDAx after one year. The public materials are not wholly unambiguous on the accrual mechanism: one passage describes end-of-epoch yield as being compounded into and reflected in the holder's balance, phrasing that could be read as a rebasing model, whereas the redemption examples, the emphasis on holding and transferring pb-tokens across DeFi, and the existence of a pb-token/underlying secondary market are all consistent with a fixed-supply, appreciating-share design. The distinction between balance appreciation and rebasing is not stated explicitly (transparency gap); the appreciating-share interpretation is the more consistent with the protocol's stated composability goals, but it is an inference rather than a documented fact.²

pb-tokens are presented as liquid and composable: holders may retain, transfer, or use them in other DeFi applications while the underlying continues to earn. The protocol directs users to a secondary market on Jupiter for trading pbUSDC against USDC; the linked market references a pbUSDC mint address of F35yYmTR6PqkbTx449P1eGhB57mRhWAdYs93eCo2dMZR. Such a secondary venue is the protocol's stated mechanism for near-instant liquidity, in contrast to the multi-epoch primary redemption path.^{2,1}

Redemption through the vault itself proceeds by burning pb-tokens against a redemption request that becomes claimable only after the epoch delay described in §2.1.3. The audit confirms a dedicated on-chain redemption account that records the redeemable underlying amount, which the claim instruction then settles to the user. The documentation does not disclose whether the pb-token mint authority and freeze authority are vested in a program-derived address — the configuration a non-custodial design requires, since it would restrict minting to the deposit path — or are retained by an administrative key. This configuration is verifiable on-chain by inspection of the mint account but is not asserted in the public materials, and it was not independently confirmed for the purposes of this section (transparency gap).⁹

2.1.3 The Epoch System

PiggyBank's vaults operate on a fixed 48-hour epoch that the protocol states is independent of Solana's native staking epochs. The epoch is the unit of both yield accrual and withdrawal processing. At each epoch boundary, accrued yield is compounded into the vault and reflected in vault positions with no manual claim step, and a new deposit is stated to begin earning only at the end of the first full epoch following the deposit. On-chain, the epoch transition is effected by the `roll_epoch` instruction, whose review in the audit shows it performing signed 128-bit integer (i128) arithmetic to adjust vault state by a quantity the code names `vault_transfer` — the mechanism by which the period's profit or loss is written into the vault's accounting (see §2.1.4).^{2,9}

Withdrawals follow the same schedule and are deliberately not continuous. A redemption request is queued until the end of the current epoch; it is then processed during the following epoch, after

which the funds become claimable. The protocol describes redemptions as being batched and processed over an epoch, with the corresponding perpetual positions unwound through execution intended to minimize market impact, and quantifies the resulting wait as roughly 48 hours for a request submitted near an epoch's end and up to roughly 96 hours for one submitted just after a new epoch begins. This corresponds to the documentation's statement that the underlying becomes claimable after two epochs have elapsed.^{1,2}

The protocol's stated rationale for an epoch-batched design in preference to continuous liquidity is execution quality: the vaults hold positions — leveraged perpetual-futures hedges and collateralized loans on external venues — that cannot be unwound instantaneously without market impact, so redemptions are aggregated and the corresponding positions closed in a controlled manner. Two further consequences follow analytically and are standard for this class of design, though they are not stated by the protocol: a discrete, delayed NAV-marking cycle prevents a redeeming user from transacting on a stale or favorable NAV ahead of a pending mark (that is, it limits NAV front-running), and it affords the operator a defined window in which to source liquidity for redemptions. The documentation does not specify the precise wall-clock boundaries of epochs, the exact moment at which `roll_epoch` is invoked, or whether epoch advancement is automated or manually triggered by the administrator (transparency gap).²

2.1.4 NAV Accounting and Trust Assumptions

Net asset value is the figure that links a vault's underlying assets to the redemption value of its pb-tokens, and PiggyBank reports it on the 48-hour epoch cadence. The decisive question for the protocol's trust profile is who computes and reports that figure, and against what assumptions. The available evidence indicates that NAV is determined off-chain by the protocol operator and written on-chain by a trusted administrator through the `roll_epoch` instruction, rather than being derived trustlessly on-chain or sourced from an independent price oracle. The audit's characterization of the protocol as managed funds under a trusted admin, the existence of an administrator-controlled pause and an `update_admin` instruction, and the `roll_epoch` mechanism that writes the period's profit-and-loss into vault state together establish that the reported NAV originates with the operator. The public documentation does not describe any oracle, attestation, or multi-party process constraining the reported value (transparency gap).^{1,9}

This matters because a vault's true NAV is not contained within the program. The assets backing the vaults are dispersed across external venues — collateral positions on Kamino or Loopscale, margin and open positions on approximately ten perpetual-futures DEXs, and, as the June 2026 episode revealed, over-the-counter holdings of illiquid tokens — whose marked value must be assembled and valued off-chain before it can be reported. NAV therefore embeds discretionary

valuation judgments. The June 2026 episode makes this concrete: the protocol elected to exclude a locked position it valued at roughly US\$1.35 million from NAV until that position's first scheduled unlock, a discretionary accounting decision that directly produced the reported drawdowns and that the protocol itself framed as the fairest treatment of depositor liquidity.^{2, 7, 8, 12}

The resulting trust assumptions are substantial. A depositor must trust that the operator (i) values external and illiquid positions accurately and in good faith, (ii) reports the resulting NAV honestly and on schedule, and (iii) does not exploit the latitude inherent in marking illiquid positions to smooth, delay, or otherwise misstate losses. Neither the documentation nor the audit report describes an on-chain bound on the magnitude of a permitted per-epoch NAV adjustment, a circuit-breaker on large markdowns, or any independent verification of the reported figure (transparency gap); the audit's relevant observations concern arithmetic-overflow safety within `roll_epoch` rather than economic limits on the value reported. In effect, the integrity of NAV — and hence the redemption value of every pb-token — rests on the honesty and competence of the administrative operator, a materially custodial trust assumption notwithstanding the protocol's non-custodial framing.⁹

2.1.5 On-Chain Verifiability and Transparency Assessment

The protocol markets its xStocks vaults as “overcollateralized and fully transparent.” A precise assessment must distinguish what a user can independently verify on-chain from what they must take on trust.²

Verifiable on-chain. The pb-token mints are ordinary Solana token accounts, so each token's total supply, decimal precision, and authority configuration can be read directly from the chain by any user, as can the pbUSDC mint identified in §2.1.2. Deposits, redemptions, and claims are on-chain transactions; the vault accounts maintained by the program — including the NAV-related state advanced by `roll_epoch` and the current epoch — are on-chain and readable; and the deployed program's bytecode is itself on-chain and, in principle, disassemblable. To this extent the protocol's reported state is transparent.⁹

Not independently verifiable. The correctness of that reported state is a different matter. Because the backing assets reside on external venues and include positions — open perpetual-futures exposure, collateralized loans, and locked over-the-counter tokens — whose value is a matter of off-chain marking, a user cannot reconstruct the true NAV from the vault account alone. DefiLlama's adapter can read three published vault state accounts and calculate an underlying amount from their on-chain fields, but that is not equivalent to reconstructing the full treasury, strategy, venue-position, and illiquid-holding picture. Doing so would require, at minimum, the complete set of the operator's treasury, strategy, venue, and position addresses, together with a

method for valuing illiquid holdings; the protocol publishes neither such a complete address set, nor a proof-of-reserves, nor an independent NAV attestation, none of which were located for this section (transparency gap). The program source is closed, so the on-chain logic cannot be independently reviewed beyond the summary in the third-party audit, and a complete official deployment/authority register was not located. Consequently, while a depositor can verify how many pb-tokens exist and what NAV the operator has reported, and while some vault state can now be read through a public adapter, they cannot independently verify that the reported NAV is correct, that the vaults are in fact overcollateralized, or that the full asset base exists as represented. The claim of full transparency is therefore accurate only as to partial reported on-chain accounting, and overstated as to the independent verifiability of the assets and valuations that underlie it. ^{10, 9, 149, 150, 151}

Key Takeaway: PiggyBank users are not exposed only to PiggyBank itself. The non-stablecoin vaults also depend on external lending venues, collateral rules, liquidation engines, oracles, and the protocol’s own undisclosed routing choices. That kind of dependency is normal in DeFi, but PiggyBank does not publish enough live routing, LTV, or venue-exposure data for users to measure it independently. ^{2, 9, 149}

2.2 Collateral-and-Borrow Flow across Kamino, Loopscale, and Jupiter Lend

2.2.1 Which Vaults Enter the Collateral-and-Borrow Flow, and How Collateral Is Routed

The collateral-and-borrow mechanism is not uniform across PiggyBank’s vaults; the protocol operates two distinct capital paths, and only one of them involves the lending venues that are the subject of this section. For the USDC vault, deposited stablecoins are deployed directly into the automated, delta-neutral funding-rate-arbitrage book — described as long-short positions across approximately ten perpetual DEXs with up to approximately 5x leverage — without any intervening collateralized loan. For the non-stablecoin vaults (the SPYx vault and the JitoSOL vault today, with TSLAx and NVDAx listed as forthcoming), the deposited asset is instead posted as collateral on a third-party Solana lending market, a stablecoin is borrowed against it, and that borrowed stablecoin is then deployed into the same funding-rate strategy used by the USDC vault. The collateral-and-borrow flow analyzed here therefore governs the SPYx vault and the JitoSOL vault (and any future xStocks vaults), not the USDC vault, which reaches the strategy book directly. ²

The venues said to receive this collateral are Kamino, Loopscale, and Jupiter Lend, but the protocol’s own materials are internally inconsistent about which of the three is actually used. The site’s “How It Works” flow states that user assets are collateralized on Kamino, Loopscale, and

Jupiter Lend; the xStocks-vault product card names only “Kamino & Loopscale”; and the FAQ refers to “trusted Solana lending protocols (such as Kamino or Loopscale),” a non-exhaustive formulation that names two venues while implying others. The GitBook documentation, for its part, adds no venue-level detail beyond the general strategy description, and the homepage “Trusted By” row displays a Jupiter Lend logo but no Kamino or Loopscale mark. The three named venues are real and well-established Solana money markets, but the discrepancy between “Kamino & Loopscale,” “Kamino, Loopscale, Jupiter Lend,” and “such as Kamino or Loopscale” means the protocol does not present a single, authoritative statement of which venues hold collateral. This is a transparency gap; one plausible reading is that the integration set expanded over time to add Jupiter Lend, but that sequencing is an inference and is unconfirmed.^{2, 1}

More consequentially, the documentation does not state whether routing is fixed or dynamic. There is no published mapping of which asset is posted to which venue, no statement of whether a given vault’s collateral sits at a single fixed venue or is split and reallocated across venues, and no description of the selection logic, allocation weights, or rebalancing triggers that would govern dynamic routing. The “such as” phrasing is consistent with operator discretion to choose among venues, but the mechanism is undisclosed. Whether collateral routing is fixed or dynamic, and on what basis, is therefore a transparency gap and remains unconfirmed.^{2, 1}

Independent verification is further constrained by the disclosure limits established in §2.1. Some PiggyBank vault state accounts are publicly visible through DefiLlama’s adapter, but the protocol does not publish a complete address set covering the program/authority, treasury, wallet, and venue-position accounts it controls across these lending venues. As a result, the existence, size, venue, and parameters of PiggyBank’s specific collateral positions could not be reconstructed from chain data for this section (a continuation of the transparency gap identified in §2.1.5). The Formal Land audit, moreover, covered only PiggyBank’s single self-contained program and did not extend to the external lending venues. The routing description in this section consequently rests on the protocol’s own representations rather than on verified venue-position observation.^{9, 149}

What can be stated with confidence is that PiggyBank’s collateral assets are eligible at all three venues, so the routing it describes is at least technically feasible. Kamino became the first major DeFi lender to accept tokenized equities as collateral, onboarding xStocks through its v2 “xStocks Market” in July 2025 with Chainlink serving as the price-feed provider. Loopscale supports more than forty collateral types including the underlying JitoSOL token and tokenized stocks, and xStocks issuer documentation lists Kamino, Loopscale, and Jupiter Lend (“JupLend”) among the Solana lending markets where xStocks can be supplied. The underlying JitoSOL liquid-staking token is a standard collateral asset on all three venues.^{13, 14, 15, 16}

2.2.2 Borrow Mechanics, Loan-to-Value, and Position Management

Against the posted collateral, PiggyBank borrows a stablecoin and routes the borrowed amount into the funding-rate-arbitrage book shared with the USDC vault. The borrowed asset is not named; USDC is the natural candidate given the USDC vault and the pbUSDC/USDC secondary market, but the protocol does not state this explicitly.²

Loan-to-value is the key missing number. PiggyBank calls the xStocks vaults overcollateralized, but it publishes no target LTV, maximum borrow ratio, or health-factor target. The venue limits show why that matters: Kamino, Loopscale, and Jupiter Lend use different collateral parameters, liquidation thresholds, and maturity structures, with Jupiter Lend advertising materially higher LTV ceilings and Loopscale adding fixed-term rollover risk. Without PiggyBank's chosen venue and operating LTV, depositors cannot judge the real liquidation cushion.^{17, 18, 19, 15, 20, 21, 22}

PiggyBank also describes automatic margin rebalancing, deleveraging, and partial TP/SL controls, but it does not disclose the target levels, triggers, keeper/admin process, or latency. It is also unclear whether those controls govern the lending-venue borrow positions, the perpetual-futures hedge book, or both.²

Finally, the borrow leg links collateral risk to strategy performance. The debt stays fixed while the borrowed stablecoins are placed at risk; if the strategy loses money, the debt does not fall. A large strategy drawdown can therefore require additional collateral, deleveraging, or repayment from other sources, and if that is not available, the position's LTV moves toward liquidation. This is the channel by which a strategy loss can propagate into the SPYx and JitoSOL vaults, as shown by the simultaneous June 2026 NAV declines across the three vaults.^{7, 11}

2.2.3 Collateral-Health Management: Liquidation Thresholds, Margin Calls, Deleveraging, and Loss Handling

Collateral health has two layers: PiggyBank's own pre-emptive controls and the liquidation systems of the venues. Only the second layer is documented in detail, because PiggyBank's public materials use broad language such as "automatic margin rebalancing," "deleveraging," and "partial TP/SL" without publishing numeric triggers, margin-call levels, keeper/admin process, or response latency.²

If PiggyBank's controls fail, the venue rules determine the outcome. Kamino, Loopscale, and Jupiter Lend all use partial liquidation mechanics, but they differ on penalties, maturity/default treatment, LTV buffers, and market-level risk sharing. Kamino can apply partial liquidation and automated deleveraging; Loopscale can liquidate loans that fail to refinance at maturity; Jupiter Lend pairs low liquidation penalties with high LTV ceilings and per-vault limits. Those

differences matter because a borrower-friendly penalty structure does not remove liquidation risk if the collateral buffer is thin.^{23, 24, 25, 26, 27, 20, 28, 29}

For depositors, the consequence is straightforward: any liquidation penalty, execution slippage, socialized bad debt, or collateral-reuse loss reduces vault assets and the redemption value of pb-tokens. PiggyBank does not publish a loss-attribution policy, reserve/insurance fund, backstop, or precise rule for how liquidation losses are reflected in NAV. Since NAV is reported through the operator-controlled 48-hour process, the treatment of such a loss remains partly discretionary.^{2, 9}

The protocol's risk disclosure mentions market volatility, possible liquidation if thresholds are breached, and failures in integrated platforms such as Kamino, but it does not provide the thresholds, margin-call process, deleveraging logic, or depositor-impact model. The collateral-health framework is therefore not assessable from public materials at the level most relevant to users.²

2.2.4 Trust and Composability Risk Introduced by Each Lending Venue

Routing collateral through external lenders adds an extra risk stack: venue smart-contract risk, oracle/liquidation risk, parameter-governance risk, market-pause/withdrawal risk, and, depending on design, socialized bad debt or rehypothecation. The Formal Land audit covers PiggyBank's own program only, so this externally routed part of depositor exposure remains outside the audited perimeter.^{9, 25}

These venues are not interchangeable. Kamino is the most established of the group, Loopscale is younger and adds fixed-term rollover/exploit-history concerns, and Jupiter Lend adds a newer high-LTV profile together with specific rehypothecation-contagion concerns. The core finding is not that any one of these venues is automatically weak; it is that PiggyBank's routing choice materially changes depositor risk, while live venue weights and exposure sizes are not published.

^{30, 31, 15, 20, 32, 33, 34, 2}

Collateral assets add another upstream layer. SPYx carries tokenized-equity issuer, custody, liquidity, and off-hours market-gap risk; JitoSOL carries liquid-staking-token risks, including depeg/slashing-related concerns and the possibility that borrowing costs outweigh staking-linked return. Those risks exist before PiggyBank's own strategy book adds leverage and funding-rate exposure.^{35, 36, 16, 25}

2.2.5 Known Incidents and Near-Misses Involving These Collateral Venues

This subsection records incidents at the lending venues themselves; PiggyBank's own June 2026 LAB episode is a strategy-layer and NAV-accounting event analyzed in §3–§4, but its interaction (or lack of it) with the collateral positions is noted at the end.

The most directly relevant venue incident is the Loopscale exploit of 26 April 2025, which pre-dates PiggyBank’s launch by roughly six months. An attacker manipulated the protocol’s pricing of RateX principal tokens to borrow against inflated collateral and drained approximately \$5.8M — 5,726,725 USDC and 1,211 SOL — from Loopscale’s USDC and SOL vaults, equal to about 12% of the protocol’s total value locked; the loss fell on vault depositors, while borrowers and loopers were unaffected. Loopscale paused its markets and froze withdrawals, and the full amount was returned within roughly 72 hours after a negotiated 10% white-hat bounty. The episode is a clean materialization of venue-level oracle, smart-contract, and withdrawal-gating dependency risk on a venue PiggyBank now names, and it illustrates the withdrawal-gating risk discussed in §2.2.4 — yet PiggyBank’s risk disclosures make no reference to any venue’s incident history.^{37, 38, 84}

On a venue PiggyBank uses and post-dating its launch, the Kamino liquidation cascade of 5–6 February 2026 (detailed in §2.2.4) seized \$19.36M of collateral across 55,649 liquidations, including \$2.14M of underlying JitoSOL collateral, without generating bad debt. It is best characterized not as a failure but as a stress test the venue passed; whether any PiggyBank position was liquidated within it is unconfirmed.³¹

The December 2025 Jupiter Lend rehypothecation controversy (also detailed in §2.2.4) is best classed as a disclosure near-miss rather than a loss event: the retraction of the “isolated” and “zero-contagion” claims, and the acknowledgement of rehypothecation, exposed a latent contagion risk in a venue PiggyBank uses, but no bad debt was reported. Its significance for this paper is evidentiary: it bears directly on the credibility of PiggyBank’s “fully transparent” framing of a flow that depends on this venue.³²

Finally, the June 2026 LAB episode should be correctly located. The loss originated in a basis trade in the strategy book — locked LAB tokens bought at a discount through an over-the-counter desk and hedged with LAB perpetual shorts — which became unsustainable amid thin liquidity and deeply negative funding, compounded by the discretionary decision to exclude the roughly \$1.35M locked LAB position from NAV until its first unlock; the result was NAV drawdowns across all vaults rather than any collateral liquidation. No public source indicates that any Kamino, Loopscale, or Jupiter Lend collateral position was liquidated in the episode; the impact reached the SPYx vault and the JitoSOL vault through the shared strategy book and the NAV mark (§2.2.2), not through a forced sale of collateral. PiggyBank’s disclosures are, however, silent on whether the borrow positions were affected at all — for instance, whether deleveraging on the lending venues was required to meet the losses on the derivatives leg — which is itself a transparency gap. A full treatment of the episode is reserved for §3–§4.^{7, 8}

In sum, no PiggyBank-specific collateral-position liquidation has been publicly disclosed or located as of the access date. The documented incidents are venue-level (LoopScale in 2025, Kamino in 2026) or disclosure-level (Jupiter Lend in 2025). They show that even reputable and widely used Solana protocols carry non-zero dependency risk under stress, but they do not by themselves prove that PiggyBank faces high counterparty/composability risk merely because it integrates with them. The stronger PiggyBank-specific finding is measurement: the protocol does not disclose which venues actually hold its collateral, at what loan-to-value, with what allocation weights, or under what routing and deleveraging rules. The venue layer is therefore a real but bounded composability risk, while the inability to quantify PiggyBank’s exact exposure belongs primarily to the transparency assessment.^{39, 31, 32, 34, 40, 41}

2.3 Delta-Neutral Funding-Rate Strategy, Leverage Mechanics, and Rebalancing Logic

The two preceding subsections established where PiggyBank’s capital sits before it reaches the strategy book: the vault and pb-token accounting layer (§2.1) and, for the non-stablecoin vaults, the collateral-and-borrow flow across Kamino, LoopScale, and Jupiter Lend (§2.2). This subsection analyzes what the protocol does with that capital once it arrives at the trading layer — the automated, delta-neutral, funding-rate book that is common to all three vaults. Because the strategy book is the ultimate source of both the protocol’s advertised yield and the loss that produced the June 2026 net-asset-value drawdowns, it is the analytical center of the protocol. As in the earlier subsections, the treatment distinguishes what the protocol asserts from what can be independently verified, flags transparency gaps and unconfirmed points explicitly, and evaluates the protocol’s claims against the established risk literature on funding-rate arbitrage rather than restating its marketing.

2.3.1 Delta-Neutrality in PiggyBank’s Specific Context: Position Construction and the Venue Set

A delta-neutral position is one whose net sensitivity to the price of the underlying asset — its delta — is approximately zero, so that the position’s profit and loss is driven not by the direction of price but by other cash flows, principally the funding rate paid on perpetual-futures contracts and any basis between the perpetual and its index. The canonical construction pairs a long position in an asset with an equal-notional short position in a perpetual on the same asset, so that a price move that gains on one leg loses an offsetting amount on the other while the short leg accrues funding. The largest production example of this design is Ethena’s USDe, which holds crypto collateral long and an equivalent short perpetual position, distributing the funding the short collects to holders; PiggyBank’s strategy belongs to the same family, and Ethena therefore serves throughout

this subsection as the relevant benchmark for what a mature implementation of the approach looks like.^{39, 40, 41}

In PiggyBank’s own description, the strategy is an “automated, delta-neutral strategy based on funding-rate arbitrage across approximately ten perpetual DEXs,” taking “long-short” positions on crypto majors and tokenized stocks with up to approximately 5x maximum leverage. The capital reaches this book by two routes established in §2.2: USDC vault deposits are deployed into the book directly, whereas deposits into the SPYx vault and the JitoSOL vault are posted as collateral on a lending venue, a stablecoin is borrowed against them, and the borrowed stablecoin is deployed into the same book. All three vaults therefore share a single strategy book — the architectural fact, demonstrated empirically in June 2026, that allows one position’s loss to mark down every vault (§2.1.1, §2.2.2).^{2, 1}

The protocol’s phrasing leaves the precise construction of the trade materially under-specified, and the ambiguity is consequential enough to set out in full. The word “long-short” across perpetual DEXs admits at least two readings. Under the first — a classic cash-and-carry, or spot-perp, basis trade — the long leg is a spot holding (an xStock or a major purchased with vault capital, or the deposited asset itself for the SPYx vault and the JitoSOL vault) and the short leg is a perpetual on that asset; the position collects positive funding while the spot leg supplies the offsetting delta. Under the second — a perpetual-to-perpetual, cross-venue funding trade — both legs are perpetuals on the same asset held on different DEXs: the position is long the perpetual on a venue whose funding is comparatively favorable to a long and short the perpetual on a venue whose funding is comparatively favorable to a short, netting to zero delta while harvesting the funding differential between venues. The phrase “long-short ... across approximately ten perpetual DEXs” is most naturally read as the cross-venue variant, but the documentation does not state which construction predominates, in what proportion, or how the choice is made; nor does it explain how a stablecoin (USDC), which carries no crypto delta of its own, becomes a delta-neutral “long-short” position without an intervening spot purchase. Which trade PiggyBank actually runs, and how the USDC vault’s deployment produces the advertised structure, is a transparency gap, and the answer is unconfirmed.⁴²

The single hard data point on what the book actually contains comes from the June 2026 episode, and it is instructive precisely because it does not match the marketing. The losing position was not a major-asset, market-neutral funding trade; it was, in the protocol’s own words, part of a “mid-cap basis trading strategy” — locked LAB tokens purchased at a discount through an over-the-counter desk and hedged with LAB perpetual shorts. That establishes three things the public strategy description does not: that the book includes spot-perp basis trades and not only cross-

venue perpetual trades; that it ranges beyond “crypto majors” into low-float, mid-capitalization tokens; and that at least one “spot” leg was an illiquid, time-locked OTC position rather than a freely tradable holding. The gap between the marketed strategy (major-asset, market-neutral, funding capture across approximately ten venues) and the observed strategy (a discretionary basis position in a thinly traded token that drew severe public criticism from ZachXBT) is the single most important finding of this subsection and is developed in §2.3.5.^{12, 7, 117}

The claim of dispersion across “approximately ten” perpetual DEXs warrants scrutiny on its own terms, because the count is presented as a strength but interacts with the structure of the Solana derivatives market in ways the protocol does not address. Solana does host on the order of ten or more perpetual venues — Drift, Jupiter Perps, Zeta Markets, Flash Trade, Adrena, Pacifica, Raydium Perps and Saros Perps (both built on Orderly), Decibel, and others appear on current directories — so a literal count of ten is plausible. Liquidity, however, is heavily concentrated rather than evenly spread: reporting on the venue mix indicates that activity clusters on Jupiter Perps, Drift, and the newer Pacifica, with the remaining venues carrying comparatively negligible volume. Spreading a book across many thin venues is not obviously risk-reducing; it can instead concentrate execution into markets where exit during stress is difficult, the very condition that the June 2026 LAB position encountered.^{43, 44}

A mechanical feature of the Solana venue set bears directly on how much of the “approximately ten” venue set is suitable for classic funding-rate harvesting, and the protocol does not address it. Solana perpetual venues do not all share the same economics. Drift’s own documentation describes a two-sided funding-rate mechanism used to align the mark price with the oracle price, with hourly updates and contract-tier clamps; that structure is compatible with a conventional funding trade in which one side receives funding. By contrast, Jupiter Perps documentation states that traders pay hourly borrow fees for leveraged positions based on position size and pool utilization, and Flash Trade documentation likewise describes continuous margin fees based on pool utilization. The implication is narrower than saying those venues are unusable: pool-style venues can still be used for hedging or execution, but they are not mechanically equivalent to order-book funding-rate venues for a classic “short receives longs-paid funding” harvest. The funding-harvest-suitable subset of the headline venue count is therefore materially smaller than the raw count implies, and PiggyBank does not disclose which venues, instruments, and fee mechanics dominate its actual book (transparency gap).^{127, 156, 157}

Finally, the claim that the book takes positions in “tokenized stocks” — necessary in particular for any genuine hedge of the SPYx vault’s S&P 500 exposure — sits uneasily with the present state of that market. Perpetuals on tokenized equities do exist: Kraken launched perpetual futures on ten

xStocks, including index products tracking the S&P and Nasdaq, with funding paid every eight hours and short positions supported, and Jupiter has announced plans for real-world-asset perpetuals beginning with a Tesla synthetic, but as a forthcoming feature rather than a live one. Aggregate stock-perpetual open interest remains small — on the order of a few billion dollars globally and described as early-stage — and the deepest venue is a centralized exchange (Kraken) rather than a Solana DEX. Where, and how deeply, PiggyBank can actually short an S&P-tracking instrument on-chain to neutralize the SPYx vault is therefore unclear, and the protocol publishes neither the venue nor the instrument used for that hedge (transparency gap); the point is left unconfirmed.^{46, 45, 47}

2.3.2 The Funding-Rate Capture Mechanic: When It Pays and When It Bleeds

A perpetual future has no expiry, so exchanges use a periodic funding payment, typically settled every eight hours, to tether the perpetual's price to its index. When the perpetual trades above the index (a premium, reflecting long-biased demand) the funding rate is positive and longs pay shorts; when it trades below the index (a discount, reflecting short-biased demand) the rate is negative and shorts pay longs. Because crypto perpetual markets are structurally long-biased — speculators more often pay up for leveraged upside — funding is positive a majority of the time, which is why a delta-neutral position that is net short the perpetual sits on the receiving side of the payment in the ordinary regime. This is the mechanism by which PiggyBank's book is intended to generate yield without directional exposure: it harvests the funding that leveraged longs pay, while the offsetting long leg (a spot holding or a long perpetual on another venue) cancels the price delta.

^{46, 40, 48}

The strategy is profitable only when the funding (and basis) captured over a holding period exceeds the full stack of costs incurred to run the position. Those costs are not trivial and are specific to PiggyBank's construction. On the order-book legs they include taker and maker fees and any borrow fee; on any pool-venue leg they include the hourly borrow fee that the position pays to the pool regardless of side; for the SPYx vault and the JitoSOL vault they additionally include the interest paid on the stablecoin borrowed against the collateral on Kamino, Loopscale, or Jupiter Lend (§2.2.2); and across all of them they include execution slippage on entry, exit, and every rebalance, plus the Solana transaction costs of maintaining the book. Net carry is the funding harvested minus this stack. The protocol advertises target net APYs of roughly 15% on the USDC vault and roughly 7% on the xStocks vaults (§1.1), figures that are presented as targets and that the protocol itself states are not guaranteed; the documentation does not publish the gross funding assumption, the cost stack, or the historical realized funding that would let a reader judge whether those targets are plausible across funding regimes (transparency gap).^{45, 2}

The conditions under which the trade ceases to pay, or actively loses money, are well documented in the literature and each maps onto PiggyBank’s design. The first and most common is a funding-rate reversal: rates can flip sign within a single interval, and a position entered for positive funding that turns negative for a sustained period transitions from earning a yield to paying one; practitioners identify the sign flip as the single most frequent way a basis book breaks. The second is basis compression and crowding: as more capital chases the same funding, the premium narrows and the expected return evaporates, and basis can widen against the position before it converges because a perpetual has no settlement date that forces convergence. The third is specific to the leveraged, lending-funded vaults: when the stablecoin borrow rate on the lending venue exceeds the funding harvested by the strategy, the position bleeds even if funding remains positive — the canonical negative-carry condition for liquid-staking-collateral leverage noted in §2.2.4 — so the SPYx vault and the JitoSOL vault carry an additional break-even hurdle that the USDC vault does not.^{39, 40}

The benchmark protocol’s response to these conditions is illuminating by contrast. Ethena treats negative funding as an anticipated state rather than an anomaly — “Negative funding rates are a feature, rather than a bug” — and manages it in two documented ways: it maintains a sized reserve fund that absorbs periods in which the combined yield turns negative, and, in practice, it has shown a preference for unwinding the perpetual hedge and rotating the backing into stablecoins rather than holding a losing short through a negative-funding regime. PiggyBank’s public materials describe no reserve or insurance fund, no published policy for handling persistent negative funding, and no commitment to unwind under defined conditions (§2.2.3); the protocol’s stated controls are confined to generalities — “automatic margin rebalancing and deleveraging based on market indicators” and a “partial TP/SL mechanism” — examined in §2.3.4. The absence of a funding-regime policy is a transparency gap with direct bearing on the durability of the advertised yield.^{41, 49, 2}

2.3.3 Leverage: Layered Application up to Approximately 5x, Amplification, and the Protocol’s Safety Claims

PiggyBank states that the funding-rate book is run with up to approximately 5x maximum leverage. Mechanically, leverage on a perpetual position means posting margin equal to a fraction of the position’s notional: at 5x, roughly one unit of margin supports five units of notional exposure on a leg, and the unrealized loss that would exhaust that margin — the distance to the leg’s liquidation price — is correspondingly about one-fifth of what it would be unlevered. The protocol does not publish the per-leg or per-vault leverage it actually targets within the 5x ceiling, the margin mode it uses (cross or isolated), or the venue-level liquidation parameters its positions

run against (transparency gap), all of which determine how close to forced liquidation a given leg sits at any time.²

A point the protocol's single leverage figure conceals is that the SPYx vault and the JitoSOL vault are subject to two stacked layers of leverage, not one. The first layer is created at the lending venue: posting the deposited asset as collateral and borrowing a stablecoin against it is itself a leveraging operation, governed by the venue's loan-to-value parameters (§2.2.2), whose ceilings range from roughly 70–80% on Kamino to as high as approximately 90% on Jupiter Lend. The second layer is the up-to-5x leverage applied to the perpetual positions funded with those borrowed stablecoins. These layers compound: a depositor's effective economic exposure to the strategy book can exceed the nominal 5x once the borrow layer is included, and a loss in the book is borne on a base that was already leveraged at the lending venue. The USDC vault, which reaches the book directly, carries only the second layer. PiggyBank does not disclose how the two layers combine, what aggregate effective leverage results, or how the borrow layer is sized relative to the perpetual layer (transparency gap); this stacking is the same coupling that, in §2.2.2, was shown to transmit a strategy loss into the collateralized vaults.

The critical and often-misunderstood property of leverage on a market-neutral book is that it does not amplify directional risk — which the hedge is designed to remove — but it does amplify precisely the risks the strategy remains exposed to. Five-times leverage multiplies the funding harvested when funding is positive (the source of the enhanced yield) and equally multiplies the funding paid when funding turns negative; it multiplies the loss from any execution slippage and from any residual delta left by an imperfect or stale hedge; and it narrows the liquidation buffer on each leg so that a sudden gap in price can liquidate a leg before the book can be rebalanced. In other words, leverage concentrates the strategy's exposure into the funding, liquidity, and execution channels that delta-neutrality does not address. A related, well-documented amplifier is cross-margin: where a leveraged short shares a margin account with other positions, the liquidation of one position can draw down the collateral backing the others, forcing the funding short out at the worst moment; the standard mitigation is to isolate the margin of the arbitrage leg. Whether PiggyBank isolates margin per position or runs cross-margin across its book is undisclosed (transparency gap) and material to how a single bad leg can propagate.^{39, 50, 51}

Against these mechanics, the protocol's safety claims are stated only at the level of marketing. PiggyBank says positions are protected by “automatic margin rebalancing and deleveraging based on market indicators,” a “partial TP/SL mechanism,” “deleveraging upon predefined thresholds,” and a “quantitative approach to protect your deposits”. None of these is accompanied by a numeric trigger, a target leverage or health level, a maximum position size or per-token concentration limit,

a description of the execution mechanism, or a statement of its responsiveness in fast markets (§2.2.3). The claim that leverage is “managed safely” therefore cannot be assessed from public materials; and the June 2026 episode demonstrates that whatever limits existed did not prevent a single mid-cap position from marking the flagship USDC vault down by roughly 15% (§2.3.5). The adequacy of the protocol’s leverage and risk controls is a material transparency gap, and the protocol’s own characterization that it acted before the position “crossed its risk limits” is difficult to reconcile with the size of the resulting loss.^{2, 7}

2.3.4 Rebalancing Logic: Triggers, Automation, and Latency

A leveraged delta-neutral book does not maintain itself. As the price of an asset moves, the notional values of the long and short legs diverge and the net delta drifts away from zero, so the legs must be resized to restore neutrality; margin must be added to or withdrawn from each leg to keep its health above the liquidation threshold as unrealized profit and loss accrues; positions must be rolled or repositioned as funding rates and cross-venue differentials shift; and the whole book must be deleveraged when conditions deteriorate. Rebalancing is the process that performs these adjustments, and for a leveraged book it is generally a high-frequency, latency-sensitive activity, because the cost of falling behind is paid in residual delta, in funding paid on a mis-sized leg, or in the forced liquidation of a leg whose margin was not topped up in time.⁵⁰

PiggyBank asserts that this is automated — “automatic margin rebalancing and deleveraging based on market indicators” — but discloses none of the operative detail. It does not state what the “market indicators” are, what thresholds trigger a rebalance or a deleverage, how often the book is evaluated, whether the controls act on the perpetual legs, on the lending-venue collateral positions, or on both, or what process executes the adjustments. Crucially, it does not clarify whether rebalancing is performed by an autonomous on-chain keeper, by an off-chain bot operated by the team, or by manual administrative action; the triggers, cadence, and execution path are all undisclosed (transparency gap) and the question of automation-versus-manual is unconfirmed. The architectural evidence assembled in §2.1 points toward substantial operator discretion rather than trustless automation: net asset value is computed off-chain and written on-chain by a trusted administrator through the `roll_epoch` instruction (§2.1.4), the program is administered by a key that controls pause and unpause (§2.1.1), and no oracle or autonomous risk module is documented. It is therefore likely, though not confirmed, that the strategy’s rebalancing is driven by the operator’s off-chain systems rather than by verifiable on-chain logic.²

Two structural features of PiggyBank’s design create latency exposures that compound the generic latency risk of rebalancing. The first is the epoch cadence. Net asset value is struck only every 48 hours and redemptions are batched and unwound over an epoch (§2.1.3), which means the

accounting and liquidity machinery operates at a 48-hour boundary frequency. Whatever intra-epoch rebalancing the protocol performs is, by its own framing, distinct from this epoch roll and undocumented; but the mismatch between a book that generally needs continuous adjustment and an accounting layer that reports every two days leaves a wide window in which a position can deteriorate without being marked, and in which a redeeming user transacts against a stale NAV (§2.1.3, §2.1.4). The second is on-chain execution latency. Solana’s sub-second slot times are an advantage for on-chain rebalancing, but pool-based perpetual venues remain exposed to oracle update lag, and a leg that must be resized or closed during a fast move competes with every other actor for blockspace and liquidity. A delta-neutral book is only as neutral as its most stale leg, and during volatility the delta can drift, and a leg can reach liquidation, faster than a cross-venue rebalance can be executed and confirmed.^{39, 50}

The June 2026 episode exposed a more fundamental limit of rebalancing than latency: rebalancing cannot neutralize a leg that cannot be traded. The long leg of the LAB position was a time-locked OTC holding that could not be sold until its first unlock; when the short leg moved against the book and funding turned costly, the protocol could adjust or close the liquid short but could do nothing with the illiquid long except hold it and exclude it from NAV. No rebalancing logic, however fast or well-automated, can keep a position delta-neutral when one of its two legs is frozen. This is a structural mismatch between the liquidity of the two legs, not a tuning problem, and it is examined as a failure mode in §2.3.5.^{7, 12}

2.3.5 Critical Analysis: Breakdown Conditions, Generic Failure Modes, and Their Application to PiggyBank

The risk literature on funding-rate arbitrage is mature and converges on a recognized set of failure modes; the strategy is delta-neutral on price but is emphatically not risk-free. The principal modes are: funding-rate reversal, in which rates flip sign and the position pays rather than receives; basis blow-out and compression, in which the spread between perpetual and index widens against the position or collapses faster than expected as the trade crowds; liquidation of the leveraged leg, in which a sharp adverse gap exhausts the short’s margin before the offsetting (and possibly less liquid) long can be realized; counterparty and venue risk, in which an exchange halts withdrawals, suffers an outage, or fails outright, stranding margin or breaking the hedge; execution, slippage, and latency risk, in which delays or thin liquidity prevent the legs from being entered, exited, or rebalanced at the prices the trade assumed; and liquidity risk in the underlying, in which a thinly traded asset cannot be exited without moving its own price. The strategy’s defining characteristic is that these residual risks are largely uncorrelated with ordinary price direction but become highly correlated with one another during market stress: the same deleveraging event that drives funding

negative also gaps prices, thins liquidity, strains venues, and crowds the exit — so the book’s several “neutral” exposures tend to fire together rather than independently.^{40, 52, 39, 51}

Mapped onto PiggyBank, every one of these modes has a concrete channel, and several are aggravated by design choices documented earlier in this section. Funding reversal is amplified by the up-to-5x leverage (§2.3.3) and, for the lending-funded vaults, by the additional negative-carry hurdle when the stablecoin borrow rate exceeds harvested funding (§2.3.2, §2.2.4). Liquidation of the leveraged leg is made more likely by leverage and by the unknown margin mode, and is the structural category that the February 2026 Kamino cascade showed PiggyBank’s underlying JitoSOL collateral, stablecoin-borrow position falls squarely within (§2.2.4). Counterparty and venue risk is elevated, not reduced, by dispersing the book across approximately ten venues of widely varying maturity, several of them early-stage and thinly capitalized (§2.3.1), and by the documented incident history of the lending venues the strategy depends on (§2.2.5). Execution and latency risk is compounded by the 48-hour epoch cadence and by reliance on operator-run, off-chain rebalancing of uncertain responsiveness (§2.3.4). Liquidity risk in the underlying is the mode that the protocol’s own conduct made acute by taking a position in a low-float token. The cumulative picture is of a strategy whose residual, non-price risks are not merely present but systematically amplified by leverage, by venue dispersion into thin markets, by a shared strategy book that propagates any single loss across all vaults (§2.1.1), and by a low-frequency, discretionary operating model.

The comparison with the benchmark sharpens the assessment. Ethena runs the same family of trade but with a markedly more conservative and transparent risk architecture: it confines its hedges to deeply liquid major assets, operates across a documented venue set with institutional off-exchange-settlement custody and a public per-venue hedge dashboard, maintains a sized reserve fund explicitly to absorb negative-funding periods, and has shown a willingness to unwind the hedge and rotate to stablecoins rather than hold a losing short. Even with those safeguards, Ethena was tested under stress: in the October 2025 deleveraging USDe briefly traded as low as roughly \$0.65 on Binance, with negative funding and exchange outages cited as the proximate strains. PiggyBank exhibits the inverse profile on each axis that mattered — it ventured into a low-float token rather than majors, publishes no reserve fund, no position addresses, and no per-venue transparency, and held an illiquid locked leg it could not unwind — which is to say it took on the family’s risks while omitting the controls the benchmark uses to survive them.^{48, 41, 49, 53}

The June 2026 LAB episode is best read as a clean, simultaneous materialization of the strategy’s principal failure modes, and its mechanics are worth tracing precisely because they show how a position marketed as “delta-neutral” produced a double-digit loss. The trade began roughly a

month earlier when PiggyBank purchased locked LAB tokens through an OTC desk for about \$100,000, a position it stated represented nearly 2% of the portfolio at the time and formed part of its mid-cap basis strategy. The hedge was a LAB perpetual short intended to neutralize the price exposure of the locked long. LAB's price then rose violently — amid alleged manipulation and against a backdrop in which ZachXBT alleged LAB-linked insider control of more than 95% of the token's supply — which drove the short into a large mark-to-market loss, while funding on the perpetual turned deeply negative, so that the short was simultaneously bleeding mark-to-market and paying funding to hold. In a textbook basis trade the offsetting gain on the long would have neutralized the short's loss; here it could not, because the long was a time-locked OTC position that could not be sold to realize its appreciation. With the hedge no longer economic to maintain, PiggyBank closed the short — crystallizing the cash loss — and was left holding the locked long, which it valued at about \$1.35 million but excluded from NAV until the first unlock scheduled for 14 August, describing the exclusion as “the fairest and most transparent way” to treat depositor liquidity. The accounting result was a marked drawdown across the shared book: approximately –15% on the USDC vault, –12% on the SPYx vault, and –9% on the JitoSOL vault.^{12, 11, 7, 8}

The episode instantiates at least four of the generic failure modes at once. Funding-rate reversal: the short paid deeply negative funding rather than collecting it. Liquidity risk in the underlying: the token was too thin to support a clean two-sided position, and the long leg could not be exited at all. Basis and hedge breakdown: the supposed offset failed because the two legs differed not in price exposure but in tradability, a liquidity mismatch between a frozen long and a liquid short that no rebalancing could repair (§2.3.4). And valuation discretion under illiquidity: the size and timing of the reported loss turned on the operator's discretionary decision to exclude the locked leg from NAV (§2.1.4), so that even the magnitude of the drawdown was a marking choice rather than a market-determined fact. Above all, the episode demonstrates the gap identified in §2.3.1 between the marketed strategy and the realized one: a vault sold to depositors as low-risk, stablecoin, market-neutral funding capture sustained a 15% loss because the book in fact held a leveraged, directional-in-effect bet on a low-float token subject to public manipulation allegations — the precise mismatch ZachXBT's critique targeted, and one that ordinary funding-rate-arbitrage risk disclosures do not contemplate. A full treatment of the incident's disclosure, governance, and aftermath is reserved for §3–§4; here it serves as the empirical confirmation of the structural fragilities this subsection has analyzed.^{7, 8}

The overall verdict on the strategy layer is therefore qualified rather than dismissive. The delta-neutral funding-rate approach is a legitimate, widely deployed strategy that can, in the ordinary positive-funding regime and with conservative parameters and liquid instruments, produce the kind of yield PiggyBank advertises; the benchmark protocol demonstrates that the design can be

run at scale. But “delta-neutral” is a statement about price exposure only, and it does not imply that the strategy is low-risk: the approach retains substantial funding, liquidity, counterparty, execution, and valuation risks that are not neutral, that leverage amplifies, and that become correlated precisely in the stressed regime when funding flips. PiggyBank’s specific implementation magnifies these residual risks through up-to-5x leverage layered atop a borrow leg, through dispersion into thin Solana venues whose deepest member does not even pay funding to shorts, through a shared strategy book that socializes any loss across all vaults, and through a discretionary, off-chain, 48-hour operating model with none of the reserve-fund, transparency, or asset-quality safeguards the mature benchmark relies on. The June 2026 loss was not an aberration external to the strategy; it was a realization of the strategy’s known failure modes, made larger by the protocol’s own design and conduct choices.

3. Security, Disclosure, and Technical Risk Surface

3.1 The Formal Land Audit — Scope, Methodology, Findings, and Limitations

Formal Land audit: scope and selected findings excerpt

Source excerpts from the audit itself: repository/commit/scope/manual review, trusted-admin characterization, and selected instruction findings.

A. Scope, method, and trusted-admin characterization

Security Audit - Piggy Bank

1 Introduction

Here is a summary of the audit scope.

- **Repository:** <https://github.com/zsociety-io/piggybank-program> (private at the time of the audit)
- **Commit hash:** 6c930b34b8e1fe2151ccb60c3b6f60f514034b91
- **Scope:** All the Rust code in the `program/` folder.
- **Methodology:** Manual review.

PiggyBank is implemented as a DeFi protocol on Solana, providing for the users managed funds by a trusted admin in order to get good performance. The code is self-contained in a single smart contract, using the Pinocchio library to make all calls and checks explicit, and potentially improve efficiency on the long run. It is generally well written and thought, and the team very responsive, so it was pleasant to iterate over the contract's code.

B. Selected finding responses on pause-state and roll_epoch arithmetic

Security Audit - Piggy Bank

b6c7f7: Also check that `name` is ASCII alphanumeric like ticker, to prevent "Unicode attacks" by weird pretty-printing in the logs.

Answer Yes we will do the change, keeping the logs as an important information to better use the transaction explorers.

063f9d: Also check if we are on "pause" for this instruction.

Answer This is fine for this kind of operation to not be put on pause. This is by design as the admin can unpause the program anyway.

7b6ce8: There is no need to add the `mint` pubkey in the data parameters, as it is also given as an account.

Answer Yes, this is a dead parameter, we will remove it.

Figure 4. Formal Land audit excerpts showing the repository/commit/scope/manual-review methodology, the trusted-admin characterization, and selected finding responses concerning pause-state and roll_epoch arithmetic. Source reviewed 15 June 2026. Source: #9.

The preceding sections have repeatedly invoked the Formal Land audit as the protocol's single published security artifact: §1.3 recorded its existence, and §2.1-§2.2 drew on it to establish the program's single-contract architecture and the boundary of its audited perimeter. This subsection examines the audit itself in detail — who Formal Land is, what the engagement actually covered, what methodology it employed, what it found, and, most importantly, what assurance it does and does not provide for a protocol that manages user funds in leveraged strategies. As elsewhere in this paper, the analysis distinguishes what can be verified from the primary document from what

the protocol’s marketing implies, and it treats the existence of an audit as a fact to be evaluated rather than as evidence of safety. The audit document was located, retrieved, and read in full for this section; it remains publicly accessible at the URL linked from the protocol’s site, so the analysis below rests on the primary source rather than on a second-hand description.⁹

3.1.1 Formal Land: Background, Verification Methodology, and Track Record

Formal Land is a formal-verification firm founded in 2021 with a public track record in blockchain and protocol-level verification. Its reputation mainly comes from work on foundational infrastructure, including Tezos-related formal-verification projects and other low-level verification engagements. That pedigree is relevant because it shows that the auditor is credible; however, it should not be overstated. A strong formal-verification background does not mean the PiggyBank engagement verified the full economic strategy, off-chain NAV process, external venue exposure, or operator-controlled strategy layer.^{54, 55, 56, 57, 58, 59, 60}

The most important limitation for this report is methodological. Formal Land’s signature strength is formal verification, but the PiggyBank report itself was a manual audit of a Rust Solana program, not a machine-checked proof of the entire protocol. Formal Land’s Rust verification tooling was also described as being under active development, which helps explain why the engagement did not take the form of complete formal verification. For PiggyBank, the audit therefore supports confidence in the reviewed on-chain program perimeter, but it does not validate the strategy book, NAV reporting, venue positions, admin-key operations, or economic risk controls.^{56, 61, 9}

3.1.2 Audit Scope: Repository, Commit, Code Perimeter, and Exclusions

The audit document states its scope concisely. The subject was the repository `github.com/zsociety-io/piggybank-program`, which the report records as private at the time of the audit, pinned to commit hash `6c930b34b8e1fe2151ccb60c3b6f60f514034b91` (the abbreviated form `6c930b34` was used in §2.1), and the scope is given as all of the Rust code in the `program/` folder. The repository is not among the public repositories of the zSociety GitHub organization as of the access date, so the source the audit examined remains closed and the audited commit cannot be independently inspected against the deployed bytecode. The report is dated 27 October 2025 — three days before the documented mainnet launch of 30 October 2025 (§1.3) — which fixes the audited snapshot firmly in the pre-launch period and compresses to roughly seventy-two hours the window in which any agreed code changes could have been made before the program went live.^{9,}

Within that perimeter the audit is comprehensive in the narrow sense that it addresses the whole on-chain program rather than a selected component. The files that drew remarks expose the program’s structure and correspond to the architecture set out in §2.1: the instruction handlers `create_vault`, `claim`, `roll_epoch`, and `update_admin`, together with the supporting modules `lib.rs` (the program entry point), `token_helper.rs` (pb-token minting and burning), and `validation.rs` (account and input validation). This is consistent with the finding in §2.1.1 that the entire protocol is contained in a single self-contained Rust program built on the low-level Pinocchio framework, a design the auditor characterized as making all calls and checks explicit.⁹

What lies outside the scope is more consequential than what lies inside it, and almost all of it is excluded by design rather than by oversight. The audited perimeter ends at the vault contract and does not extend to: the external Solana lending venues — Kamino, Loopscale, and Jupiter Lend — through which the non-stablecoin vaults route collateral and borrow stablecoins, so that a substantial share of depositor risk sits in third-party code the audit did not examine (§2.2.1, §2.2.4); the protocol’s off-chain components, including the process that computes net asset value, the systems that rebalance and deleverage the strategy book, the over-the-counter desk through which positions such as the June 2026 LAB holding were acquired, and the operator’s wallets and open positions across approximately ten perpetual venues (§2.1.4, §2.3.4); the strategy and economic logic itself — the funding-rate-arbitrage construction, the up-to-5x leverage, position sizing, and per-token concentration limits (§2.3); the custody and governance of the administrative key, which the report treats as a given rather than an object of review; and the correctness and manipulation-resistance of any price oracle or NAV mark. The deployed program’s on-chain identifier is, moreover, not stated anywhere in the report (§2.1.1). The scope is thus accurately described as all the Rust code in the program folder; the difficulty, developed in §3.1.5, is that the on-chain program is precisely the portion of the protocol a user can already inspect on-chain, while the portions a user cannot independently verify lie entirely outside the audited code (§2.1.5).

3.1.3 Methodology: Manual Review, Not Formal Verification

The single most important fact about this audit is its stated methodology. Under the heading that summarizes the engagement, the report gives the methodology in two words: “Manual review”. There is no formal verification, no machine-checked proof, and no specification artifact of the kind that constitutes Formal Land’s published portfolio (§3.1.1); nor does the report describe any fuzzing, symbolic execution, or automated static-analysis campaign. The only tooling the report mentions is a recommendation that the team adopt cargo fmt and cargo clippy and wire them into a continuous-integration script — advice directed at the team’s own future practice rather than a technique the auditor applied (§3.1.4).⁹

The gap between this and the firm’s reputation is the crux of the section. Formal Land’s name, its self-description, and its entire public record are built on formal verification — the production of machine-checked proofs that a property holds for all inputs (§3.1.1). The PiggyBank engagement delivered none of that. The two forms of assurance are not interchangeable: by Formal Land’s own account, formal verification covers behaviors that are difficult to validate exhaustively with tests, whereas a manual review is an expert inspection of a code snapshot, inherently non-exhaustive and bounded by the reviewer’s attention and the time available. The capability the protocol’s marketing evokes when it cites “the formal verification work for Tezos” was, on the evidence of the report itself, not applied to PiggyBank’s code.⁵⁶

Several considerations plausibly explain the choice of a manual-only engagement, though none is stated in the report and each is therefore offered as inference rather than fact. Formal Land’s Rust-to-Rocq pipeline was still maturing in October 2025 (§3.1.1), so the firm was less equipped to formally verify a Rust Solana program than an EVM or OCaml target; the program’s mutation-oriented, low-level Pinocchio style is of the kind the firm has flagged as needing rewriting before verification (§3.1.1); and the roughly seventy-two-hour window before launch (§3.1.2) left no room for proof engineering, which is measured in weeks rather than days. The point is not that manual review is illegitimate — it is a standard and valuable engagement type, and Formal Land’s own materials list a one-time audit of code at a fixed point in time among its offerings — but that the assurance actually delivered is a manual review, and should be read as such rather than as the formal verification the firm is known for.⁵⁶

The limitations of a manual review in this setting follow directly. The approach cannot exhaustively cover the program’s execution paths or input space; the arithmetic-overflow remark discussed in §3.1.4 is precisely the class of defect that formal verification is designed to exclude with certainty and that manual review can only flag heuristically, on the strength of a reviewer noticing it. The review produced no severity taxonomy and no verified record of remediation (both developed in §3.1.4). And the document is brief — six pages including the title page and table of contents, with the substantive findings occupying roughly three pages — which is consistent with a light-touch review rather than the exhaustive, line-by-line audit one would expect to underwrite a protocol entrusted with managing user funds. In fairness, the review was evidently competent and collaborative: the auditor judged the code generally well written and well conceived, found the Pinocchio design’s explicit calls and checks helpful, described the team as very responsive, and in at least one instance modified the code directly during the engagement (the deserialization change recorded in §3.1.4). The critique that follows is structural — about the form and reach of the engagement — and not a comment on the reviewers’ skill.⁹

3.1.4 Findings: Inventory, Severity, and Remediation Status

Two structural observations frame the findings before any individual item is considered. First, the audit assigns no severity ratings. It presents what it calls a deduplicated list of findings, discussed directly with the team, each tagged with a generated identifier and grouped by source file; there is no critical, high, medium, low, or informational classification anywhere in the document. This is a departure from contemporary practice, in which Rust and Solana audit engagements typically deliver severity-ranked findings and re-test math-heavy code paths for latent integer overflow after fixes. The absence of a severity scheme means the reader cannot triage the findings or gauge residual risk from the document itself, and must instead reconstruct relative significance independently.^{9, 62}

Second, the document records the team's responses to each remark, and almost all of them are written in the future tense — the team will remove a parameter, will add a check, will make a change. The report does not record any verification that these changes were implemented, contains no re-audit, and gives no commit hash for a remediated version of the code. Read against the roughly seventy-two-hour pre-launch window (§3.1.2), this means the audit captures the team's stated intentions to remediate rather than confirmed resolutions present in the deployed program. Strictly, then, no finding can be characterized as verified-resolved on the basis of this document; the most that can be said of the agreed items is that they were acknowledged with a stated remediation intent. Two findings were declined outright, and one was applied by the auditors themselves during the review.

Applying an analyst's severity reading — expressly the present author's, not the auditor's — the findings divide into a small tier of genuine substance and a larger tier of code hygiene and defensive hardening. Based on the published audit descriptions, the findings were not presented as critical or high severity, and they do not appear from the report text to describe a direct loss-of-funds exploit. The more material items cluster in two places: the program's accounting arithmetic and its account-consistency validation.

Most material (accounting and validation). The remark with the clearest security weight, identifier fb87b4 in roll_epoch.rs, is that the instruction performs signed 128-bit (i128) arithmetic whose additions and subtractions can overflow and should be replaced by checked operations. Its significance derives from the role of roll_epoch established in §2.1.3–§2.1.4: this is the instruction that writes each epoch's profit-and-loss into vault state and therefore governs the reported net asset value on which every pb-token's redemption value depends. An overflow in this path is an accounting-corruption risk rather than a cosmetic one. The team agreed to address it, either by adding invariants that keep values out of the dangerous range or by re-expressing the arithmetic in

unsigned 128-bit form with guard conditions; the auditor added the technical observation that Rust has no native 256-bit integer types, so that a fully general fix requires either checked arithmetic with invariants or an external big-integer crate at additional runtime cost. This is exactly the category of defect that formal verification eliminates exhaustively and that a manual review can only flag, and the agreed fix is recorded as an intention rather than a verified change. A second material cluster concerns missing validation on the user-facing redemption path: identifier 62a8e8 in `claim.rs` asks whether the supplied vault account should be checked with the corresponding validation function, and identifiers 52e482 and 641271 in `validation.rs` ask whether the vault and redemption identifiers should be validated and whether the instruction data and the program-derived-address bump should be deserialized and checked. On Solana, absent account-consistency and bump validation is a recognized vulnerability class (account substitution), and these checks sit on the path that settles funds to users. The auditor downplayed the first — noting it would merely be at the user’s own loss to supply an incorrect vault — but observed that the same gap recurs across the other user instructions that involve the vault, indicating a systematic pattern at the audited commit; the team agreed that all three are additional security checks to be added. Taken together these four items (one overflow, three validation gaps) are the audit’s substantive content, and all were acknowledged with stated but unverified remediation (cross-referenced to the trust-model discussion in §2.1.1–§2.1.2).⁹

Defensive hardening (low). Identifier b6c7f7 in `create_vault.rs` recommends constraining the vault name to ASCII alphanumeric characters, as the ticker already is, to prevent “Unicode attacks” in which misleading pretty-printing of the name in transaction-explorer logs could deceive a reader; this is a log- and interface-spoofing concern rather than a fund-threatening one, and the team agreed to it. Two findings were declined. Identifier 063f9d, also in `create_vault.rs`, asks that vault creation honor the program’s paused state; the team declined, stating that exempting this operation is by design because the administrator can in any case unpause the program — an answer that itself confirms the administrative key’s blanket pause-and-unpause authority over the entire program (§2.1.1). Identifier cdb377 in `lib.rs` asks whether the program identifier should be checked; the team declined, reasoning that reliance on the deployment address suffices and that the check carries a small performance cost. Finally, the section the report labels “Copy operations” records a change the auditors made themselves, replacing a pattern that deserialized a parameter by first copying a slice and then translating it to an integer with a direct translation from the data slice — a minor correctness-and-efficiency improvement applied during the engagement rather than left as a recommendation.⁹

Code hygiene and clarity (informational). The remainder are maintainability items. The general advice recommends adopting `cargo fmt` and `cargo clippy` in continuous integration and preferring

pub(crate) over pub to surface dead code such as unused setters; the clippy recommendation can in principle catch real defect classes, but as framed it is advisory and carries no specific commitment. Three findings flag redundant or dead parameters — `underlying_amount` in `claim.rs`, which duplicates a value already held in the redemption account (cb5626); the mint public key in `create_vault.rs`, which is also passed as an account (7b6ce8); and an unused data parameter in `update_admin.rs` (0669b4) — each of which the team agreed to remove. One finding, 3bf4a4 in `roll_epoch.rs`, asks what happens when the period's `vault_transfer` is zero given that state has already been mutated earlier in the instruction; the team confirmed the behavior is intentional and agreed to add an empty else branch purely for clarity. And ef5640 in `token_helper.rs` suggests using the key already defined in the constants of `validation.rs` as a single source of truth, which the team agreed would simplify the logic.⁹

In aggregate the report contains on the order of thirteen findings and remarks. The auditor did not assign severity labels, so any severity grouping in this report is an analytical reading of the published descriptions rather than a substitute for the auditor's own classification. On that basis, the more significant items appear concentrated in the accounting and validation paths, while the remaining items read as lower-severity code hygiene or defensive-hardening issues. Two findings were declined as design choices, one was applied by the auditors during the review, and every remaining agreed item is recorded as a stated intention to remediate rather than as a confirmed, re-audited fix in the deployed program.

3.1.5 Critical Evaluation: What the Audit Guarantees, What It Does Not, and Sufficiency for a Leveraged-Strategy Protocol

Taken on its own terms, the audit fairly establishes a modest set of positives. A competent reviewer with deep formal-methods experience inspected the entire on-chain program at a defined commit, judged the code generally well written, surfaced a real arithmetic-overflow risk in the core accounting instruction and a recurring account-validation gap on the redemption path, and documented — candidly — that the protocol provides users with funds managed by a trusted administrator. The engagement was iterative and the reviewer improved the code directly in at least one instance. Within its perimeter, the review is a reasonable, if light-touch, examination, and there is no basis in the document for impugning the reviewers' competence.⁹

What the audit does not guarantee is more extensive, and falls into four heads. It is a manual review and not a formal verification, so it offers no exhaustive coverage of inputs and no machine-checked proof; its assurance is bounded by the reviewer's attention on a single snapshot (§3.1.3). It assigns no severity ratings, so it provides the reader no means of triage or of weighing residual risk (§3.1.4). It records no verified remediation, so the program actually deployed at launch is not

confirmed to contain the agreed fixes; the audited commit is pre-launch and the fixes are stated intentions (§3.1.2, §3.1.4). And its perimeter is narrow — the on-chain Rust program only — which, by the analysis of §2.1.5, is the very part of the protocol that a user can already read from the chain, while the parts a user cannot independently verify (the off-chain NAV process, the existence and value of the backing assets, and the strategy) lie wholly outside the audited code.

The decisive question is whether a code audit of this kind is sufficient for a protocol that manages user funds in leveraged strategies, and the answer is that it cannot be — not because of any deficiency in execution, but as a matter of what such an audit can attest. A review of the vault contract can establish, at most, that the on-chain bookkeeping is implemented as written. It cannot establish that the strategy is prudent, that leverage is sized conservatively, that positions are diversified rather than concentrated in an illiquid micro-cap, that the operator will mark illiquid holdings honestly and on schedule, or that the administrative key is held safely and cannot be misused. These are the dimensions on which a leveraged-strategy yield protocol actually succeeds or fails, and none of them is a property of the Rust in the program/ folder.

The June 2026 episode is the empirical demonstration of this disjunction. The loss originated entirely outside the audited perimeter: a discretionary basis trade in an illiquid token subject to public manipulation allegations, acquired off-chain through an over-the-counter desk, hedged with a perpetual short, and marked at the operator's discretion when the locked leg was excluded from net asset value (§2.3.5, and the remainder of §3 and §4). No conceivable manual review of the vault program at commit 6c930b34 could have anticipated, flagged, or prevented it, because the relevant risk lived in the strategy and the off-chain valuation rather than in the on-chain code. The audit and the loss concern disjoint parts of the system. It is telling, too, that the independent scrutiny the episode attracted — most prominently from the on-chain investigator ZachXBT — was directed at risk management, position sizing, and the decision to expose depositor capital to a token he characterized as a scam, and that none of the incident coverage located for this paper made any reference to the Formal Land audit at all. The audit was simply orthogonal to what went wrong. Even read at face value, and even on the counterfactual assumption that every finding was remediated, it provides assurance over a small — and arguably the least hazardous — fraction of the protocol's risk surface.^{7,8}

The areas the audit leaves untouched are uncoverable by this kind of engagement by design, and each would require a different assurance mechanism. The off-chain components — the NAV computation, the rebalancing and deleveraging systems, the over-the-counter interactions — the strategy and economic logic, the custody and governance of the administrative key, the third-party lending venues, and the integrity of any price oracle all sit outside a static review of the on-chain

program. Addressing them would call for machine-checked verification of the accounting invariants (the very capability Formal Land specialises in but did not apply here), an on-chain or independently attested proof-of-reserves and NAV process, published position addresses, explicit and enforced risk and concentration limits, a multi-signature or time-locked arrangement over the administrative key, and an operational and economic review of the strategy itself — none of which the protocol publishes (§2.1.4, §2.1.5, §2.3).

This sharpens into a final observation about how the audit is used as a trust signal. The protocol advertises that it has been audited by Formal Land, which it describes as the auditing team behind the formal-verification work for Tezos with Nomadic Labs. The claim is literally true and the pedigree is real, but it invites two inferences the document does not support: that PiggyBank’s code was formally verified, when in fact it was manually reviewed (§3.1.3); and that being audited implies the protocol is safe, when the audit covers a fraction of the risk surface and is silent on the very dimensions that produced the only material loss to date (§3.1.4–§3.1.5). The presence of the audit is a genuine but narrow positive; deployed as evidence for the safety of the protocol as a whole, it is substantially overdrawn.²

In sum, the Formal Land audit is best read as a limited, manual, pre-launch review of the on-chain bookkeeping — useful within its perimeter, refreshingly candid about the protocol’s reliance on a trusted administrator, but neither a formal verification nor an assurance that reaches the strategy, valuation, key-management, and external-venue risks that constitute the bulk of a depositor’s exposure. Its existence should raise a reader’s confidence in the narrow correctness of the vault program’s code, and in nothing more.

Key Takeaway: The Formal Land audit is useful evidence that PiggyBank’s on-chain vault program received a competent pre-launch manual review, but it should not be treated as protocol-wide assurance. It reviewed a defined Rust code snapshot and identified issues around accounting, validation, and admin-controlled design, but it did not formally verify the program or confirm that all intended fixes were deployed.⁹

For depositors, the main point is narrower: the audit can increase confidence that the vault bookkeeping was inspected, not that PiggyBank’s strategy was safe. It does not validate the off-chain NAV process, live venue positions, OTC trades, leverage sizing, asset concentration, liquidation routing, oracle chain, admin-key custody, or the external lending and perpetual venues used by the protocol. Those are the areas where most of the economic risk sits.^{2, 9}

This distinction matters because the LAB loss did not appear to come from a bug in the audited vault program. It came from strategy selection, position structure, liquidity, funding conditions, and discretionary NAV treatment — all outside the audit perimeter. The audit is therefore a

narrow positive for code hygiene, but it does not materially resolve PiggyBank’s larger transparency, strategy, valuation, and governance risks. 7, 8, 9

3.2 Protocol-Disclosed Risks vs. Actual Observed Risk Exposure

Section 3.1 established that PiggyBank’s single published security artifact is a narrow, pre-launch review of the on-chain bookkeeping and does not by itself address strategy, valuation, key-management, or external-venue risk. This subsection therefore tests a narrower question: whether the protocol’s own risk disclosures adequately describe the risks created by the mechanics analyzed in Section 2 and by the observed LAB and Drift episodes. Broader transparency questions — team identity, public metrics, partnership claims, NAV attestation, and first-mover marketing — are reserved for Section 5.2 to avoid duplicating that analysis.

3.2.1 The Disclosed Risk Surface: An Inventory of PiggyBank’s Risk Representations

PiggyBank’s risk disclosures are distributed across three locations of very unequal substance, and a reader assembling the protocol’s complete risk representation must consult all of them, because no single document is comprehensive. The three are the marketing site (a homepage FAQ entry and a footer disclaimer), the Terms of Service (a separate PDF), and the GitBook documentation. A fourth document, the Privacy Policy, governs data handling rather than investment risk and is treated only in passing.

The most user-facing disclosure is the homepage FAQ answer to the question “What are the risks?”, which is also the disclosure a depositor is most likely to read. It opens by characterizing the vaults as employing conservative strategies and automated risk management while conceding that all DeFi activity carries inherent risk, and then enumerates four specific hazards: that market volatility can reduce the value of collateral and lead to possible liquidation if thresholds are breached; that smart-contract vulnerabilities or failures in integrated platforms such as Kamino may affect funds; and that stablecoin depegging or network disruptions could affect yield performance or withdrawals. It closes by reasserting that the delta-neutral strategy features tight risk management, including automatic margin rebalancing, deleveraging upon predefined thresholds, and a partial take-profit/stop-loss mechanism. The footer disclaimer repeats the same catalog in compressed form — liquidation risk, smart-contract vulnerabilities, stablecoin depegging, and network disruptions — and adds three further representations: that yields are generated through delta-neutral funding-rate arbitrage and are not guaranteed, that past performance does not guarantee future results, and that the protocol is unavailable to United States residents and other restricted jurisdictions. It directs the user to “read our full risk disclosure before

depositing”. Notably, no standalone “full risk disclosure” document is linked anywhere on the site; the resources the footer and navigation expose are the GitBook documentation, the audit PDF, the FAQ, the Terms of Service, and the Privacy Policy. The operative “full risk disclosure” is therefore, in practice, the FAQ answer together with the Terms; the reference to a discrete fuller document overstates what is published, and the discrepancy is a minor transparency gap.²

The substantive risk-allocation disclosures reside in the Terms of Service, dated to an initial publication of 27 November 2025 and last updated 3 April 2026. Section 6 of the Terms is the protocol’s full risk catalog, and it is extensive in the manner characteristic of a DeFi front-end template: it disclaims all representations and warranties and provides the Site on an “AS IS” and “AS AVAILABLE” basis (Terms §6.2); excludes liability for consequential, incidental, punitive, and economic damages, including lost profits (Terms §6.1); enumerates a broad set of blockchain and smart-contract hazards under Terms §6.7, including the novelty and experimental character of the technology, asset-price volatility, regulatory uncertainty, cryptography and quantum risk, chain-fork handling, dependence on third-party wallet and node software, and — material to §3.2.5 below — upgrade-authority risk; warns that users “may be exposed to a risk of total loss and forfeiture of all relevant digital assets” in the event of malfunction, bug, hack, or adverse exercise of upgrade authority (Terms §6.7); disclaims responsibility for the authenticity, legality, or value of any token transacted (Terms §6.3); and disclaims any professional, accounting, or legal advice (Terms §6.4). The Terms further state that the Site Operator is not registered with or supervised by any financial regulator (Terms §2.9), reserve to the operator sole and absolute discretion to discontinue, limit, or deny access to the Site (Terms §§2.5–2.6), and impose a suite of liability-limiting and procedural provisions: a six-month survival period for claims (Terms §6.5), a cap on the operator’s aggregate liability of one hundred United States dollars (Terms §6.11), mandatory confidential individual arbitration seated in the British Virgin Islands under British Virgin Islands governing law, and an express waiver of class actions and jury trial (Terms §7). Terms §5.5 has each user represent that they are sophisticated, have conducted their own independent investigation, and have not relied on any statement of the operator. Taken together, Terms §§5–7 construct a comprehensive legal shield and disclose, at the level of generality typical of the genre, that a depositor may lose everything.⁶³

The GitBook documentation, by contrast, contains essentially no risk disclosure at all. Its introduction describes the vault and pb-token mechanics and the epoch/NAV cadence, and asserts that the protocol is “built around: transparency, risk management, and resilient infrastructure” with a team experienced in blockchain, financial engineering, and quantitative research, but it offers no dedicated risk section, no enumeration of hazards, and no governance disclosure. The asymmetry is itself worth recording: the document a user would most naturally treat as the protocol’s technical

reference is silent on risk, while the risk content lives in marketing copy (the FAQ) and a legal instrument (the Terms) whose framing, examined next, does not match the product.¹

A structural feature of the Terms deserves emphasis at the outset because it conditions everything that follows. The Terms do not describe PiggyBank as an actively managed, leveraged yield fund. They describe a passive, informational front-end to a generic decentralized exchange: PiggyBank is defined as a “liquidity provision protocol through which tokens can be deposited into decentralized exchanges,” the Site is said merely to “aggregate and publish publicly available third-party information” and to draft transaction messages for the user to sign, and the operator is said not to “own, operate or control” the smart-contract system, to hold no tokens, and to be incapable of transacting on a user’s behalf. The template origin is visible in residual artefacts — the document references a token ticker, “\$HBB,” that has no connection to PiggyBank, and its prohibited-use and token-list provisions contemplate spot swapping and automated-market-making rather than vault deposits into a discretionary strategy book. This passive-front-end framing is in direct tension with the protocol’s actual operation as established in §2.1 and §2.3: a single program in which depositor funds are “managed funds by a trusted admin,” whose net asset value is computed off-chain and written on-chain by an administrative key, and whose capital is deployed by the operator into leveraged perpetual positions and over-the-counter holdings. The legal instrument that purports to govern the user relationship therefore mischaracterizes the very thing the user is exposed to. The consequence for disclosure adequacy is developed throughout the remainder of this subsection.^{63, 9}

3.2.2 Conspicuous Absences: Risks the Mechanics Imply but the Disclosures Omit

The disclosed catalog — liquidation, smart-contract failure, depeg, network disruption, and a generic total-loss warning — is the standard risk set of a passive on-chain product. It is largely accurate as far as it extends, but measured against the mechanics analyzed in §2 it omits, or actively obscures, the risks that are specific to PiggyBank’s managed, leveraged, discretionary model. Several of these omissions are material.

First, leverage is disclosed as a source of yield but never as a source of risk. The homepage’s “Where does the yield come from?” answer states that capital is deployed with up to approximately 5x maximum leverage, and the footer names funding-rate arbitrage as the yield engine; but no risk disclosure anywhere mentions that leverage amplifies losses, that the SPYx vault and the JitoSOL vault carry a second, stacked layer of leverage created at the lending venue (§2.3.3), or that a leveraged leg can be liquidated before an offsetting leg can be realized. A depositor reading only the risk disclosures would not learn that the product is leveraged at all. This is a material transparency gap.

Second, and most consequentially, the disclosures are silent on net-asset-value-marking discretion. As established in §2.1.4, a vault’s NAV is determined off-chain by the operator and written on-chain through the `roll_epoch` instruction, and because the backing assets include positions whose value must be marked rather than read from a price feed, NAV embeds discretionary valuation judgments. Nothing in the FAQ, footer, GitBook, or Terms discloses that the operator marks the book, that the operator may exercise judgment over the value of illiquid holdings, or that the operator may exclude a position from NAV entirely. This is the single mechanic that determined the size and timing of the LAB drawdown (§3.2.3), and it is undisclosed.

Third, the disclosures contain no concentration or position-sizing limit and no statement that the book may take idiosyncratic, single-name positions. A depositor is told the strategy is “market-neutral” and diversified across approximately ten venues, but is given no per-token concentration cap, no maximum position size, and no indication that vault capital could be concentrated in one thinly traded asset (§2.3.3). Relatedly, the disclosed strategy scope — long-short positions on “crypto majors” and “tokenized stocks” — does not disclose that the book may acquire low-float or mid-capitalization tokens, may transact through over-the-counter desks, or may hold time-locked positions that cannot be sold. The realized scope, examined in §3.2.4, materially exceeded the disclosed scope.

Fourth, the disclosures omit the liquidity-mismatch and funding-regime risks that are intrinsic to the strategy. The frozen-leg failure mode — in which one leg of a nominal hedge cannot be traded, so that rebalancing cannot restore neutrality (§2.3.4) — is nowhere disclosed. Nor is the funding-regime risk that a delta-neutral funding book can bleed when funding turns persistently negative or when the borrow rate exceeds harvested funding (§2.3.2); on the contrary, “delta-neutral” is deployed as a synonym for low risk. The benchmark protocol treats negative funding as an anticipated state and maintains a sized reserve fund against it (§2.3.2); PiggyBank discloses neither a reserve or insurance fund, nor a loss-attribution policy, nor any backstop, even as the FAQ affirms that the strategy is designed “to protect your deposits”. The silence on loss handling is, as §2.2.3 noted, itself a finding.²

Fifth, the epoch lock-up is disclosed as a feature but not framed as a liquidity risk. The 48-to-96-hour redemption delay is presented as a convenience of batched, low-impact execution; the disclosures do not acknowledge that the same mechanism means a depositor cannot exit a deteriorating position in real time, that redemptions transact against a NAV struck only every 48 hours, or that the operator could in principle pause the program (§2.1.1, §3.2.5).

Finally, the venue-dependency risks documented in §2.2 are disclosed only by the single example “failures in integrated platforms like Kamino.” The report should not treat the use of reputable

Solana infrastructure as inherently severe; the relevant omission is that PiggyBank does not state which venues actually hold collateral or at what loan-to-value, does not acknowledge material venue-level incident history where relevant, and does not explain whether a venue pause, liquidation, or oracle failure would be absorbed by the operator or transmitted to depositors through NAV. The missing disclosure therefore supports a transparency and risk-management criticism more strongly than a blanket claim that every integrated venue is itself high risk.²

3.2.3 Disclosed-versus-Observed: Mapping the Representations onto the Record

The adequacy of a disclosure is tested not in the abstract but against what actually happens. Two loss events are on the record, and they map onto the disclosures in opposite ways.

The April 2026 Drift exposure is the case in which disclosure, outcome, and remedy align. During the 1 April 2026 exploit of Drift Protocol — a governance and admin-key compromise in which an attacker seized administrative control through pre-signed durable-nonce transactions, whitelisted a worthless artificially priced token as collateral, and drained approximately US\$285 million — PiggyBank disclosed that it had roughly US\$106,000 of exposure through its delta-neutral strategy, which had served as margin for a 750 SOL long position on Drift hedged against a 750 SOL short position on another venue (Lighter), out of approximately US\$3.95 million then deployed in the strategy book. This exposure falls squarely within a disclosed risk — “smart-contract vulnerabilities or failures in integrated platforms” — and the protocol’s response was to cover the loss from team funds, keeping depositors whole. In this episode the disclosed risk set was adequate to the event, and the operator absorbed the loss rather than passing it to depositors. The disclosed figure also incidentally corroborates the shared-book architecture of §2.1.1: the roughly US\$3.95 million strategy book exceeds the approximately US\$3.06 million USDC vault, consistent with stablecoins borrowed against the SPYx and underlying JitoSOL collateral being commingled into a single book.^{64, 65, 66, 67, 146}

Drift Exploit Update

- PiggyBank had a \$106k exposure to Drift, out of the 3.95M USDC deployed in our delta-neutral trading strategy. This served as margin for a 750 SOL Long position on Drift, against a 750 SOL Short position on Lighter, reduced to 273 SOL before the exchange was paused.

- The worst case scenario drawdowns would have been: 2.67% on USDC vault, 1.60% on SPYx, 1.52% on pbJITOSOL. This is exactly the kind of counterparty risk priced in the high APY that such strategies provide.

- Nevertheless, the team is so confident in PiggyBank's vision — extra yield on xStocks — that it has unanimously decided to **personally reinject \$106k to cover for users' losses**, even though Drift had passed our deployment risk analysis.

- Today's NAV will be rolled as usual within 2 hours, with 0% APY for the epoch.



From solscan.io

11:39 PM · Apr 1, 2026 · 46K Views

Figure 5. PiggyBank communication about the Drift exploit update and its decision to absorb the related loss. Screenshot captured / source reviewed 13 June 2026. Source: #146.

The June 2026 LAB episode is the opposite case, in which the loss that actually occurred fits none of the disclosed risk categories. Testing the event against each enumerated FAQ hazard:

The disclosed liquidation risk — that market volatility can reduce collateral value and trigger liquidation if thresholds are breached — did not describe the loss. No collateral position on Kamino, Loopscale, or Jupiter Lend was liquidated; no public source records any such liquidation, and the impact reached the SPYx vault and the JitoSOL vault through the shared strategy book and the NAV mark rather than through a forced sale of collateral (§2.2.5). The loss arose on the derivatives and valuation side, a channel the disclosed liquidation language does not capture.

The disclosed smart-contract and integrated-platform risk did not describe the loss either. Contemporaneous reporting was careful to distinguish the episode from an exploit: it was “a trading and risk-management event rather than a verified token-contract hack”. No PiggyBank or third-party contract failed. The disclosed depeg and network-disruption risks were likewise inapplicable.⁶⁸

What actually produced the loss were four factors, none of which appears in the disclosed risk set: concentration in a single illiquid, low-float token subject to public manipulation allegations; a hedge whose two legs were mismatched in tradability; deeply negative funding on the short leg; and a discretionary decision to exclude the locked long leg from NAV. The mechanics are now well established from multiple independent reports. PiggyBank had, roughly a month earlier, purchased about US\$100,000 of locked LAB tokens at a discount through an over-the-counter desk — a position it stated represented nearly 2% of the portfolio at the time — and opened a LAB perpetual short as a hedge. When LAB underwent what the protocol described as violent manipulation amid thin liquidity and deeply negative funding, the short accrued a large mark-to-market loss while paying to be held; the protocol judged maintaining the hedge “economically irrational” and closed the short, crystallizing the loss. The locked long, valued at roughly US\$1.35 million at then-current prices, was excluded from NAV until its first scheduled unlock on 14 August because the tokens could not be sold, an accounting choice that directly produced reported drawdowns of approximately 15% (USDC vault), 12% (SPYx vault), and 9% (JitoSOL vault).^{11, 7,}

12

The most pointed disclosure failure concerns the affirmative characterizations rather than the omissions. The disclosures describe “conservative strategies,” “automated risk management,” and “tight risk management” with “deleveraging upon predefined thresholds,” and the protocol’s own account of the episode is that it acted “before the position crossed its risk limits”. That account is difficult to reconcile with the outcome: a single position representing about 2% of the portfolio produced a roughly 15% markdown of the flagship stablecoin vault. Either the risk limits permitted a position whose unwind could cost the vault 15%, in which case the descriptors “conservative” and “tight” materially overstate the controls; or the limits were in fact exceeded, in which case the claim of acting within them is inaccurate. On-chain investigator ZachXBT pressed precisely this point, questioning how a single strategy error could produce so large a loss in a stablecoin vault and pointing to apparent gaps in position limits and safeguards. Independent commentary made the disclosure mismatch explicit, observing that for depositors in a stablecoin vault a 15% markdown is far beyond what typical risk disclosures suggest, and that stablecoin denomination does not eliminate strategy risk.^{7, 8, 69, 70}

A further inadequacy emerges from comparing the two episodes. In April, a roughly US\$106,000 loss arising from a disclosed risk (integrated-platform failure) was absorbed by the team; in June, a position of comparable initial size (a US\$100,000 OTC purchase) produced a markdown borne entirely by depositors. The differential treatment is not, on the present record, explicable from any published rule: the protocol discloses no loss-attribution policy, no reserve or insurance fund, and no criteria distinguishing losses it will socialize to depositors from losses it will absorb (§2.2.3). Whether a given loss falls on the team or on depositors is therefore discretionary and unpredictable from the disclosures — a disclosure inadequacy in its own right, independent of the merits of either decision. To the protocol’s credit, it disclosed the LAB drawdown publicly and promptly on its own channels, acknowledged the impact to depositors, and promised a detailed report; the criticism here concerns the adequacy of the ex ante risk disclosures and the consistency of loss handling, not the candour of the post hoc acknowledgement. The outstanding questions a depositor would need answered — the over-the-counter terms, the position sizing, the hedge execution, the funding losses, the NAV treatment, the counterparty, the unlock schedule, and above all whether the exposure matched the advertised vault strategy — were, as one outlet cataloged, still unresolved at the time of reporting.^{69, 68}

3.2.4 The Delta-Neutral Narrative versus Observed Behavior: Was the LAB Hedge Delta-Neutral?

The protocol’s central marketing claim is that its strategies are “delta-neutral” and “market-neutral” and generate yield “without taking directional market exposure”. The question this subsection isolates is whether the LAB position was consistent with that claim. Answering it requires separating two senses of the term. In its technical sense, a position is delta-neutral if its net sensitivity to the underlying price is approximately zero, achieved by pairing offsetting long and short legs (§2.3.1). In its colloquial marketing sense, “delta-neutral” is used — including by PiggyBank — as a near-synonym for low-risk capital preservation. The two are not equivalent, and the LAB episode is where they came apart.²

It should first be acknowledged, in fairness, that the book did run genuine delta-neutral trades. The clearest evidence is the SOL position disclosed during the Drift episode: a 750 SOL long on one venue hedged by a 750 SOL short of equivalent size on another, in a deeply liquid major asset across two venues. That is a textbook delta-neutral, cross-venue construction of exactly the kind the marketing describes, and its existence shows the protocol was not simply mislabelling a directional book wholesale.⁶⁵

The LAB position, by contrast, was delta-neutral in form but not in substance. In form, it was assembled with hedging intent and a nominal net delta near zero at inception: a long in locked

LAB offset by a LAB perpetual short. But a hedge functions as a hedge only if both legs are tradable, so that a loss on one can be realized as an offsetting gain on the other. The LAB long was a time-locked over-the-counter position that could not be sold until 14 August. With one leg frozen, the structure did not behave as a hedge at all; it decomposed into a liquid short whose losses were fully and immediately realizable and an illiquid long whose offsetting gains existed only on paper and could not be monetized. When LAB rose, the short bled — mark-to-market loss compounded by deeply negative funding — while the “neutralizing” long gain was unrealizable. The net economic exposure was thus asymmetric and directional in effect, the opposite of neutrality, and the discretionary exclusion of the locked long from NAV then converted that paper gain into a non-entity for accounting purposes while leaving the realized short loss in the books (§2.1.4, §2.3.5). No rebalancing could repair this, because the defect was a liquidity mismatch between the legs rather than a drift in their relative sizing (§2.3.4).⁷

Two features of the underlying asset deepen the divergence from a true delta-neutral trade. A delta-neutral funding strategy presupposes a functioning two-sided market in which the perpetual tracks a fair index, funding behaves as a transfer between longs and shorts, and a short can be entered and exited without being squeezed. LAB was not such a market on the public record reviewed here. ZachXBT’s May 2026 LAB material alleged opaque over-the-counter deals, unilateral vesting changes, unclear distribution, and insider control exceeding 95% of supply, with the token reaching a multibillion-dollar fully diluted valuation amid alleged market-maker coordination and hidden supply mechanics. Those claims remain allegations, but the thin liquidity and violent price action PiggyBank itself reported are consistent with a market in which a nominal hedge could become structurally fragile. In that setting, the price against which the “neutral” position is marked may not represent reliable executable liquidity, and the short can become vulnerable to the squeeze and negative-funding dynamics that materialized. ZachXBT’s criticism is therefore relevant as a risk-management critique, not as a legal finding about PiggyBank.^{8, 7, 117}

The conclusion on item four is therefore a qualified negative. The LAB hedge was constructed with the formal attributes of a delta-neutral position, and it would be inaccurate to call it a naked directional bet at inception; but it was not delta-neutral in any operative sense, because a frozen long leg cannot neutralize a liquid short, and because the underlying lacked the functioning two-sided market a funding-capture hedge requires. It is more accurately described as a leveraged, illiquid, single-name basis bet wearing the form of a hedge. The deeper finding is the gap between the marketed narrative and the realized conduct: a strategy sold as conservative, market-neutral funding capture on majors and tokenized stocks in fact ranged into an OTC-acquired, locked, low-float token. PiggyBank itself framed the position as falling “within its strategy scope” — a framing that, if accurate, establishes that the disclosed strategy scope was materially broader and riskier

than the “delta-neutral, conservative” marketing conveyed to depositors. The mismatch is not between the strategy and a one-off deviation; it is between the strategy as marketed and the strategy as practiced (§2.3.1, §2.3.5). And it underscores the elementary point the disclosures elide: “delta-neutral” is a statement about price exposure only, and does not entail that a position is low-risk, liquid, prudently sized, or honestly markable.⁸

3.2.5 Admin-Key and Governance Risk: Who Controls the Strategy Layer?

The final dimension concerns who controls PiggyBank’s strategy layer, whether the team can unilaterally change strategy parameters or deploy capital into undisclosed assets, and how adequately these facts are disclosed. The control facts are established by the audit and the mechanics of §2, and they are unambiguous. The protocol is a single self-contained program governed by an administrative key. The audit confirms that the key holds pause and unpause authority over the entire program — the team declined a recommendation that vault creation honor the paused state precisely on the ground that the administrator can in any case unpause the program — and that an `update_admin` instruction exists to rotate the administrative authority. Net asset value is computed off-chain and written on-chain through `roll_epoch` (§2.1.4), and the strategy itself — the opening and closing of perpetual positions, the over-the-counter acquisitions, the rebalancing and deleveraging — is executed by the operator’s off-chain systems rather than by verifiable on-chain logic (§2.3.4). There is no on-chain governance, no token-holder vote, no oracle constraining the reported NAV, and no disclosed multi-signature or time-lock arrangement over the administrative key. The strategy layer is, in short, controlled entirely and discretionarily by the operator.⁹

Against this, the disclosures say little, and what they say is generic. The GitBook is silent on governance and key management; the homepage markets the product as non-custodial and the xStocks vaults as “overcollateralized and fully transparent” without addressing who controls the keys. The only governance disclosure of substance is §6.7 of the Terms, which — to the protocol’s credit, and more explicitly than many DeFi front-ends — discloses the category of upgrade-authority risk in some detail. It states that smart contracts deployed to Solana may be upgradable or modifiable by holders of designated private keys or multi-signature arrangements; that such authority may be held by a single key, a multi-signature arrangement, or any other governance mechanism, and that its structure may change over time without notice to users; and it enumerates risks including the compromise of upgrade-authority keys through hacking or social engineering, the malicious or negligent use of upgrade authority by its holders, and changes to smart-contract logic that alter the economic terms, risk parameters, collateral requirements, or operational behavior of the protocol in ways adverse to users.^{2, 63}

The limitation is that this language is boilerplate disclosing a category, not a disclosure of PiggyBank’s own arrangement. It does not state whether PiggyBank’s program is in fact upgradeable; who holds PiggyBank’s administrative or upgrade authority; whether that authority is a single key, a multi-signature scheme and of what threshold, or a time-locked arrangement; or what the deployed program identifier is, by which a user might verify any of these facts on-chain (§2.1.1, §3.1.2). Indeed, Terms §6.7 expressly shifts the burden to the user, stating that users are “solely responsible for independently verifying the governance structure, key management arrangements, timelock mechanisms (if any), and identity and independence of upgrade authority holders” — a responsibility that is, in practice, difficult to discharge, because the program identifier and the operator’s wallet and position addresses are unpublished (§2.1.5, §2.2.1). The category of admin-key risk is therefore disclosed; the specific facts a depositor would need to assess it are withheld, and the verification the disclosure demands of the user is obstructed by the protocol’s own non-disclosure. This is a material transparency gap.

On the operative question — can the team unilaterally change strategy parameters or deploy capital into undisclosed assets? — the answer does not require inference, because it has been demonstrated. The LAB position is itself the proof: the operator unilaterally deployed depositor capital into an undisclosed, off-chain-acquired, time-locked, low-float token, through an over-the-counter desk, with no prior disclosure to depositors, no apparent on-chain limit, and full discretion over how the resulting holding was valued and when it was excluded from NAV. The SOL position deployed across Drift and Lighter similarly evidences unilateral selection of venues and instruments. Nothing in the on-chain program or the public disclosures constrained these decisions; the only operative constraints are the operator’s own unpublished internal risk limits, whose adequacy the episode placed in question (§3.2.3). The team thus exercises unilateral control over venue selection, asset selection (including assets entirely outside the disclosed scope), leverage within the stated ceiling, position sizing, rebalancing, and — through `roll_epoch` — the reported NAV on which every pb-token’s redemption value depends. This is a custodial degree of operational control, notwithstanding the non-custodial framing of both the marketing and the Terms.

The April 2026 Drift exploit sharpens the significance of this. The specific failure vector that Terms §6.7 describes in the abstract — the compromise or abusive exercise of administrative or upgrade authority — is not hypothetical in this ecosystem; it is the same class of failure seen in the Drift exploit, where an attacker obtained admin control and drained roughly US\$285 million, and PiggyBank was a downstream casualty of that very event. The episode demonstrates that an undisclosed key arrangement is not a benign omission: a depositor cannot judge whether PiggyBank is more or less exposed than Drift was to the same class of failure, because

PiggyBank’s arrangement — single key versus multi-signature, time-locked versus immediate, custody and independence of the holders — is not disclosed. It is also worth noting, without asserting any causal link, that the most recent revision of the Terms is dated 3 April 2026, two days after the Drift exploit; whether the upgrade-authority disclosure was strengthened in response to that event cannot be established from the available record and is unconfirmed.^{64, 66}

A governance assessment must finally consider what recourse a depositor has if a loss results from the operator’s exercise of this control. The operating entity is itself undisclosed (§1.3), and the Terms are governed by British Virgin Islands law, mandate confidential individual arbitration, waive class actions and jury trial, impose a six-month claim-survival window, disclaim consequential damages, and cap the operator’s aggregate liability at one hundred United States dollars. The combination is notable: the protocol concentrates discretionary control over strategy and valuation in an anonymous operator while contractually minimizing that operator’s accountability and disclaiming the very managed relationship the operator in fact conducts. A depositor who believed the LAB position or its NAV treatment reflected operator misjudgment would find the disclosed avenues of redress narrow by design.⁶³

In sum, the gap between PiggyBank’s disclosed and observed risk is primarily a matter of incompleteness and under-specification rather than a finding that every disclosure is false. The disclosed risk surface is a generic catalog of a passive on-chain product — liquidation, smart-contract failure, depeg, network disruption — wrapped in a comprehensive AS-IS, total-loss disclaimer whose template even characterizes the service as an informational front end rather than as an actively managed leveraged strategy. That is not enough to describe the actual product. The observed record shows discretionary asset selection, off-chain NAV marking, cross-vault strategy exposure, external-venue dependencies, and operator-determined loss treatment. The issue is therefore not that PiggyBank discloses no risk; it is that the public disclosures do not specify the economically material risks at the level a depositor would need to evaluate them.

Key Takeaway: The audit and vault code are only one part of the risk. PiggyBank also depends on price feeds, lending markets, perpetual DEXs, and Solana-level infrastructure. The issue is not that these venues are automatically weak; it is that users cannot see PiggyBank’s live venue weights, position sizes, routing, or concentration, so they cannot judge how much risk sits in each external system.^{2, 9}

3.3 Smart Contract Risk, Oracle Risk, and Counterparty/Composability Risk

Sections 3.1 and 3.2 examined the protocol’s single published security artifact and the gap between its disclosed and observed risk; both concluded that the bulk of a depositor’s exposure

lies outside the audited on-chain program, in the strategy, the off-chain valuation process, the administrative key, and the external venues through which capital is deployed. This subsection develops that conclusion along its technical axes. It treats, in turn, the smart-contract risk of the on-chain program itself; the oracle and net-asset-value-marking risk that governs how vault assets are priced; the dependency risk presented by the three integrated lending venues — Kamino, Loopscale, and Jupiter Lend — through which the non-stablecoin vaults route collateral; the dependency risk presented by the approximately ten perpetual venues across which the strategy book operates; and finally the degree to which all of these risks are correlated rather than independent, and what a Solana network outage or a major Solana DeFi failure would mean for PiggyBank’s depositors. As throughout this paper, the analysis rests on primary sources and contemporaneous reporting, distinguishes what can be verified from what is asserted, marks transparency gaps and unconfirmed points explicitly, and gives the protocol and its integrated venues credit where the record supports it. The recurring structural finding is that PiggyBank’s risk surface is both wider than its own audit reaches and more tightly correlated than its “diversified across ten venues” framing implies.

3.3.1 Smart Contract Risk: Vault-Program Vulnerabilities, Upgrade Authority, and the Boundary of Audited Assurance

The smart-contract risk of a Solana vault protocol divides into two questions: what classes of defect the on-chain program could contain, and what the audit did and did not establish about their presence in the deployed code. On the first question, Solana programs are exposed to a well-characterized family of vulnerability classes that differ from their Ethereum analogues because of Solana’s account model, in which a program receives the accounts it operates on as inputs to each instruction and must itself verify that those accounts are the ones it expects. The canonical classes are account-substitution or “account confusion” (a caller supplies a valid-looking but incorrect account that the program fails to check for consistency), missing signer and ownership checks, unvalidated program-derived-address (PDA) derivation and bump seeds, arithmetic overflow and precision loss in fixed-point accounting, and unsafe cross-program-invocation (CPI) into other programs. Each of these is a recognized Solana audit category, and several appeared directly in the Formal Land review of PiggyBank’s program (§3.1.4).

The audit’s substantive findings map onto exactly these classes. The remark with the clearest security weight, identifier fb87b4 in roll_epoch.rs, concerned signed 128-bit (i128) arithmetic in the instruction that writes each epoch’s profit-and-loss into vault state, whose additions and subtractions could overflow and were recommended to be replaced with checked operations. Because roll_epoch is the instruction that governs the reported net asset value on which every pb-

token's redemption value depends (§2.1.4), an overflow here is an accounting-corruption risk rather than a cosmetic one. A second cluster, identifiers 62a8e8 in claim.rs and 52e482 and 641271 in validation.rs, concerned missing account-consistency and PDA-bump validation on the user-facing redemption path — precisely the account-substitution class, on the path that settles funds to users — and the auditor observed that the same gap recurred across the other user instructions involving the vault, indicating a systematic pattern at the audited commit. These are not exotic; they are the standard failure modes of Solana programs, and the audit flagged them heuristically rather than excluding them exhaustively.⁹

The decisive limitation, established at length in §3.1, is what the audit's methodology and perimeter could attest. The engagement was a manual review, not a formal verification: it produced no machine-checked proof, no severity taxonomy, and no verified record that the agreed fixes were implemented in the deployed program, and the audited commit (6c930b34) predates the documented mainnet launch by roughly seventy-two hours. The arithmetic-overflow remark is the canonical illustration of why this matters: overflow in fixed-point accounting is exactly the defect class that formal verification eliminates with certainty and that a manual review can only notice, and the agreed remediation is recorded as an intention rather than a confirmed change. By contrast, the integrated lending venues that hold PiggyBank's collateral have each been reviewed by several firms and, in two cases, formally verified (§3.3.3) — a disparity in assurance posture that is itself a risk-relevant fact, because the single least-audited contract in the dependency chain is the one that mints and redeems the depositor's claim.⁹

Upgrade-authority risk is the second dimension of smart-contract risk and, on the public record, the least assessable. Solana programs deployed through the upgradeable BPF loader can have their bytecode replaced by the holder of the program's upgrade authority, which may be a single key, a multi-signature arrangement, or a governance mechanism; an upgrade can change the economic terms, risk parameters, or operational behavior of a program instantaneously and without user consent. PiggyBank's Terms of Service disclose this category in some detail, warning that contracts on Solana may be upgradable by holders of designated keys, that the authority's structure may change over time without notice, and that its compromise or malicious exercise could alter the protocol adversely or expose users to total loss. As §3.2.5 established, however, this is a disclosure of the category rather than of PiggyBank's own arrangement: the Terms do not state whether PiggyBank's program is in fact upgradeable, who holds its upgrade authority, whether that authority is a single key or a multi-signature scheme of some threshold, or whether any time-lock applies, and the deployed program's on-chain identifier is not published anywhere, so a user cannot verify the arrangement on-chain (§2.1.1, §3.1.2). Whether the program is immutable or upgradeable, and under what controls, is therefore unconfirmed from public sources — and it is

the single fact a depositor would most need to assess upgrade-authority risk. What the audit did confirm is the related administrative authority: the program is governed by an administrative key holding blanket pause-and-unpause power over the entire program, and an `update_admin` instruction exists to rotate that authority.^{63, 9}

That this class of risk is not hypothetical in the Solana ecosystem is demonstrated by the two most relevant recent incidents, both of which sit at the boundary of, or just outside, an audited perimeter. The April 2026 compromise of Drift Protocol was an administrative-control failure: an attacker seized admin authority through pre-signed durable-nonce transactions, whitelisted a worthless artificially priced token as collateral, and drained approximately US\$285 million. The April 2025 exploit of Loopscale, examined further in §3.3.3, was a business-logic and pricing failure in a deployed, audited program: the protocol mispriced a class of collateral, allowing undercollateralized borrowing that drained roughly US\$5.8 million. The lesson the pair carries for PiggyBank is that an audit certifying the static correctness of a program’s bookkeeping — which is the most the Formal Land review attempted — does not bound the risk arising from administrative authority over that program or from the pricing of the assets it accounts for, and that both of those are, for PiggyBank, either undisclosed or handled off-chain. The portions of the smart-contract risk surface the audit explicitly did not cover — the external lending venues, the off-chain net-asset-value process, the strategy logic, and the custody and governance of the administrative key (§3.1.2, §3.1.5) — are addressed in the remaining subsections, because they are where the consequential risk lives.^{64, 38}

3.3.2 Oracle Risk: How PiggyBank Prices Its Vaults, the Oracle Dependency Chain, and NAV-Marking Failure Modes

Oracle risk in the context of PiggyBank must be analyzed in two layers, because the protocol’s pricing architecture is unusual: the price of a depositor’s claim is not set by an oracle at all, while the prices that govern the venues holding the depositor’s collateral are. Each layer carries a distinct failure mode.

At the first layer — the valuation of the pb-token itself — PiggyBank does not use a price oracle. As established in §2.1.4 and confirmed by the protocol’s documentation, each vault operates on an epoch-based system in which net asset value is reported every 48 hours, computed off-chain by the operator and written on-chain through the `roll_epoch` instruction. The redemption value of every pb-token is therefore the operator’s reported mark, struck twice a week, rather than a value any external party reads from a price feed; §3.2.5 found that no oracle constrains the reported NAV. The salient feature of this design is the absence of an on-chain check on the mark. The contrast with a comparable institutional Solana vault is instructive: Upshift’s Solana vault architecture

computes its share price on-chain directly from the vault-state account, requires no external oracle, and — critically — enforces asymmetric increase and decrease limits on the deployed-assets figure together with a maximum-value error, expressly so that no single update can move the reported NAV by an implausible amount, “whether from operator error or attempted manipulation”. PiggyBank discloses no analogous cap, circuit breaker, or bound on the magnitude of a single NAV update. The first-layer oracle risk is thus better described as NAV-marking risk: there is no oracle to manipulate because the operator’s discretion has replaced the oracle, and the only constraint on the reported value is the operator’s own judgment (§3.2.2, §3.2.5).^{1, 71}

The June 2026 LAB episode is the realized instance of this failure mode, and it is the operator-discretion analogue of a valuation-failure event. The mechanics, established in §3.2.3–§3.2.4, were that a thinly traded, low-float token subject to public manipulation allegations was held as a locked long against a perpetual short, that the locked long was excluded from NAV because it could not be sold, and that this discretionary exclusion produced reported drawdowns of approximately 15% (USDC vault), 12% (SPYx vault), and 9% (JitoSOL vault). Two oracle-adjacent observations follow. First, the asset against which the “neutral” position was marked did not have a fair, manipulation-resistant price to begin with: the token’s price action was, on the protocol’s own account, violent and thin, so any mark — whether by an oracle or by the operator — would have inherited that unreliability. Second, because the mark was the operator’s rather than an oracle’s, the magnitude and timing of the loss were a discretionary valuation choice (which leg to include, and when), not a mechanical consequence of a price feed. The general failure mode this exposes for any NAV-reporting vault is that discretionary marking of illiquid holdings transfers the entire integrity of the share price to the operator’s honesty and competence, with no external check; and the specific failure mode is that exposure to assets without a reliable price makes even a well-intentioned mark unreliable.⁷

At the second layer — the venues holding PiggyBank’s collateral and carrying its positions — oracles are central, and PiggyBank inherits their oracle risk indirectly. The Solana lending venues price collateral through external feeds: Kamino prices collateral using Pyth and Switchboard feeds aggregated through its own Scope oracle layer, and the pool-based perpetual venues price positions directly from oracle feeds, with Jupiter Perps quoting against the Pyth-based Dove oracle and Drift integrating Pyth for low-latency pricing. Pyth is the dominant Solana oracle, publishing first-party prices on a sub-second cadence sourced from more than eighty institutional publishers. The collateral and perp legs of PiggyBank’s strategy therefore depend, at one remove, on the correctness, liveness, and manipulation-resistance of these feeds.^{30, 45, 72, 73}

The known failure modes of oracle manipulation in this context are well documented on Solana, and two are directly relevant. The first is the manipulation of an oracle that references a low-liquidity market to inflate the apparent value of collateral, enabling undercollateralized borrowing. This is precisely the Loopscale exploit: an attacker manipulated the protocol’s pricing of RateX principal tokens — likely by pushing the price on a thin trading pair the oracle referenced — and used the inflated value to withdraw roughly US\$5.8 million, about 12% of the protocol’s then-total value locked, in a venue PiggyBank uses for collateral. The canonical Solana instance of the same class is the October 2022 Mango Markets attack, in which Avraham Eisenberg opened large leveraged positions in the MNGO token and then rapidly purchased MNGO across the exchanges feeding the oracle, pumping its reported price roughly more than tenfold in roughly twenty minutes and using the artificially inflated collateral to drain approximately US\$110 million; the protocol’s US\$10 million insurance fund was insufficient to cover depositor losses, and Eisenberg was convicted by a jury in 2024 of commodities fraud and manipulation, though the trial court set that verdict aside and entered a judgment of acquittal in 2025, a ruling prosecutors have since appealed. The lesson for a NAV-reporting protocol that holds idiosyncratic, thinly traded assets is direct: the same dynamics that let a manipulator inflate an oracle-priced collateral value are the dynamics that make an operator’s discretionary mark of an illiquid holding unreliable, and PiggyBank’s book has demonstrably ranged into exactly such assets (§3.2.4).^{74, 75, 76}

The second failure mode is staleness or liveness failure, which is correlated with network conditions and is therefore developed in §3.3.5. In brief, pull-style oracles such as Pyth depend on price updates being posted on-chain within a freshness window, and during Solana congestion or an outage those updates can lag, leaving venues to act on stale prices; commentary on Solana perpetual venues notes that illiquid markets can gap through stop levels during congestion or oracle staleness. For PiggyBank, an oracle that is stale across the venues simultaneously is not an idiosyncratic event but a correlated one, because the venues share the same oracle infrastructure — a point that connects oracle risk to the concentration analysis below.⁷⁷

3.3.3 Integrated-Lending Dependencies: Kamino, Loopscale, and Jupiter Lend

The non-stablecoin vaults post collateral on Solana lending venues and borrow stablecoins into the shared strategy book (§2.2.1, §2.2.4). PiggyBank names Kamino and Loopscale explicitly and, as discussed in §2.2, also references Jupiter Lend. The relevant point is not that these are low-quality venues; each is a real DeFi venue and each has more external security review than PiggyBank’s own program. The risk is that PiggyBank depositors inherit each venue’s code, oracle, liquidation, parameter, pause, and bad-debt dynamics, without seeing PiggyBank’s live routing weights or position sizes.^{2, 9}

Kamino. Kamino is the most mature and systemically important of the three: it is one of Solana’s largest DeFi protocols, has multiple audits and formal-verification work, and has not been identified in the sources reviewed here as having suffered a major exploit through 2026. Its residual risks are still material for PiggyBank because a large Kamino event would be a Solana-wide lending-market stress event, not just a single-venue problem.^{30, 78, 79, 80, 17}

Loopscale. Loopscale is the smallest and riskiest venue profile. It launched in April 2025 and was exploited roughly two weeks later for about US\$5.8 million due to a pricing/oracle-business-logic failure involving RateX principal-token collateral. The funds were later returned after a bounty negotiation and users were made whole, and the protocol added hardening measures, but the episode remains relevant because it shows how exotic collateral and pricing logic can break in a lending market.^{81, 82, 83, 84, 85, 86}

Jupiter Lend. Jupiter Lend is the newest and fastest-scaling profile. It has multiple 2025 audits and large TVL, but it also uses high LTV ceilings and confirmed rehypothecation through Fluid’s shared liquidity architecture. The December 2025 controversy over “isolated” vault messaging matters for PiggyBank because collateral posted to Jupiter Lend may not be economically isolated from unrelated market stress in the way a casual reader might assume.^{87, 88, 89, 90, 91, 92, 93, 94, 95}

Taken together, Kamino looks strongest, Loopscale has the clearest exploit-history concern, and Jupiter Lend has the clearest rehypothecation/contagion concern. PiggyBank’s user-facing materials do not provide live venue allocation, exposure size, or routing logic, so a user cannot know whether their collateral risk is concentrated in the most robust venue or in the venue carrying the highest marginal risk.

3.3.4 Perpetual-DEX Dependencies: Insolvency, Liquidity Fragmentation, and Smart-Contract Failure Across Approximately Ten Venues

The strategy book’s directional legs are opened on perpetual venues, and the protocol says it operates funding-rate arbitrage across approximately ten perpetual DEXs, using long-short positions on crypto majors and tokenized stocks at up to roughly 5x leverage. The venue list is not published. Only Drift and Lighter are confirmed from the incident record, where a 750 SOL long on Drift was hedged with a 750 SOL short on Lighter (§3.2.3). This venue non-disclosure is another transparency gap: users cannot enumerate the counterparties carrying their margin or assess concentration by venue.²

The likely venue universe spans materially different models. Drift, Jupiter Perps, Zeta Markets, Flash Trade, and Lighter use different combinations of order books, liquidity pools, oracle pricing, and settlement layers. Lighter is especially important because it is not Solana-native; it is an

Ethereum-anchored zero-knowledge rollup. That gives partial diversification against a Solana-only outage but adds cross-chain margin and second-settlement-layer risk.^{77, 43, 72, 96}

Three dependency risks follow. First, a venue can fail, become insolvent, or suffer a smart-contract/oracle failure, leaving PiggyBank exposed to the margin held there. Second, liquidity can fragment across venues, so a hedge leg may be expensive or impossible to adjust during stress. The LAB episode is the realized example: a short leg on a thin market with deeply negative funding and violent price movement could not be maintained economically and was closed at a loss.⁷⁷

Third, the strategy depends on both sides of each hedge remaining tradeable at the same time. A venue does not need to collapse to harm PiggyBank; it only needs to become unavailable, illiquid, or mispriced when a leg must be adjusted. Because PiggyBank does not publish a weighted live-position register, outsiders cannot distinguish a broadly diversified venue book from a concentrated book hidden behind the phrase “approximately ten perpetual DEXs”.²

3.3.5 Concentration and Correlated Risk: Shared Dependencies, Network-Outage Exposure, and Blast Radius

The individual dependencies above are not independent. PiggyBank’s lending venues, perpetual venues, oracles, collateral assets, and settlement layer are heavily concentrated in the Solana ecosystem. The phrase “diversified across approximately ten venues” is therefore accurate only at the level of individual counterparties, not at the level of shared systemic risk.

At the lending level, Kamino and Jupiter Lend are not merely two unrelated counterparties; they are systemically central Solana venues. The collateral leg routes through Kamino, Loopscale, and Jupiter Lend, while much of the perpetual leg appears to use Solana-native venues such as Drift and Jupiter Perps. A stress event large enough to impair one major venue would likely coincide with stress across the same ecosystem, collateral set, and liquidity base.^{97, 78, 88}

The shared dependencies are straightforward: Solana network liveness, common oracle infrastructure, recurring collateral and settlement assets such as SOL, JitoSOL, and USDC, and cross-protocol collateral reuse. Solana’s outage history remains relevant, even though reliability improved after the major confirmed February 2024 outage and the chain later recorded a long period without another major confirmed halt. During an outage or oracle dislocation, PiggyBank may be unable to update NAV, rebalance, deleverage, close hedge legs, or process redemptions normally.^{98, 99, 100, 95}

The shared-book architecture widens the blast radius. Stablecoins borrowed against SPYx and JitoSOL collateral are mixed with USDC vault capital into a common strategy book, so a shock to

one leg can affect all vaults. The LAB episode showed this pattern in practice, producing simultaneous NAV declines across the USDC, SPYx, and JitoSOL vaults. A correlated Solana, oracle, USDC, JitoSOL, or major-venue event would likely hit several parts of the structure at once rather than one isolated position.^{66, 7, 11}

There is partial diversification: Lighter gives at least some non-Solana perpetual exposure, and Solana’s reliability has improved relative to earlier years. Those facts reduce probability in some scenarios, but they do not remove the core concentration problem. PiggyBank’s external risk is still dominated by a small set of Solana venues sharing common infrastructure, collateral assets, and oracle dependencies, while users cannot see live concentration.^{96, 98, 99, 100}

4. LAB Token Incident and Protocol Response

4.1 The LAB Token Hedge: Public-Source Reconstruction of Events

4.1.1 The LAB Token: Origin, Purpose, and Market Characteristics

The asset at the center of the incident is the LAB token (ticker LAB; styled \$LAB), the native asset of a multi-chain trading-terminal project operated under the brand “The Lab” / “LABtrade” (X handle @LABtrade_). The project markets itself as multi-chain trading infrastructure that aggregates spot, limit, and perpetual-contract execution across decentralized venues behind a single interface, paired with an “AI research engine” for transaction optimization and strategy generation. These are project representations, not independently audited facts, and they matter here only because LAB became the asset behind PiggyBank’s loss.^{101, 102}

For this report, the important background is not the full founder history of LAB, but the token’s risk profile. Independent coverage and ZachXBT’s public LAB allegations connected LAB to prior founder and market-making concerns, disputed supply conditions, and affiliated OTC/private-sale activity. These points are included only insofar as they help explain why a locked LAB basis position was materially different from a conservative major-asset funding trade; they are not treated as legal findings or as findings about PiggyBank insiders.^{103, 104, 105}

LAB public-source risk flags relevant to the position

Prepared as an analytical summary; allegations are not treated as findings about PiggyBank.

Risk flag	Public-source status	Why it matters for the PiggyBank trade
Low float / supply opacity	Reported by data providers and public commentary; exact float disputed.	A locked spot leg is harder to value and harder to exit when tradable float is thin.
Insider / OTC allocation allegations	Alleged by ZachXBT and repeated in coverage; not independently adjudicated here.	Preferential OTC terms and concentrated control can make headline market prices unreliable.
Short-squeeze / funding warning	ZachXBT publicly warned against treating the LAB findings as a simple short signal.	A nominal hedge can become costly if the short pays negative funding while the long leg is locked.

Sources: #103, #104, #105, #117. This figure summarizes source status; it is not an independent forensic finding.

Figure 6. LAB public-source risk flags relevant to PiggyBank's locked-basis position. Author-prepared analytical summary from Sources #103, #104, #105, and #117; allegations are not treated as findings about PiggyBank.

Market characteristics at the time of the incident. Three features of LAB's market structure are essential to understanding the PiggyBank loss: an extreme low float, opaque and disputed supply, and a parabolic price subject to public manipulation allegations.

- **Low float and supply opacity.** Total supply is reported at 1.0 billion LAB, but estimates of circulating supply diverged sharply across data providers around the incident window — CoinGecko reportedly listed roughly 76.5 million while CoinMarketCap showed roughly 309.9 million, with the project itself providing no clear breakdown. Near launch, only about 7.7% of supply was estimated to be in free float. ZachXBT alleged that insiders and affiliated market makers controlled more than 95% of supply through undisclosed OTC sales, private-sale allocations, airdrop wallets, and team holdings. These remain allegations; LAB did not substantively rebut them in the cited coverage.^{108, 107, 105, 104}

- **Price trajectory.** LAB surged approximately 364% to about \$3.18 on 3 May 2026, topping CoinGecko's gainers with roughly \$253 million of 24-hour volume, then reversed by roughly 70% within the same period. Over the following weeks it climbed steeply: an exchange snapshot dated 7 June 2026 quoted LAB near \$19.02 with roughly \$45.6 million of 24-hour volume and a market-cap rank near 29. During the early-May pump, ZachXBT's investigation put the token's fully diluted valuation at approximately \$6 billion — an early-May, manipulation-era figure rather than a peak, since with the reported 1.0 billion token supply the later 7 June mark of \$19.02 would imply a fully diluted valuation several times larger. The \$6 billion should therefore be read against the float and supply opacity described above rather than as a precise supply-times-price product. Because these marks were printed during a period of alleged manipulation on thin liquidity, every

LAB valuation in this section — including the \$1.35 million figure discussed below — should be treated as paper value that may diverge materially from realizable value.^{107, 102, 101, 105, 103}

- **Manipulation allegations and a specific short-selling warning.** Across May 2026 ZachXBT published escalating findings alleging opaque private loans (one draft agreement reportedly carrying interest of up to 7.5% per month, repayable in LAB at market price), unilateral extension of the public-sale lock-up from three to nine months, preferential OTC pricing to key opinion leaders and whales (discounts reportedly in the 60–80% range in exchange for promotional content), and market-maker coordination. Critically, ZachXBT cautioned that his findings should not be read as encouragement to short LAB: he argued that the concentrated insider supply could instead “become fuel” for squeezes against bearish traders. As developed in 4.1.3, this is precisely the trap into which PiggyBank’s hedge fell.^{106, 109, 104, 105}

4.1.2 Chronological Reconstruction of the Incident

The reconstruction below is assembled from PiggyBank’s own 6 June community disclosure, its later 10 June compensation/drawdown communication, and contemporaneous third-party reporting. Where the public record fixes only an approximate date, this is stated explicitly; the protocol had not, as of the updated review date, published the on-chain trade records that would allow precise dating.^{119, 152, 155}



PiggyBank 🟡 @piggybank_fi · Jun 6



Dear Community,

- One month ago, PiggyBank got into a new position for \$100k (~2% of the portfolio at the time), as part of the strategy's scope of deploying into mid-cap basis trades.
- The strategy involved purchasing locked \$LAB tokens at a significant discount through a Tier-1 OTC desk and hedging that exposure through perpetual shorts. During the holding period, \$LAB experienced violent manipulation, including chaotic price movements on thin liquidity, and deeply negative funding rates.
- Maintaining the hedge became economically irrational and would have caused us to cross our risk thresholds. We therefore decided to close our short to cap the downside.
- In accounting terms, the total value of our \$LAB locked position is \$1.35m at current prices. However, due to the illiquid nature of this position, Piggybank is excluding it from the NAV until the first unlock on August 14. Even though this is an evolving situation that could still generate substantial returns, this is the fairest and most transparent way to let users manage their liquidity.
- Accordingly, today's NAV will show a ~15% drawdown on the USDC vault, ~12% on SPYx and ~9% on JitoSOL.
- A detailed report will be published next week with next steps.

💬 209

↺ 185

❤ 234

📊 412K



Figure 7. PiggyBank public community update disclosing the LAB position and projected vault NAV drawdowns. Screenshot cropped to the official PiggyBank post only; source reviewed 13 June 2026. Source: #119.

Context (October 2025 – April 2026). LAB launched in October 2025; PiggyBank itself launched on Solana mainnet on 30 October 2025 (see Section 1). The two launches are roughly contemporaneous, but no operational link between the projects is established by the available evidence, and none is asserted here.

Early-to-mid May 2026 — ZachXBT's LAB allegations become public. On 7 May 2026, ZachXBT first publicly accused the LAB founder of centralized-exchange manipulation said to harm retail investors. On 10 May, investor Simon Dedic publicly warned holders to exit the token. On 14 May, ZachXBT published a comprehensive investigation alleging more than 95% insider control, the BVI corporate structure, altered vesting, and the approximately \$6 billion fully diluted valuation.^{110, 104, 105}

Approximately early May 2026 — PiggyBank enters the position. PiggyBank later stated that it opened the position “about one month” before its 6 June disclosure, which places entry in roughly the first week of May 2026. The exact on-chain entry date is not part of the public record. The

deployment was \$100,000 in cash, described as roughly 2% of the portfolio at the time (implying a portfolio of about \$5.0 million, broadly consistent with the self-reported aggregate TVL of roughly \$4.5 million noted in Section 1). The trade structure was a discounted-lock-up basis trade: the purchase of LAB tokens, locked, through an over-the-counter desk at a significant discount, hedged by a short position in LAB perpetual futures. PiggyBank characterized this as falling within the strategy’s scope for deploying into “mid-cap basis trades”. The timing is material: entry coincided with — and the holding period ran straight through — the most prominent phase of ZachXBT’s public manipulation allegations against the very asset PiggyBank had shorted.^{7, 12, 11}

Mid-May to early June 2026 — the position deteriorates. Over the holding period LAB’s price rose several-fold, from low-single-digit dollars in early May toward approximately \$19 by 7 June, on thin liquidity. PiggyBank reported that LAB then experienced violent price manipulation, an evaporation of liquidity, and deeply negative perpetual funding rates. For a short position this combination is doubly punitive: the mark-to-market loss grows as the price rises, and deeply negative funding means the short pays the long side at each funding interval simply to hold the position.^{107, 101, 11, 7}

6 June 2026 (Saturday) — unwind and disclosure. PiggyBank closed the perpetual short, stating that maintaining the hedge had become economically unjustifiable and risked breaching its internal risk limits. It published a “Dear Community” transparency thread on X (@piggybank_fi) disclosing the closure and the expected NAV impact. It valued the remaining locked LAB at approximately \$1.35 million at then-current prices but removed that holding from NAV until the first scheduled unlock on 14 August 2026, citing illiquidity. The accounting consequence was an expected drawdown of about 15% for the USDC vault, about 12% for the SPYx vault, and about 9% for the JitoSOL vault.^{7, 12, 11}

LAB incident timeline and evidence status

Public-source reconstruction; exact wallet-level trade records were not published in the materials reviewed.

Date / window	Public-source event	Evidence status
Early May 2026	PiggyBank later stated it entered a discounted locked-LAB position hedged with perpetual shorts.	Protocol statement; exact on-chain entry record not published.
7-14 May 2026	ZachXBT and others published LAB-related manipulation and supply-control allegations.	Third-party allegations about LAB, not findings about PiggyBank.
6 June 2026	PiggyBank closed the short and disclosed estimated NAV drawdowns of ~15% / ~12% / ~9%.	Protocol disclosure and secondary reporting.
10 June 2026	PiggyBank released a later official compensation-related statement / drawdown explanation.	Official protocol statement; payment-level verification not independently established here.
14 August 2026	First scheduled unlock of the locked LAB position.	Future contingent event at the report's access date.

Sources: #7, #11, #119, #120, #198.

Figure 8. Public-source timeline of the LAB incident, including the later compensation-related PiggyBank statement.

Author-prepared analytical summary from Sources #7, #11, #119, #120, #152, and #155.

6 June 2026 — ZachXBT's response. The same day, ZachXBT quote-posted PiggyBank's update with screenshots and used pejorative language to argue that the protocol had exposed user funds to LAB, tying the loss directly to his earlier LAB findings. The Crypto Times reported that it had requested comment from PiggyBank and had not received a response by the time of publication.^{8, 7}

7-8 June 2026 — secondary coverage and a not-a-hack clarification. A range of outlets reported the event over the following two days. Several noted that the episode had initially circulated on social media as a possible LAB "exploit," but that the confirmed facts pointed to a trading and risk-management loss rather than a verified smart-contract hack.^{11, 69, 70, 68}

10 June 2026 — follow-up drawdown and compensation communication. PiggyBank later issued an official compensation/drawdown communication. That communication matters because it moved the response beyond the initial 6 June disclosure by stating a compensation intention, but it still did not provide the trade-record-level evidence that would allow outside verification: no entry or exit transaction signatures, no OTC counterparty, no perpetual venue, no total realized loss, no per-user schedule, and no payment proof. The 14 August 2026 first unlock therefore remained a future contingent event, and the realized user outcome still depended on LAB's liquid market value at and after that date.^{152, 155}

4.1.3 Mechanics of the Loss: From Hedge to NAV Drawdown

The central technical question is how a hedged, nominally market-neutral trade produced a roughly 15% drawdown in a stablecoin vault. Three mechanisms compound: a liquidity and

settlement mismatch between the two legs; a directional short into a multi-fold price rise paired with negative funding carry; and a NAV accounting convention that excludes the only offsetting asset.

(1) The trade, and why it was not delta-neutral in practice. In a discounted-lock-up basis trade, the protocol buys locked spot tokens at a discount and shorts an equal token quantity of perpetual futures, intending to capture the discount as the lock rolls off and to earn funding carry, while directional price risk nets out between the legs. The flaw realized here is that the two legs had asymmetric liquidity. The spot leg was contractually locked: it could not be sold, posted as margin, or delivered against the short until 14 August. The short leg, by contrast, was fully liquid, marked to market continuously, and required USDC margin. As LAB rose, the short generated mark-to-market losses and margin demands that the locked spot could not service. A position that is delta-neutral in token terms is not risk-neutral when only the losing leg can be transacted: the hedge protects economic value on paper, but it cannot prevent a liquidity-driven, one-sided cash loss.

(2) The directional and funding losses. PiggyBank covered the short to cap further losses, crystallizing the mark-to-market loss accumulated as LAB appreciated, plus the cumulative negative funding paid over roughly thirty days. As 4.1.1 notes, ZachXBT had warned specifically that shorting LAB invited a squeeze given its concentrated supply; the realized loss is, in effect, the materialization of that warning. When funding flips persistently negative on a thinly traded token, a short hedge becomes prohibitively expensive to maintain, which is the trap PiggyBank described.^{7, 70}

(3) The NAV accounting asymmetry. PiggyBank's decision to exclude the locked LAB (about \$1.35 million in paper value) from NAV — defensible on illiquidity grounds — means the spot leg's appreciation is not credited to NAV, and therefore cannot net against the realized short loss. The gain leg is rendered invisible to NAV while the loss leg flows through in full. This is the accounting fulcrum on which a notionally neutral trade tips into a large reported loss.^{7, 11}

Worked reconstruction (illustrative; only the endpoints are sourced). The sourced anchors are: \$100,000 of cash deployed; locked LAB later valued at approximately \$1.35 million; that holding excluded from NAV until 14 August; reported drawdowns of 15% (USDC vault), 12% (SPYx vault), and 9% (JitoSOL vault); and the self-reported vault TVLs from Section 1 of roughly \$3.06 million (USDC vault), \$823,000 (SPYx vault), and \$639,000 (JitoSOL vault). Applying the drawdowns to those TVLs implies an aggregate, USDC-equivalent markdown of approximately $(0.15 \times \$3.06\text{M}) + (0.12 \times \$0.823\text{M}) + (0.09 \times \$0.639\text{M}) \approx \$459,000 + \$99,000 + \$58,000 \approx \$615,000$.

The token-quantity algebra reconciles the headline figures. Let Q be the quantity of LAB acquired and p_{buy} the discounted lock price, so that $Q \times p_{\text{buy}} = \$100,000$. The later paper mark $Q \times p_{\text{now}} \approx \1.35 million implies a ratio $p_{\text{now}} / p_{\text{buy}} \approx 13.5$. That ratio decomposes into two distinct effects: the OTC and lock-up discount, and genuine price appreciation during the hold. ZachXBT's reporting put OTC discounts in the 60–80% range. Assuming, purely illustratively, an effective discounted entry near \$1.40 per token — consistent with a deep OTC and lock-up discount within ZachXBT's reported 60–80% range applied to LAB's early-May trading range — then $Q \approx \$100,000 / \$1.40 \approx 71,000$ LAB, and at $p_{\text{now}} \approx \$19$ the locked stake $\approx \$1.35$ million — reconciling the published numbers. Under the same assumptions, a short hedging this token quantity carried a notional of roughly \$250,000 at entry, not \$100,000: the cash outlay understated the hedged exposure by roughly two-and-a-half times. The realized short loss equals $Q \times (p_{\text{cover}} - p_{\text{entry}})$ plus funding; for that loss to approximate the roughly \$615,000 aggregate markdown, the implied cover price falls in the high-single to low-double digits — consistent with PiggyBank closing the short partway up the run — with cumulative negative funding contributing the remainder. These intermediate values (entry price, discount, short notional, cover price, and funding paid) are not disclosed and are presented only to demonstrate that the sourced endpoints are mutually consistent. The exact decomposition awaits PiggyBank's promised trade records.¹⁰⁹

Two analytic conclusions survive the parameter uncertainty. First, the cash figure of “\$100,000, roughly 2% of the portfolio” materially understated the economic exposure, because the hedged notional and the locked face value were several times larger than the cash outlay. Second, the loss — roughly \$615,000 in aggregate across the three vaults, about six times the cash deployed — is the characteristic signature of a short into a multi-fold price move compounded by negative funding, with no offsetting credit because the long leg was illiquid and excluded from NAV. The leverage that turned a 2% position into a 15% vault drawdown came not from the position's headline size but from the short's exposure to a multi-bagger move and the accounting exclusion of the only asset that could have offset it.

Why the USDC vault bore the largest percentage. The losses from a single LAB book were apportioned across all three vaults, producing the 15% / 12% / 9% split. The exact allocation key is not disclosed; the differential implies either that the USDC vault held the largest share of the position relative to its size, or that the loss-to-NAV ratio was highest there. The structural observation matters more than the arithmetic: a single speculative book was shared across products, so the vault marketed as the most conservative — USDC, a capital-preservation product — absorbed the largest proportional hit, an outcome difficult to reconcile with the vaults' differentiated risk positioning.

The epoch dimension. PiggyBank reports NAV on a 48-hour epoch cycle, and redemptions clear two epochs after a request is submitted. The drawdown therefore surfaced to users in the NAV print of the epoch following the unwind. Because the locked LAB is excluded from NAV, any later recovery — if LAB holds value through 14 August — would accrue to whoever holds vault shares at that future NAV print, not necessarily to depositors who exit at the marked-down value in the interim. This distributional consequence between exiting and remaining depositors was not addressed in the disclosure.¹

4.1.4 Disclosure to Users: Timeline, Content, and Omissions

Timeline of disclosure. The protocol's first and, as of the access date, principal disclosure was the X thread of 6 June 2026. The disclosure was reactive: it was published at the moment of unwinding, after a month-long deterioration and amid pre-existing public scrutiny of LAB, rather than at the initiation of the position or as the risk built. There is no evidence in the record of any contemporaneous disclosure to depositors that a discounted-lock-up basis trade in an illiquid altcoin was being carried in the vaults during May.^{7,8}

What was said. The thread disclosed the existence and approximate size of the position (\$100,000, roughly 2% of the portfolio at entry); the structure (discounted locked OTC spot plus a perpetual short, framed as a mid-cap basis trade); the proximate causes of the loss (manipulation, illiquidity, and deeply negative funding); the decision to close the short; the approximately \$1.35 million locked valuation; the exclusion of that holding from NAV until 14 August; and the expected per-vault drawdowns of 15%, 12%, and 9%. Relative to a bare loss announcement, this is a moderately detailed disclosure: it named the asset, the structure, and the accounting treatment.⁷

11, 12

What was omitted. The omissions are nonetheless substantial. Absent from the disclosure were: any on-chain evidence or trade records (entry and exit transactions, the OTC counterparty, the perpetual venue, and the funding actually paid); a quantified breakdown of the loss between its directional and funding components; the cross-vault allocation methodology that produced the 15% / 12% / 9% split; whether any compensation or make-whole was contemplated; clear guidance on whether, and at what value, users could withdraw given the marked-down NAV; the rationale for a stablecoin vault bearing the largest proportional loss; and any acknowledgement that the asset had been publicly flagged for alleged manipulation at or before the time of entry. Two of these gaps were noted directly in contemporaneous reporting: PiggyBank did not announce compensation or explain whether users could withdraw at the revised values, and observers specifically flagged the absence of a post-mortem detailing position limits, stop-losses, or

diversification rules and why they failed to prevent a 15% drawdown. The promised detailed report had not been published as of the access date.^{7, 69}

Comparison to best practice. Mature DeFi incident disclosure converges on a recognizable template: rapid acknowledgement; a detailed post-mortem with a documented causal chain and on-chain transaction references; a quantified loss accounting; an explicit remediation or compensation decision; clear and immediate user guidance on funds and withdrawals; and, where possible, independent verification. A useful contemporaneous benchmark is the Lazy Summer / Summer.fi Arbitrum USDC vault post-mortem of December 2025, which set out an explicit, step-by-step causal chain, named the external dependencies that drove the impairment, and discussed the reimbursement dynamics in detail. Measured against that template, PiggyBank’s disclosure was timely in the narrow sense that it was issued at the unwind, and it was commendably specific about the accounting treatment and the headline numbers. But it fell short on verifiability (no trade records or transaction hashes), on completeness (no loss decomposition and no allocation key), on remediation (no compensation decision and no withdrawal guidance), and on candour regarding the asset’s prior public risk flags. The protocol’s own documentation foregrounds “transparency” and “risk management” as design pillars; the episode is in visible tension with the first and, as 4.1.5 argues, with the second.^{111, 1}

A secondary framing issue. Because the event initially circulated as a possible exploit or hack, PiggyBank’s characterization of it as a contained, rules-bounded trading decision — the protocol stated that it had acted before the position crossed its risk limits — is more accurate than “hack”. Yet the same framing recasts a discretionary bet on a publicly flagged token as routine risk management. The trade records that PiggyBank withheld are exactly what would be needed to test that characterization — in particular, whether the internal risk limits the protocol invoked were defined in advance and were consistent with a delta-neutral stablecoin mandate, or whether they were invoked only after the loss had materialized.^{7, 68}

4.1.5 Consistency with the Protocol’s Stated Strategy

The stated mandate. As documented in Section 1 and in PiggyBank’s own materials, the protocol operates asset-specific Solana vaults running automated, delta-neutral funding-rate arbitrage across approximately ten perpetual DEXs, taking long and short positions on crypto majors and tokenized stocks at up to approximately 5x leverage, with collateral routed through Kamino, Loopscale, and Jupiter Lend; the USDC vault is marketed for an approximately 15% target net APY of stablecoin yield, and the protocol describes itself as built around transparency, risk management, and resilient infrastructure.^{2, 1}

PiggyBank’s defense, on its own terms. PiggyBank framed the LAB trade as falling within the strategy’s scope for deploying into mid-cap basis trades. A basis trade — long spot, short perpetual — is, in the abstract, consistent with a delta-neutral mandate, and the documentation describes broad strategy latitude. On a literal reading of a broadly worded mandate, the protocol can argue that the trade type was permitted.⁷

Four considerations nonetheless place the position outside the substance of the mandate. First, the liquidity profile was incompatible with delta-neutrality: delta-neutral execution presupposes the ability to transact both legs in order to maintain neutrality and to exit, but a locked, illiquid spot leg cannot be rebalanced or closed, so the position could never be neutralized once it moved — it was structurally incapable of behaving in a delta-neutral manner, whatever its nominal construction. Second, the “mid-cap” label flatters the asset: LAB’s standing rested on an approximately \$6 billion fully diluted valuation atop a roughly 7.7% float, with disputed circulating supply and alleged insider control above 95%, which describes a low-float token subject to public manipulation allegations rather than a liquid major or a tokenized equity — the asset classes the USDC vault’s marketing emphasized. Third, the risk was knowable from essentially the outset of the position and throughout it: ZachXBT’s manipulation allegations emerged on 7 May and his explicit squeeze warning followed on 14 May — at or immediately after PiggyBank’s early-May entry — and they persisted throughout the hold, so maintaining (rather than exiting) a short into precisely the squeeze dynamic a prominent investigator had flagged is difficult to square with “risk management” as a design pillar. Fourth, the capital-preservation product bore the largest loss: the USDC stablecoin vault, the product most strongly associated with capital preservation, absorbed the largest proportional drawdown from a discretionary altcoin bet, contradicting the differentiated risk profile that a depositor would reasonably infer from a “stablecoin yield” vault.^{107, 108, 105, 110}

On balance, the LAB hedge sits in the gap between PiggyBank’s broadly permissive internal mandate and its conservative external positioning. The trade may have been technically inside the letter of an expansively worded strategy scope, but it was inconsistent with the delta-neutral, capital-preserving proposition marketed to USDC depositors — both because an illiquid, locked leg cannot be delta-neutral, and because public manipulation allegations and squeeze warnings existed around the asset at the time of entry. The episode therefore reads less as an unforeseeable market accident than as a discretionary, directional exposure to a flagged token: an exposure whose risks were knowable in advance and whose losses were borne disproportionately by the protocol’s most conservative vault.

4.2 ZachXBT’s Public Critique, On-Chain Claims, and Community Response

4.2.1 ZachXBT: Profile, Methodology, and Standing

ZachXBT is a pseudonymous on-chain investigator whose reports are widely followed in crypto markets. His relevance here is not biography; it is that his LAB claims materially shaped public interpretation of PiggyBank’s loss. Public accounts credit his investigations with tracing scams and stolen funds, and his reports commonly include wallet-level transaction evidence, timestamps, and open-source corroboration.^{112, 113, 114}

His standing has limits. He operates pseudonymously and his work is not a court judgment, regulator finding, or audit. Claims from his LAB thread should therefore be treated as allegations unless independently verified. Still, because his work frequently prompts market and exchange responses, his remarks about PiggyBank were relevant to the incident’s reputational impact and the public narrative that followed.^{115, 116, 113}

4.2.2 The Specific Claims Regarding PiggyBank

Scope of the public critique. The ZachXBT–PiggyBank nexus consists of one public X item, published on 6 June 2026, in which he quote-posted PiggyBank’s own transparency thread while linking the episode to his earlier LAB allegations. His language was pejorative and reputationally significant, but the item was not a formal audit, regulator finding, or full forensic report on PiggyBank. Reporting characterizes the substance of the remark as a criticism of PiggyBank’s risk management, a question as to why depositor funds were exposed to LAB at all, and — in one account — the pointed question of how a single strategy error could produce so large a loss in a stablecoin vault.^{117, 8, 7, 69}

ZachXBT material: confirmed, alleged, and not alleged

The report treats public claims as relevant to risk interpretation, not as legal findings.

Category	Treatment in this report	Reason
PiggyBank’s LAB trade and drawdown figures	Treated as PiggyBank self-disclosure, not as ZachXBT proof.	The existence and size of the position come from PiggyBank’s own public update.
LAB insider-control and market-manipulation claims	Treated as allegations about LAB unless separately corroborated.	They are relevant to asset-risk context but are not findings about PiggyBank.
PiggyBank fraud / insider connection to LAB	Not alleged by this report.	The public record reviewed does not establish that PiggyBank insiders were connected to LAB insiders.

Sources: #117, #103, #104, #105. The figure is an analytical boundary map, not a screenshot.

Figure 9. Claim-status boundary for ZachXBT’s PiggyBank/LAB-related public critique. Author-prepared analytical summary from Sources #117, #103, #104, and #105; third-party claims are treated as allegations unless independently verified.

The confirmed–alleged boundary. Three distinctions must be held firmly. First, the factual substrate of the episode — the existence, size, and structure of the trade, the unwind, the per-vault drawdowns, the exclusion of the locked tokens from net asset value, and the roughly \$1.35 million paper mark — is “confirmed” only in the sense that PiggyBank self-disclosed it; ZachXBT did not independently establish any of it but took the protocol’s own statement at face value and editorialized upon it (see 4.1.2 and 4.1.4). Second, ZachXBT’s harsh characterization was directed at LAB, the asset, and rests on his prior LAB allegations rather than on any forensic finding about PiggyBank. It is important to record that he did not allege that PiggyBank was itself fraudulent, nor that any PiggyBank insider was connected to LAB; the criticism attaches to the token PiggyBank purchased and to the protocol’s risk management, not to an established PiggyBank fraud finding. Third, the underlying LAB allegations — insider control exceeding 95% of supply, a British Virgin Islands corporate structure, unilaterally altered vesting, preferential OTC pricing, market-maker coordination, and an approximately \$6 billion fully diluted valuation — remain allegations and pertain to LAB; they are reconstructed in 4.1.1 and are not findings about PiggyBank.

An evidentiary correction. At least one outlet reported that ZachXBT alleged a “PiggyBank insider” controlled more than 95% of LAB’s supply. This appears to be a conflation. The primary post and the more careful coverage attribute the 95%-insider-control allegation to LAB-linked insiders, with no claim of a PiggyBank insider. The error is worth flagging because it materially misstates the relationship between the two projects, for which no operational link is established anywhere in the record (4.1.2).^{11, 105, 104}

4.2.3 On-Chain Evidence and the Limits of Independent Verification

No PiggyBank-specific on-chain evidence was cited. Despite the prominence the section title gives to on-chain forensics, ZachXBT’s PiggyBank post identified no PiggyBank wallet addresses, no Solana transaction signatures, and no specific contract interactions. Its evidentiary basis was PiggyBank’s own screenshots together with his prior LAB work. The on-chain analysis in that prior work concerned LAB’s supply structure and flows — clustered insider and market-maker holdings, OTC movements, and vesting allocations — and not PiggyBank’s vault activity (4.1.1).⁸

Independent verification on Solana explorers is not currently possible. On the public record, PiggyBank’s LAB position cannot be independently verified, for three reasons. First, no identifiers have been published: neither PiggyBank, nor ZachXBT, nor any outlet has disclosed the entry or exit transaction signatures, the vault or treasury addresses involved, the OTC counterparty, or the perpetual venue. Solana’s public explorers permit any address or signature to be inspected, so the obstacle to verification is the absence of disclosed identifiers, not the availability of tooling.

Second, the composition of the position lies largely off the Solana ledger: a locked OTC spot purchase and a perpetual short are precisely the kinds of transactions that need not appear as native on-chain events, since OTC deals are typically settled bilaterally and perpetual futures execute on venues whose internal fills cannot be reconstructed from a block explorer. Even with the vault addresses in hand, a third party could at best observe USDC margin flows, not the economic terms — the discount obtained, the short notional, or the funding actually paid. Third, self-disclosure is the sole source: every quantitative anchor (the \$100,000 of cash, the \$1.35 million paper mark, the 15% / 12% / 9% drawdowns, the 14 August unlock) traces back to PiggyBank’s 6 June thread and its NAV reporting, none of it corroborated against on-chain data by an independent party (4.1.2, 4.1.4).¹¹⁸

Implication. The episode is atypical of the events that ordinarily attract ZachXBT’s attention precisely because it lacks the on-chain evidentiary spine that characterizes his investigations; it rests on a protocol’s voluntary admission. For the paper’s risk assessment this is consequential: a depositor cannot today verify the size, the terms, or even the existence of the LAB book independently, and must rely on PiggyBank’s representations — a transparency gap developed further in 4.2.5 and 4.1.4.

4.2.4 The Community Response

What the record shows. The publicly documented reaction has three components. Before PiggyBank’s disclosure, the loss circulated on social media as a possible LAB “exploit” or hack; that framing was superseded once the confirmed facts pointed to a trading and risk-management loss rather than a contract compromise. Following ZachXBT’s post, his comments gained rapid traction, with community members responding through memes and criticism and several observers remarking on the irony of a yield optimiser losing principal on a single high-risk bet. Commentators then broadened the discussion into a debate over whether yield vaults should run directional hedges in low-float, retail-driven tokens at all, noting that depositors frequently fail to grasp their underlying exposure until losses surface.^{68, 8, 70}

What the record does not show. There is no documented evidence in the cited coverage of an organized depositor action — coordinated redemptions, a governance proposal, a legal complaint, or a formal compensation demand — as of the access date. So far as it can be sourced, the public response is confined to social-media reaction and press commentary; claims beyond that cannot be substantiated and are not asserted here.

4.2.5 PiggyBank's Official Response and Its Adequacy

The response, with respect to ZachXBT and later compensation materials. PiggyBank's first substantive public communication on the episode was the 6 June "Dear Community" thread on X, whose content is reconstructed in 4.1.4. The protocol later issued a 10 June compensation/drawdown communication, now included as Source #152. These materials addressed the mechanics of the trade, the NAV exclusion, and compensation intention, but they did not directly rebut the substance of ZachXBT's LAB-risk criticism and did not publish trade records, transaction signatures, counterparty identity, venue identity, funding paid, or loss-allocation calculations. The 10 June material therefore improves the remediation record compared with the initial 6 June silence, but it does not convert the public record into a verifiable post-mortem.^{119, 152, 155}

Adequacy. Measured against the mature-disclosure template set out in 4.1.4 — rapid acknowledgement, a documented causal chain, on-chain transaction references, a quantified loss accounting, an explicit remediation decision, clear withdrawal guidance, and independent verification — the response is mixed. It was adequate on the narrow dimensions of timeliness, broad narrative explanation, and the later existence of an official compensation statement. It remains inadequate on verifiability, because the public record still lacks the trade identifiers, loss decomposition, funding figures, cross-vault allocation key, withdrawal consequences, and payment-level evidence needed for independent reconstruction. PiggyBank's own materials foreground transparency, a "quantitative approach to protect your deposits," and a Formal Land audit; that audit pertains to contract correctness rather than discretionary trading-risk governance, and the LAB episode sits in visible tension with the transparency claim. The key missing item is therefore no longer merely "a response," but a verifiable, trade-record-bearing post-mortem and reimbursement record.^{2, 9, 152, 155}

4.2.6 Beyond the LAB Incident: Structural Risk-Management Concerns

Separating ZachXBT's contribution from the broader critique. It is necessary to distinguish what ZachXBT himself surfaced from what the wider reaction raised, because the section title risks crediting him with a structural audit he did not perform. His scrutiny of PiggyBank was narrow and reactive (4.2.2). His prior LAB investigation, however, supplied the one genuinely structural point bearing on PiggyBank's risk management: the asset was publicly and prominently flagged for alleged manipulation, alleged insider control, and opaque supply across 7–14 May 2026 — contemporaneous with PiggyBank's early-May entry and persisting throughout its hold (4.1.1, 4.1.2). The implication concerns process rather than misfortune: a delta-neutral, capital-preservation-oriented vault either failed to screen for, or chose to disregard, a high-salience public

red flag on the very token it was shorting, including ZachXBT’s explicit warning that LAB’s concentrated supply could fuel a squeeze against short sellers — the precise mechanism that subsequently materialized (4.1.1, 4.1.3). His framing of the episode as a speculative posture in a low-float hype token, rather than a hedge on a liquid major or a staking derivative, locates the failure in risk selection and not merely in execution.⁷⁰

What the episode surfaced more broadly. Independent of ZachXBT, the facts of the disclosure and contemporaneous reporting raised several structural concerns that outlast the LAB trade. The USDC stablecoin vault — the product most associated with capital preservation — absorbed the largest proportional drawdown, at roughly 15%, from a discretionary altcoin bet, inverting the risk ordering a depositor would reasonably infer (4.1.3, 4.1.5). A single speculative book was apportioned across all three vaults in the 15% / 12% / 9% split, so a product-specific mandate did not contain a product-specific bet (4.1.3). The trade may sit inside an expansively worded internal mandate for “mid-cap basis trades” while falling outside the conservative, delta-neutral proposition marketed externally, and a locked, illiquid leg is in any event structurally incapable of behaving in a delta-neutral manner (4.1.5). Observers specifically flagged the absence of a post-mortem detailing position limits, stop-losses, or diversification rules and why they failed to prevent a 15% drawdown; PiggyBank invoked internal risk limits but has not shown that they were defined in advance. The reliance on self-disclosure with no on-chain trade records (4.2.3) is not specific to LAB but is a standing transparency limitation of how the protocol reports, one that prevents independent verification of any vault strategy. And to the extent LAB’s 95%-insider-control allegation is ever substantiated, the episode reframes from an execution error into a failure to price concentration and market-structure risk into a vault exposure — though this remains contingent on allegations that are, to date, unproven (4.1.1).^{11, 70, 69}

Net structural reading. ZachXBT’s direct contribution was a high-visibility critique that crystallized attention rather than a forensic finding. The durable structural concerns — an inversion of the risk ordering across products, a gap between a permissive internal mandate and a conservative external proposition, undisclosed and possibly post-hoc risk limits, and a self-disclosure regime that resists independent verification — are supported by the aggregate record and by the analysis in 4.1. They would persist as open questions about PiggyBank’s risk governance even if the specific LAB position ultimately recovers value at the 14 August 2026 unlock (4.1.2, 4.1.5).

4.3 Protocol Response, NAV Drawdown Analysis, and User Impact

This section consolidates and extends the disclosure record reconstructed in 4.1.4 and 4.2.5, carrying it forward to the 15 June 2026 targeted update. Sections 4.1 and 4.2 established the

position, its mechanics, and the adequacy of the protocol’s first disclosure; the purpose here is to fix the full official-response timeline, including the later compensation-related communication, to reconstruct the NAV drawdown as precisely as public data permits, to delimit user impact and recourse, to test for verifiable post-incident change, and to assess the response against incident-handling norms. Throughout, a recurring methodological constraint from 4.2.3 applies: every quantitative anchor traces to PiggyBank’s own representations and cannot be independently verified on-chain, because no entry or exit transaction signatures, vault or treasury addresses, OTC counterparty, or perpetual venue have been disclosed.^{152, 155}

4.3.1 The Official Response: Statements, Actions, Timeline, and Remediation

Structure of the response. As of the updated review date, PiggyBank’s official response comprised two protocol communications, both issued on its X account (@piggybank_fi): the “Dear Community” transparency thread of 6 June 2026, reconstructed in 4.1.4 and 4.2.5, and a follow-up compensation/drawdown communication dated 10 June 2026. Treating these two communications as a single, evolving response is the correct frame: the second did not remove the evidentiary limitations of the first, but it added a narrative explanation and a remediation pledge.^{119, 152}

The 6 June actions and statements. The operative action taken on 6 June was the closure of the LAB perpetual short, which crystallized the accumulated mark-to-market and funding losses; the protocol stated that maintaining the hedge had become, in its own word, “economically irrational” and risked breaching its internal risk limits. The accompanying statement disclosed the position’s existence and approximate size (\$100,000, roughly 2% of the portfolio at entry), its structure (discounted, locked OTC spot in LAB hedged by a perpetual short, framed as a mid-cap basis trade), the proximate causes (manipulation, evaporating liquidity, and deeply negative funding), the residual locked LAB valued at approximately \$1.35 million, the exclusion of that holding from NAV until the first scheduled unlock on 14 August 2026, and the resulting per-vault drawdowns of approximately 15% (USDC vault), 12% (SPYx vault), and 9% (JitoSOL vault). PiggyBank characterized the NAV-exclusion decision as the “fairest and most transparent way to let users manage their liquidity” and undertook to publish a detailed report the following week.^{7, 11, 12, 8}

What the 6 June response omitted. As detailed in 4.1.4, the first disclosure announced no compensation and gave no guidance on whether, or at what value, depositors could withdraw against the marked-down NAV. Multiple outlets reported that they had sought comment and received none by publication. The protocol did not engage with the substance of ZachXBT’s risk-management critique (4.2.5).^{7, 8}

The 10 June compensation/drawdown communication. On 10 June 2026 PiggyBank released a further official statement explaining how the trade went wrong and, for the first time, pledging

compensation to affected users. Secondary reporting of that communication states that affected users would receive USDC compensation based on actual losses, with proposed funding sources including NAV discrepancies, future LAB sales after unlock, and a share of future platform revenue, and that users recorded in the 6 June snapshot would be eligible. Because the primary post is an X source and no independently verifiable payout ledger is available, those details should be treated as PiggyBank's stated remediation plan rather than as proof that reimbursement has been completed.^{152, 155}

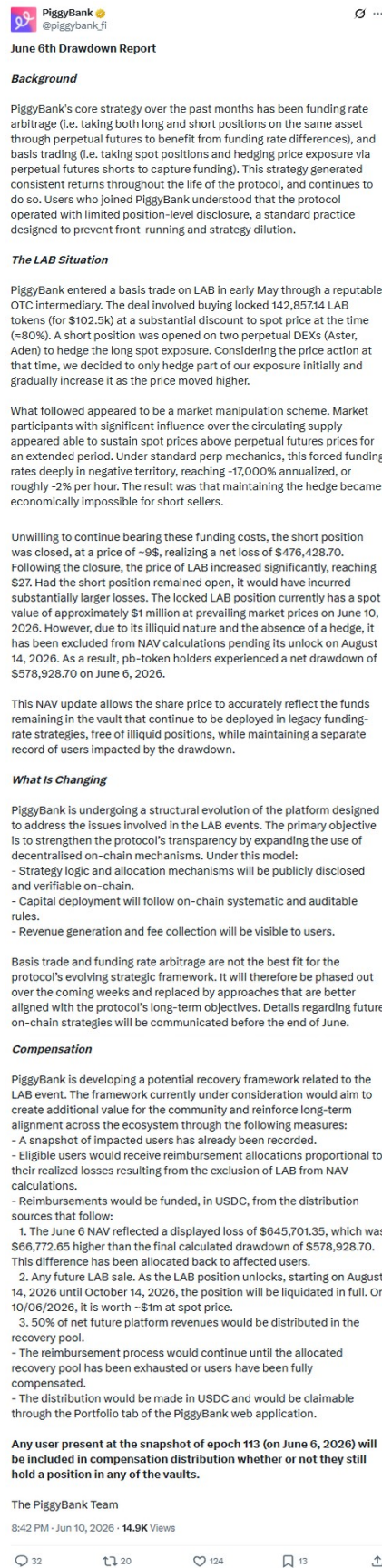


Figure 10. Full stitched screenshot of PiggyBank's 10 June 2026 compensation/drawdown communication. Source reviewed 15 June 2026. Source: #152.

Precise characterization of the 10 June report relative to 4.1/4.2. The follow-up communication must be characterized carefully, because it does not correspond to the trade-record-bearing post-mortem whose absence is recorded in 4.1.2, 4.1.4, and 4.2.5. Those sections anticipated a report disclosing trade records, risk thresholds, loss calculations, and a plan for the unlock; the 10 June drawdown report supplied none of these. By the protocol's own 10 June account, it did not disclose the total loss figure or a reimbursement timeline, and it did not publish entry or exit transaction signatures, the OTC counterparty, the perpetual venue, the funding actually paid, a decomposition of the loss between its directional and funding components, or the cross-vault allocation key that produced the 15% / 12% / 9% split. In the narrow sense used in 4.1.4 and 4.2.5, therefore, the substantive post-mortem remained outstanding as of 13 June: what had appeared by 10 June was a narrative explanation accompanied by an unquantified remediation pledge, not the verifiable trade-level accounting that observers had called for.^{120, 69}

Compensation and remediation: official statement, not payment-level verification. The compensation statement is the single material remediation addition after the initial 6 June silence. It is a genuine improvement because it moves the response from explanation to proposed recourse. Its operative limitation remains evidentiary: the public materials reviewed here still do not disclose a total realized loss figure, a per-user amount, a binding payment date, a completed reimbursement transaction trail, or independent confirmation that affected depositors have been made whole. For that reason, the statement should be recorded as PiggyBank's public compensation position, but not as independent verification of completed compensation.^{152, 155}

4.3.2 Quantitative NAV Drawdown Analysis

Nature of the drawdown: an accounting markdown, not a peg or contract loss. The reported drawdown is not a depeg of the underlying stablecoin or a smart-contract impairment; it is an accounting consequence of removing the locked LAB (approximately \$1.35 million in paper value) from NAV while the realized short loss flows through in full — the accounting asymmetry analyzed in 4.1.3. The event was initially circulated on social media as a possible LAB exploit, but the confirmed facts point to a trading and risk-management loss rather than a verified hack (4.1.2).

^{11, 68}

Per-vault drawdown magnitudes. The sole quantitative drawdown figures in the public record are the protocol's own estimates of approximately 15% for the USDC vault, approximately 12% for the SPYx vault, and approximately 9% for the JitoSOL vault, reported consistently across all secondary coverage. PiggyBank explicitly framed these as estimates reflecting the announced accounting treatment, subject to change when the locked LAB begins unlocking on 14 August.^{7, 12,}

^{68, 11}

Limits on a finer reconstruction. A reconstruction of absolute NAV — a per-share index or a time series of vault NAV before, during, and after the event — is not possible from public data. PiggyBank reports NAV to depositors through its application on a 48-hour epoch cycle and does not, so far as can be located, publish a historical NAV series or the underlying share-price marks. The only published quantitative anchors are therefore the three percentage drawdowns and the \$1.35 million paper mark; all four trace to PiggyBank’s own disclosure and remain uncorroborated on-chain (4.2.3). The reconstruction below is consequently expressed as a structured statement of what is known, what is derived, and what is unavailable, rather than as a continuous NAV curve.^{2, 1}

Reconstructed NAV path. Before the event, each vault’s NAV reflected its full asset base, including the LAB book carried at cost or at a hedged mark; the self-reported vault sizes from Section 1 were approximately \$3.06 million (USDC vault), \$823,000 (SPYx vault), and \$639,000 (JitoSOL vault). During the holding period (roughly early May to 6 June), the economic position deteriorated as LAB rose several-fold on thin liquidity and funding turned deeply negative, but this deterioration was not necessarily surfaced to depositors in full while the hedge remained open and the spot leg was carried as an offset (4.1.2). The discrete, observable NAV step occurred at the unwind: closing the short crystallized the loss, and excluding the locked spot leg from NAV removed the only offsetting asset, producing the one-sided markdown of 15% / 12% / 9%. After the event, NAV remains at the marked-down level; because the offsetting asset is excluded until 14 August, the drawdown does not reverse within the reporting in the interim, and any later recovery — if LAB holds value through the unlock — would be credited only at that future NAV print (4.1.3).

Timing of recognition under the epoch mechanism. The drawdown surfaced to users in the NAV print of the epoch following the 6 June unwind, consistent with the 48-hour epoch cycle; redemption requests are batched and clear over the subsequent epoch, so withdrawals typically settle 48 to 96 hours after submission. This cadence matters for the user-impact analysis below: a depositor reacting to the marked-down NAV print could not exit instantaneously, and the value realized on exit is the value at the epoch in which the redemption clears.^{2, 1}

Peak drawdown, persistence, and recovery. On the public record the peak reported drawdown is approximately 15%, borne by the USDC vault; it is an accounting peak fixed by the NAV-exclusion decision rather than a continuously re-marked figure. Its persistence is structurally bounded below by the 14 August unlock: while the locked LAB is excluded, the markdown cannot be reversed through NAV, so it persists at least until the first unlock and, depending on LAB’s realizable price thereafter, potentially beyond it. As of the access date there is no evidence of NAV

recovery; recovery has not occurred and cannot occur within the current accounting treatment before the unlock, and its eventual magnitude is contingent on LAB’s price at and after 14 August — an uncertain prospect given LAB’s thin liquidity, alleged insider control, and the additional sell pressure that the unlock itself is expected to introduce (4.1.1).^{70, 7}

Total-value-locked decline, distinguished from NAV. TVL and NAV are distinct: NAV is the per-vault value attributable to existing shares, whereas TVL is the aggregate capital remaining in the protocol, which falls as depositors redeem. SolanaFloor reported that PiggyBank’s TVL declined by approximately 49.4%, from about \$5.4 million to about \$2.73 million, after the LAB disclosure; a 15 June DefiLlama spot check showed PiggyBank TVL at approximately \$2.5 million. (These third-party, protocol-wide figures are measured at different dates and on a different basis than the roughly \$4.5 million self-reported vault aggregate in Section 1, and are not a single time series.) These figures should still be treated as time-sensitive, but they are directionally consistent: the post-incident contraction in capital appears to have exceeded the reported per-vault NAV drawdowns, indicating redemption-driven flight of capital rather than merely the accounting markdown itself. That distinction matters for recourse, because a shrinking deposit base and lower ongoing fee base reduce the resources from which any platform-revenue-based compensation could be funded.^{154, 153}

4.3.3 User Impact: Scope, Aggregate Loss, and Recourse

Number of users affected: not disclosed. Neither PiggyBank nor any located secondary source has published the number of depositors affected, whether in aggregate or by vault; this figure is unavailable on the public record and is not estimated here. What can be stated is the scope by product: the loss was apportioned across all three vaults — the USDC vault, the SPYx vault, and the JitoSOL vault — so depositors in each were affected, with the proportional impact ordered as the 15% / 12% / 9% drawdowns indicate.⁷

Aggregate dollar loss: undisclosed by the protocol; only a derived estimate exists. PiggyBank has not published a total loss figure, either in the 6 June thread or in the 10 June drawdown report. The only quantification available is the derived estimate constructed in 4.1.3, which applies the reported per-vault drawdowns to the self-reported vault sizes to yield an aggregate, USDC-equivalent markdown of approximately $(0.15 \times \$3.06\text{M}) + (0.12 \times \$0.823\text{M}) + (0.09 \times \$0.639\text{M}) \approx \$459,000 + \$99,000 + \$58,000 \approx \$615,000$. This figure is a derivation from sourced endpoints, not a disclosed or independently verified loss; it depends on the accuracy of the self-reported vault sizes and on the allocation that produced the percentage split, neither of which has been published. It should be read alongside the structural observation from 4.1.3 that the headline “\$100,000, roughly 2% of the portfolio” materially understated the economic exposure, because the hedged

notional and the locked face value were several times the cash outlay, and the realized loss — on the order of six times the cash deployed — reflects a short into a multi-fold price move compounded by negative funding, with no offsetting credit because the long leg was illiquid and excluded from NAV.¹²⁰

Recourse available to affected depositors. Four avenues are available, none complete on the public record. First, depositors may withdraw at the marked-down NAV, subject to the 48-to-96-hour epoch batching; doing so crystallizes the loss and may affect participation in any later recovery depending on how the compensation plan and future NAV treatment are implemented. Second, depositors may remain in the vaults and await the 14 August unlock, accepting illiquidity and the risk that LAB's realizable value at unlock is far below its paper mark. Third, depositors may rely on PiggyBank's official compensation statement, which refers to USDC compensation for affected users and future funding from LAB sales and platform revenue, but which remains unaccompanied by independent payment-level evidence. Fourth, formal or legal recourse is limited and uncertain on the available record: no governance proposal, coordinated claim process, or legal complaint is documented here, and the protocol's operating legal structure is not clearly disclosed. There is no third-party insurance or external backstop in the record from which depositors could otherwise be made whole.^{1, 2, 152, 155}

4.3.4 Post-Incident Changes and Their Verifiability

As of the updated review date, no verifiable structural remediation had been published. The 6 June disclosure and 10 June compensation/drawdown communication explain the LAB position and state a compensation intention; secondary reporting also describes planned structural reforms, including greater transparency around on-chain mechanisms, strategy logic, and fund allocation, and a gradual phase-out of basis trading and funding-rate arbitrage. Those statements are material, but they remain plans or representations unless and until reflected in public documentation, address disclosures, position limits, eligible-asset rules, real-time position disclosure, a complete strategy/treasury/venue-position register, proof-of-reserves, or independent NAV attestation. The protocol's public documentation reviewed here had not yet incorporated such verifiable changes.

^{152, 155, 2, 149}

PiggyBank's public materials also continued to describe the same automated, delta-neutral funding-rate strategy and to foreground risk management without incorporating the incident into the core documentation or the public news page. This section records the historical response only; the current-state implications of that absence are assessed in Section 5.1. The caveat is that lack of externally visible change is not proof that no internal control changed — it only means that no such change could be verified from the public record.²

4.3.5 Assessment of the Response Against Industry Standards

The benchmark. The assessment applies the mature-disclosure template set out in 4.1.4 — rapid acknowledgement; a detailed post-mortem with a documented causal chain and on-chain transaction references; a quantified loss accounting; an explicit remediation or compensation decision; clear and immediate user guidance on funds and withdrawals; and, where possible, independent verification — for which the Lazy Summer / Summer.fi Arbitrum USDC vault post-mortem of December 2025 is a useful contemporaneous benchmark, having set out a step-by-step causal chain, named the external dependencies, and discussed reimbursement in detail.¹¹¹

Dimension-by-dimension evaluation, updated to 10 June. On timeliness, the response was reasonably prompt: the first disclosure coincided with the unwind on 6 June, and the follow-up materials arrived within the “next week” window the protocol had set — though the disclosure was reactive, issued at and after crystallization rather than at entry or as the risk built (4.1.4). On transparency, the response was specific about the accounting treatment and headline drawdowns but opaque on everything required for verification: no trade records, transaction signatures, counterparty, venue, funding paid, loss decomposition, or cross-vault allocation key, leaving the account uncorroborable on-chain (4.2.3). On loss accounting, the standard is not met: no total loss figure has been published. On remediation, the official compensation statement is a real improvement over the initial 6 June silence, but without amounts, deadlines, funding source, or payment-level verification it still does not satisfy the standard of an explicit, independently verifiable compensation scheme. On user guidance, the standard is not met: depositors were given no clear direction on whether, or at what value, to withdraw against the marked-down NAV. On independent verification, the standard cannot be met on the current disclosures at all.¹⁵²

The accounting choice, weighed fairly. The decision to exclude the illiquid locked LAB from NAV is defensible on its own terms — marking an unsellable, thinly traded asset into NAV could overstate realizable value — and PiggyBank presented it as the fairest treatment for users managing liquidity. The countervailing observations are that the same convention minimizes the headline NAV optics while the economic loss persists, that it can obscure real-time solvency signals, and that it creates the distributional asymmetry between exiting and remaining depositors noted in 4.1.3 and 4.3.3.^{8, 70}

PiggyBank’s defensible points. In fairness to the protocol, several features of its conduct are creditable. It characterized the episode accurately as a contained, rules-bounded trading loss rather than a hack, and stated that it had acted before the position crossed its risk limits. It disclosed the asset, the structure, and the accounting treatment with more specificity than a bare loss announcement, issued a follow-up within the promised window, and moved — between 6 and 10

June — from silence on remediation to a public commitment to compensate. These are not the hallmarks of an exit or a cover-up, and the analysis should not treat them as such.^{7, 68}

Net assessment. On balance, the response improved over the week but remained incomplete on the dimensions that matter most for a product marketed on capital preservation. It was adequate on timeliness and on specificity about the accounting treatment and headline numbers; it improved on remediation by placing an official compensation statement on the public record; and it remained insufficient on loss accounting, verifiable causal detail, user withdrawal guidance, and independent verification. The decisive tests — a quantified and dated compensation scheme, payment-level evidence if compensation has occurred, and a trade-record-bearing post-mortem against which ZachXBT’s critique and the protocol’s own “within-scope” framing could be tested rather than merely asserted — remained unmet in the public record reviewed here. Until those materials exist, the episode stands as one in which the protocol communicated promptly and pledged remediation, but did not yet furnish the evidence or executable detail that the sector’s better incident-handling practice, and its own stated commitment to transparency and risk management, would require.^{2, 152}

5. Risk Posture and Transparency as of the Review Date

5.1 Ongoing Risk Management Concerns Post-LAB Incident

This section assesses PiggyBank’s risk posture as of the public-source access date of 13 June 2026, with targeted updates through 15 June 2026 where stated. Current-state claims in this section are dated observations, not permanent market conditions. The purpose is to determine whether the conditions that produced the loss persisted in the public record at that time: whether the protocol retained capacity to repeat a similar position (5.1.1); whether the controls governing asset and strategy selection were on-chain and verifiable or off-chain and trust-based (5.1.2); whether the then-prevailing funding-rate environment supported or undermined the strategy (5.1.3); how leverage risk looked under those market conditions (5.1.4); and how Solana-ecosystem concentration transmitted counterparty risk to PiggyBank (5.1.5). The recurring methodological constraint from 4.2.3 applies throughout: selected vault state accounts are visible through a public adapter, but the protocol’s live positions, treasury accounts, strategy accounts, perpetual venues, lending accounts, and authority/signature structure are not disclosed as a complete map. Forward exposure is therefore assessed from PiggyBank’s own representations and from observable public data about its counterparties at the review date, not from independently verifiable on-chain position reconstruction.¹⁴⁹

5.1.1 Persistence of the Structural Conditions That Enabled the Incident

Section 4.3.4 established the historical response finding: no externally verifiable control change followed the LAB incident by the access date. The current-state question is narrower and more practical: whether PiggyBank’s public framework now contains controls that would prevent another position of the same character — a discretionary, illiquid, OTC-sourced basis trade in a low-float token — from being taken with depositor capital.

On the public record, the answer is no. PiggyBank’s materials still describe the same broad automated, delta-neutral funding-rate strategy across approximately ten perpetual DEXs, with non-stablecoin vaults collateralized on Solana lending venues to borrow stablecoins for the same strategy. The documentation still describes allocation to funding-rate arbitrage and liquidity provision without publishing an eligible-asset whitelist, a position-limit schedule, concentration constraints, an OTC-deal policy, or a rule excluding illiquid locked-token basis trades.^{2, 1}

That matters because PiggyBank did not present LAB as a breach of its mandate; it presented the trade as a calculated mid-cap basis position within the strategy’s scope. In the absence of a newly published boundary around eligible assets, liquidity, position size, and OTC sourcing, the same structural permission appears to remain. The protocol’s own news page likewise did not

incorporate the incident into its governing materials as of the access date, leaving prospective depositors with the pre-incident public framework.^{69, 121}

5.1.2 The Current Risk-Management Framework: Controls and Their Verifiability

PiggyBank represents that it uses automated margin rebalancing, deleveraging based on market indicators, predefined thresholds, and a partial take-profit / stop-loss mechanism. Those claims are relevant, but they do not disclose the values that would make the framework auditable: leverage ceilings by strategy, maximum position size, drawdown triggers, asset-eligibility rules, concentration limits, liquidity requirements, or escalation procedures.²

The gap is not simply that the thresholds are unpublished. The represented controls address margin, liquidation, and execution risk, while the LAB failure mode also concerned mandate risk: the decision to enter a low-float, illiquid, OTC-sourced asset in the first place. A stop-loss or deleveraging rule may limit losses after a position deteriorates, but it does not by itself prevent the operator from selecting an asset that depositors would not have understood as part of a conservative, market-neutral strategy.^{2, 69}

The trading-risk framework is therefore best characterized as off-chain and trust-based. The strategy is administered by the operator across external venues, NAV is reported every 48 hours rather than through a continuously verifiable on-chain index, and the treasury addresses, venue-position feeds, strategy accounts, and trade records needed to test compliance are not published as a complete set. The on-chain controls that do exist — such as token-standard and transfer-hook mechanics — govern share issuance, custody, or compliance boundaries, not the trading mandate that produced the LAB exposure.^{1, 2, 149, 150}

Net assessment. PiggyBank may have internal controls, but the public framework does not allow a depositor to verify that those controls prevent a repeat of the LAB-style failure mode. The framework remains operator-administered, parameter-light, and substantially unverifiable from outside the protocol.

5.1.3 The Funding-Rate Environment and the Conditions for Sustained Loss

The strategy's dependence on the sign of funding. PiggyBank's core return source is the funding rate on perpetual futures. In the canonical cash-and-carry configuration consistent with its marketing — long spot (or borrowed collateral) hedged by a short perpetual — the position earns when funding is positive, because longs pay shorts, and the short leg collects that payment as market-neutral yield. The favorability of the environment for the core strategy therefore turns on whether perpetual funding on the traded assets is positive and sufficient to clear costs.²

At the time of the source check, the prevailing regime appeared unfavorable. Through the first half of 2026 the funding environment had been structurally adverse to a short-perpetual carry strategy on the majors. Bitcoin perpetuals posted a negative 30-day average funding rate for 46 consecutive days as of mid-April 2026, described as the longest sustained negative-funding streak since the post-FTX period of November 2022. Solana futures funding ran negative for sixteen consecutive days earlier in the year, with an open-interest-weighted eight-hour average near minus 0.0042, indicating that shorts were paying longs to maintain exposure. Into June the picture remained negative-skewed and the price trend deteriorated further: SOL was reported near 81 dollars in late May with funding skewed slightly negative, before breaking below the 80-dollar support level and declining toward the 60s during June. A negative-funding regime inverts the carry: a long-spot / short-perpetual book pays funding rather than receiving it, producing NAV erosion even when the position is delta-neutral and even without a LAB-style concentrated loss. The protocol's advertised target net yields of approximately 15% on the USDC vault and approximately 7% on the xStocks vaults are predicated on positive carry that this dated market regime did not reliably supply.^{122, 123,}

124, 125, 126, 2

Conditions that would cause sustained losses. Generalizing the LAB mechanism and overlaying the then-current funding environment, the strategy would generate sustained losses under five conditions, which can compound. First, persistently negative funding on the shorted legs, so that the strategy pays rather than collects — the condition observable on the majors during the source-check window. Second, a sharp upward price move in a shorted asset combined with thin exit liquidity, which crystallizes a basis loss on unwind — the precise LAB pattern (4.1.2). Third, a basis divergence the hedge fails to track, as where the spot leg is locked or illiquid, or where a tokenized-stock leg gaps against a stale reference during hours when the underlying equity market is closed. Fourth, funding-clamp dynamics on lower-tier markets: Drift clamps the hourly funding magnitude by contract tier, at 0.125% for the highest tiers but as much as 0.4167% per hour for markets below the C tier, so a low-float market can sustain a funding charge on the order of 0.42% per hour, or roughly 10% per day, against a short position. Fifth, an inversion of the lending leg, where the interest rate on the borrowed stablecoin exceeds the carry the strategy earns, turning the collateralized-borrow structure of the xStocks vaults and the JitoSOL vault into a net cost.¹²⁷

A fairness qualification, and the net judgment. In fairness to the protocol, a flexible long-short desk can in principle position on the favorable side of funding — receiving negative funding by holding the long perpetual and shorting or borrowing the spot — and the protocol's materials describe a long-short rather than a strictly one-directional strategy. Two considerations limit the weight of this qualification. The reverse cash-and-carry is materially harder to execute on-chain, because spot-borrow and short availability are limited and tokenized equities are difficult to short,

and PiggyBank has not disclosed that it operates the reverse leg. The posture consistent with its marketing and with the LAB trade is the conventional one of shorting the perpetual to collect funding, and that posture was disadvantaged by the funding regime observed at the review date. This should not be treated as a permanent market condition; it is a dated assessment of the environment visible in the public record during the source-check window.²

5.1.4 Leverage Risk Under Prevailing Market Conditions

The two layers of leverage. PiggyBank’s leverage is represented at up to approximately 5x on the USDC funding-arbitrage strategy. For the xStocks vaults and the JitoSOL vault a second layer of leverage is embedded in the structure itself: depositors’ assets are posted as collateral on Kamino, Loopscale, or Jupiter Lend, stablecoins are borrowed against them, and the borrowed stablecoins are deployed into the perpetual strategy, so these vaults carry both the borrow-against-collateral leverage of the lending leg and the leverage of the perpetual leg, each with its own liquidation threshold.²

Why 5x was more dangerous under the review-date conditions than the headline implies. A 5x cap is moderate relative to venue maxima — Jupiter permits up to 100x and Drift up to 20x — but leverage risk is conditional on volatility and liquidity, both of which appeared adverse at the review date. SOL was reported in a steep downtrend, having broken the 80-dollar support and fallen toward the 60s, and Solana on-chain liquidity had thinned markedly, with monthly DEX volume falling from a peak near 145 billion dollars in October 2025 to roughly 42 billion dollars by the end of April 2026. In that setting, a position at a given leverage has materially less margin buffer than in a calm, liquid market, and automated deleveraging remains exposed to the documented failure modes of illiquid Solana markets, where positions can gap through stops during network congestion or oracle staleness.^{43, 128, 77}

Two-sided liquidation exposure and cross-protocol cascade. The leverage risk is two-sided. On the perpetual leg, a delta-neutral book is largely insensitive to price, but its margin can still be stressed by funding outflows in the negative-funding regime described above and by adverse marks on the short during an upward squeeze — the LAB pattern. On the lending leg, a falling SOL price erodes the value of underlying JitoSOL collateral and a falling equity tape erodes the value of xStocks collateral, pushing each borrow toward its liquidation loan-to-value; and because the Kamino xStocks lending market has been reported at roughly 92% utilization, the liquidity available to unwind or top up a position at the moment of stress may be constrained, while high utilization simultaneously raises borrow rates and compresses or inverts net carry. Because the same collateral assets back the borrows that fund the perpetual positions, a liquidation on the lending leg can force closure of the hedge leg and vice versa — a cross-margin amplification

analogous to the dynamic Drift documents, in which a single impaired position can drain collateral backing otherwise-healthy ones. The epoch mechanism compounds this exposure: because redemptions clear only over a 48-to-96-hour window (4.3.2), the protocol cannot delever instantaneously to meet a wave of withdrawals, and a forced unwind into thin liquidity would realize slippage on top of any market loss.^{16, 129, 77}

Net assessment of leverage risk. Nominal leverage of approximately 5x is moderate, but the effective leverage risk under prevailing conditions is elevated. The headline figure understates the risk because it omits the second, lending-leg layer of leverage in the xStocks vaults and the JitoSOL vault, and because it is stated without reference to the volatility and liquidity conditions that determine how much buffer that leverage actually provides. In the current regime of elevated SOL volatility, contracting on-chain liquidity, high lending-market utilization, and demonstrated fragility of automated deleveraging in illiquid Solana markets, the protocol's leveraged positions carry a higher risk of stress-driven loss than the 5x figure alone would suggest.

5.1.5 Systemic Risk from Solana-Ecosystem Concentration

The state of Solana DeFi at the access date. At the review date, Solana DeFi appeared both contracting and concentrated. Total value locked across the chain stood in the range of roughly 4.6 to 5.5 billion dollars, down approximately 56% from the August 2025 peak above 11.5 billion dollars; monthly active users had fallen to a two-year low near 34 million; chain fees were down roughly 50% since January; and monthly DEX volume had fallen from a peak near 145 billion dollars in October 2025 to about 42 billion dollars by late April 2026 as memecoin activity cooled. A countervailing nuance is that SOL-denominated TVL reached an all-time high near 80 million SOL in the first quarter of 2026, indicating that native holders had not fully exited and that part of the dollar contraction reflects price rather than capital flight. The ecosystem was also highly concentrated at that review date: Kamino was the dominant lending market at roughly 2 to 3 billion dollars, Jupiter spanned lending, perpetuals, and liquid staking across separate multi-hundred-million-dollar lines, and the gap between the top three protocols and the tenth was on the order of twentyfold.^{128, 78, 130, 97}

Why concentration is a first-order risk for PiggyBank specifically. PiggyBank's strategy is not diversified away from Solana; it is constructed on a small set of Solana venues that are themselves the ecosystem's points of concentration. Its counterparty surface comprises, on the trading side, approximately ten perpetual DEXs led by Drift; on the financing side, three lending markets in Kamino, Loopscale, and Jupiter Lend; and at the asset level, JitoSOL as a Jito liquid-staking token and SPYx as a tokenized-equity certificate issued through xStocks and Backed. Each is a single point of failure whose impairment the protocol's self-disclosure regime (4.2.3) would not surface

to depositors in advance, and the concentration means that diversification across approximately ten venues provides less protection than the count implies, because those venues share the same chain, the same dominant lending counterparty, and correlated liquidity conditions.²

The realized precedent and its direct relevance. This is not a hypothetical exposure. Public reporting described the 1 April 2026 Drift incident as an exploit of the largest perpetual DEX on Solana, involving approximately 285 million dollars, a North Korea-linked actor, and a months-long infiltration pattern. Public reporting also described user losses estimated near 295 million dollars and contagion across more than twenty protocols, including Carrot Protocol pausing mint and redeem functions after roughly half its TVL was affected, Pyra Protocol disabling withdrawals, and Prime Numbers Fi reporting losses in the millions. PiggyBank stated that it lost approximately 106,000 dollars in that event and reimbursed affected users from its own team treasury. The episode is directly probative for three reasons: it demonstrates that a major Solana perps venue used by PiggyBank had been exploited in a severe incident; it confirms that PiggyBank carried live exposure to that venue; and it shows that the protocol's backstop for counterparty losses is its own treasury — the same unquantified resource standing behind the still-unexecuted LAB compensation pledge (4.3.1), and a resource that the reported post-incident TVL contraction (4.3.2) would have eroded.^{64, 131, 132, 67, 146}

The ongoing transitional dimension at Drift. The Drift exposure was also transitional rather than steady state at the access date. Following the exploit the protocol secured a rescue package of approximately 148 million dollars led by Tether, was relaunching as a perpetuals venue that settles in USDT rather than the USDC it previously used, and was re-auditing its components with OtterSec and Asymmetric Research, with a relaunch targeted tentatively for the May-to-June 2026 window. For a strategy that had routed flow through Drift, this meant either re-onboarding to a freshly relaunched protocol with a changed settlement asset — itself a risk event — or concentrating more flow into remaining venues and raising single-venue exposure elsewhere.^{131, 132}

Tokenized-equity dependency risk in the SPYx vault. The SPYx vault adds a distinct layer of issuer, custody, and redemption dependency risk. SPYx is a tokenized tracker certificate backed one-to-one by the underlying security held with a regulated custodian, conferring economic exposure without ownership or voting rights and reinvesting dividends into token value; the instrument is unregistered under United States securities law and is offered only to qualified investors. Two structural risks follow. First, an off-hours basis risk: xStocks trade continuously on-chain and around the clock on certain venues while the underlying equity market does not, so the token can move against a stale or absent reference price, stressing any hedge and any collateral valuation during periods when the cash market is closed. Second, an issuer, custody, and

redemption-gating risk: the certificate’s peg depends on Backed’s custody arrangements and redemption mechanics, a trust-based dependency entirely outside PiggyBank’s control. The reported 92% utilization of the Kamino xStocks lending market compounds the liquidity dimension of this exposure on the financing leg.^{129, 133, 16}

Net assessment of systemic exposure. As of the review date, PiggyBank’s counterparty exposures were concentrated in the Solana venues that appeared thinner and more concentrated than a year earlier and that had already demonstrated tail events. The protocol’s loss-absorption mechanism for counterparty failures was a discretionary, unquantified team treasury rather than ring-fenced insurance or an external backstop (4.3.1). This systemic-risk profile was therefore elevated at the source-check date, but it should not be treated as a permanent market condition. Consistent with the rest of this assessment, the risk also could not be independently verified or bounded from PiggyBank’s disclosures, since the specific venues, accounts, and per-counterparty exposures were not published (4.2.3).

5.2 Transparency Gaps: Anonymous Team, Self-Reported Metrics, and Unverified Claims

Section 3.2 assessed whether PiggyBank’s formal risk disclosures match the product’s actual mechanics. This section addresses a different transparency question: whether the protocol’s public claims about identity, scale, integrations, NAV production, and first-mover status can be corroborated independently. Where PiggyBank makes material claims about transparency, scale, integrations, NAV, or risk controls, the absence of public corroboration is treated as analytically relevant rather than as neutral silence.^{149, 150}

5.2.1 Team Anonymity

What is known about the team. PiggyBank attributes its operation to no named individual. The application and marketing site carry no team page, no founder names, and no link to any LinkedIn, GitHub, or other professional profile; the only points of contact are a generic address, hi@piggybank.fi, and an unnamed “Private Deals” contact form. The documentation refers to the operators only collectively, asserting that “the team has strong expertise in blockchain, financial engineering, and quantitative research” without identifying any person, entity, or prior project. Third-party coverage of the protocol corroborates the gap, recording that its “funding details and backers have not been publicly disclosed”. Searches of company registries and professional networks return only unrelated namesakes — a United States children’s-allowance application and the Nigerian savings platform PiggyVest — none of which is the Solana protocol under examination. The single named external party associated with PiggyBank is its auditor, Formal Land, addressed in 5.2.4.^{2, 1, 5}

What is not known. No operating entity, operating jurisdiction, named principals, verifiable track record, or identity of the treasury and program-authority signers is disclosed. The deficiency compounds the disclosure gap recorded in 4.2.3: although selected vault state accounts can be read through a public adapter, the complete treasury, signer, authority, strategy, and venue-position map is unpublished, so the protocol remains opaque at both the human layer and the economically material address layer.¹⁴⁹

Why anonymity is a heightened risk in this specific configuration. Anonymity is common in the sector and is not, in itself, evidence of bad faith; pseudonymous teams have shipped durable code for years, and the foundational precedent of Bitcoin demonstrates that an anonymous origin is compatible with legitimacy. The risk is conditional on the operating model. PiggyBank is not a passive, immutable, fully on-chain primitive whose behavior is fixed by code; it is an actively managed, discretionary, leveraged strategy in which an off-chain operator selects positions, sources assets over the counter, and reports NAV (5.1.1–5.1.2). Where an anonymous operator exercises custodial-style discretion over leveraged depositor capital, the accountability that substitutes for “code is law” — the reputational and legal exposure of identifiable principals — is absent precisely where it would matter most. Prevailing due-diligence guidance treats a team with no verifiable identity as a primary red flag, observing that in a large majority of documented rug pulls the responsible team had no verifiable identity, no professional history, and no real names anywhere in their materials. This is not an assertion that PiggyBank is such a scheme; it is the narrower observation that the protocol lacks the very attribute by which the two cases would be distinguished *ex ante*.^{134, 135}

A countervailing observation on demonstrated accountability. Disclosure and accountability are distinct, and PiggyBank has exhibited conduct inconsistent with a pure exit posture: it absorbed the approximately 106,000-dollar loss it sustained in the April 2026 Drift exploit from its own treasury and issued a compensation pledge following the LAB incident (4.3.1, 5.1.5). An anonymous team can behave responsibly, and this record is a meaningful, if partial, offset. It remains a weaker assurance than the combination of identifiable principals and ring-fenced funds, however, because voluntary reimbursement from an unquantified discretionary treasury depends on the continued solvency and goodwill of the same undisclosed party, and the LAB compensation pledge was not independently payment-verified in the public materials reviewed here.

Comparison with disclosure norms in the sector. DeFi disclosure spans a spectrum, from fully identified teams (the Ethereum model, in which named public figures anchor accountability) through pseudonymous-but-track-recorded builders to wholly anonymous launches with no history. Products that intersect securities, custody, and real-world-asset expectations cluster toward

the identified end of that spectrum. Judged against its own category — a real-world-asset-adjacent, leveraged, custody-style product handling tokenized equities — PiggyBank discloses materially less than its peers and counterparties, each of which operates as an identifiable entity: Kamino, Backed (the xStocks issuer), and Jito are all publicly attributable. PiggyBank’s disclosure posture is closer to that of an anonymous token launch than to the tokenized-equity infrastructure it is built upon.¹³⁶

Net assessment. The identity of the team is unknown and unverifiable on the current record; the risk that this poses is amplified, not mitigated, by the discretionary and leveraged nature of the operating model; the disclosure falls below the norm for the protocol’s category; and while demonstrated reimbursement conduct provides a partial offset, it does not substitute for the accountability that identification and ring-fenced funds would supply.

5.2.2 TVL and User Metrics

The figures as presented and updated. PiggyBank self-reported its scale through figures rendered on its own interface at the original source-review date: the vault table listed per-vault TVL of approximately 3.06 million dollars (USDC vault), 823,000 dollars (SPYx vault), and 639,000 dollars (JitoSOL vault), an aggregate on the order of 4.5 million dollars; a headline “Protocol Metrics” panel reported more than 2,000 total users and three asset classes; and the completed Season 0 was recorded as having 1,832 unique depositors. A later 15 June spot check of DefiLlama’s PiggyBank protocol page listed TVL at approximately \$2.5 million. These figures should not be blended as if they were measured at the same time: the official-interface figures are a reviewed historical snapshot, while the DefiLlama figure is a later third-party live-data point reflecting the post-incident state.^{2, 153}

Internal inconsistency on a single surface. The reviewed PiggyBank interface did not reconcile cleanly with itself. The hero banner displayed APY figures that differed from the approximate target APYs discussed elsewhere in the materials, and the metrics, vault table, and user-count claims were stated on different bases that the page did not let a reader reconcile. None of this is necessarily evidence of bad faith, and some discrepancies may reflect a stale or partially loaded data feed; but a self-reported metrics surface that does not internally cohere is, at minimum, unreliable as a measurement without external cross-checking.²

Partial third-party tracking and why it matters. PiggyBank is no longer a case of pure operator-only TVL reporting. DefiLlama’s PiggyBank protocol page currently reports TVL and its adapter repository lists three PiggyBank vault state accounts associated with the vault-share tokens pbUSDC, pbSPYx, and pbJitoSOL. The methodology states that the adapter reads on-chain accounts and calculates underlying value by summing managed and epoch-deposit values and

subtracting redemptions. That is materially better than relying only on the protocol interface, because a third party can observe specific vault accounting state on-chain. The limitation is equally important: the adapter reads vault-account accounting fields, not the operator’s full treasury, strategy allocation, borrowing venues, perpetual positions, locked OTC holdings, or NAV methodology. The result is partial independent TVL visibility, not full independent risk verification.^{153, 149}

What DefiLlama visibility does and does not prove

Adapter visibility narrows the address gap; it does not independently verify the full strategy book.

Visible through public adapter / dashboard	Not established by that visibility
Three PiggyBank vault state accounts associated with pbUSDC, pbSPYx, and pbJitoSOL.	Complete treasury, signer, authority, venue-position, and strategy-account register.
On-chain vault-account fields used to calculate underlying amounts for the listed vaults.	Per-venue exposures, leverage, collateral routing, funding paid, or open perpetual positions.
Partial third-party tracking of vault-level scale.	Independent NAV attestation, proof-of-reserves, or independently verified user-count figures.

Sources: #138 and #195. This figure replaces a general dashboard screenshot with a verification-boundary summary.

Figure 11. DefiLlama visibility boundary: what the public adapter and dashboard can read versus what they do not independently verify. Prepared from Sources #137, #149, and #153.

How self-reported TVL can be manipulated. Several vectors are generic to the category and bear directly on PiggyBank’s structure. The first is multiple-counting of recycled capital: the xStocks vaults and the JitoSOL vault post depositor assets as collateral, borrow stablecoins against them, and redeploy those stablecoins into the perpetual strategy (5.1.4), so the same underlying dollar can be represented at the collateral layer, the borrowed layer, and the deployed layer at once. The second is reporting gross notional rather than net depositor equity, which a leveraged book inflates by the leverage multiple. The third is self-funded or circular deposits that lift a headline figure without representing independent demand. The fourth is mark-to-model valuation of illiquid or off-hours legs (5.1.5). Independent trackers mitigate these by running open-source adapters that read vault balances directly and apply consistent rules for double-counting and for distinguishing collateral from open interest — the same discipline that leads DefiLlama to split a composite protocol across separate line items and to report derivatives “TVL” as collateral rather than notional.^{97, 137}

Why the gap remains structural. The existence of the DefiLlama protocol page and adapter narrows the verification gap but does not close it. It shows that selected vault state accounts can be identified and read, and it provides a third-party TVL point, but it does not expose the complete

wallet, treasury, venue-position, and strategy-allocation map needed to reconcile self-reported NAV, leverage, collateral routing, counterparty concentration, or illiquid holdings. A researcher should therefore treat TVL as partly externally checkable at the vault-account level, while still treating user counts, APYs, NAV marks, live-position exposure, and strategy allocation as unaudited or operator-dependent claims unless separately substantiated.^{153, 149, 150}

Net assessment. PiggyBank’s scale metrics are no longer wholly unsupported by third-party infrastructure, because DefiLlama provides both a current protocol TVL page and an adapter with partial on-chain vault-state visibility. They are still not equivalent to an audited proof of reserves, independent NAV attestation, or full position-level disclosure. The correct treatment is therefore “partly externally checkable, time-sensitive, and materially incomplete,” not “fully verified.”^{153, 149,}

150

5.2.3 Unverified Partnership and Integration Claims

The claim surface. PiggyBank’s homepage carries a “Trusted By” row displaying the marks of Solflare, xStocks, Jupiter Lend, Binance Wallet, and Jet. The “Trusted By” framing connotes that those parties have vetted or endorsed the protocol. The evidence supports a more differentiated picture, in which the displayed marks fall into three distinct relationship types and only one is a reciprocal endorsement.²

Substantiated and reciprocal: xStocks. The xStocks relationship is real and bilateral at least at the marketing and operational level. xStocks publicly promoted PiggyBank, presenting it as a means to earn “extra yield on top of xStocks through PiggyBank’s delta neutral strategies”, and PiggyBank in turn depends operationally on the xStocks issuer’s restricted-country list, which it uses to geofence access. This is the one displayed relationship for which reciprocal acknowledgement could be located.^{138, 2}

Infrastructure PiggyBank consumes, one-directionally: Jupiter Lend. Jupiter Lend, by contrast, is infrastructure that PiggyBank uses rather than a party that has endorsed it. Jupiter Lend is explicitly permissionless, designed “as a core piece of Solana’s financial infrastructure, enabling developers and partners to build new applications on top of its architecture”; PiggyBank borrowing through it is PiggyBank trusting Jupiter, not the reverse. No Jupiter announcement, listing, or page referencing PiggyBank could be located. The same one-directional logic applies to Kamino and Loopscale, named in the strategy description as borrowing venues (5.1.4); the marks evidence a dependency that PiggyBank has on those protocols, not an endorsement those protocols have given PiggyBank.^{139, 140}

Permissionless wallet connectivity: Solflare and Binance Wallet. Solflare and Binance Wallet are Solana wallets that connect to essentially any decentralized application without the application’s relationship being bilateral or vetted. No Solflare or Binance Wallet announcement, campaign, or listing referencing PiggyBank could be located, notwithstanding that both maintain active and well-documented partnership and campaign channels covering numerous other projects. The most defensible reading of these two marks is wallet-connect compatibility, which is universal and permissionless and does not constitute an endorsement.^{141, 142}

An unexplained and probably defunct mark: Jet. The “Jet” mark is unexplained in PiggyBank’s materials. The most prominent Solana protocol of that name, the lending protocol Jet Protocol, announced its shutdown effective 1 February 2024, instructing users to withdraw all funds — well before PiggyBank’s October 2025 launch — which would make a current “trusted by Jet Protocol” relationship implausible were that the referent. The identity cannot be confirmed from the mark alone, but no live entity of that name within PiggyBank’s documented stack could be identified.¹⁴³

Net assessment. The “Trusted By” framing overstates the relationships it depicts. Standard due-diligence practice is to verify any claimed partnership on the counterparty’s official channels; applied here, only the xStocks relationship survives as a reciprocal endorsement. The remaining marks represent either infrastructure PiggyBank consumes (Jupiter Lend, and by extension Kamino and Loopscale) or permissionless wallet connectivity (Solflare, Binance Wallet), neither of which is an endorsement, together with one mark (Jet) whose referent is unverified and most plausibly defunct. No reciprocal acknowledgement of PiggyBank by Solflare, Jupiter, or Binance Wallet could be found in any of those parties’ own materials.¹³⁵

5.2.4 NAV Reporting Integrity

Who produces NAV, and on what basis. NAV is produced by the protocol operator and reported to depositors through the application on a 48-hour epoch cycle, with deposits beginning to accrue and redemptions clearing against these epoch prints rather than against a continuously verifiable on-chain index. The marks that determine NAV derive from the operator’s own positions across approximately ten perpetual venues and three lending markets (5.1.5), positions that are not independently observable because the underlying accounts are undisclosed (4.2.3).^{1, 2}

What prevents manipulation. On the present record, no external mechanism does. There is no proof-of-reserves attestation, no on-chain NAV or proof-of-reserves oracle, and no independent administrator or auditor of the periodic NAV; the operator marks its own book and publishes the result. Proof-of-reserves exists precisely to permit third parties to verify holdings against liabilities, and its absence or incompleteness is a recognized vector for distorted NAV in fund-like

products. The structural difficulty is that the same anonymous party (5.2.1) trades the book, effectively custodies the assets, and reports their value, while depositors transact at a figure they cannot independently check at the moment of deposit or redemption and, given the 48-to-96-hour batched-withdrawal window (4.3.2), cannot exit instantly should they doubt it.¹⁴⁴

Why the audit does not close the gap. The Formal Land report is a real public security artifact and should not be treated as merely an unverified marketing claim. It establishes a dated pre-launch review, a repository and commit, and a code perimeter for the on-chain program. Its limits are different: it does not verify the accuracy of off-chain position marks, reported NAV, live venue positions, treasury holdings, collateral routing, admin-key custody, or post-launch economic behavior. The engagement therefore supports the existence of external code review within a defined perimeter, but it does not answer the transparency questions that matter most for depositors: whether the live vault backing, leverage, positions, and NAV marks can be independently reconstructed or attested.^{9, 149, 150}

A fairness qualification on what is verifiable. The share-accounting layer is on-chain and inspectable. Public materials describe compatibility with Token-2022 and transfer-hook mechanics, but they do not clearly specify the token standard and authority configuration for every live pb-token. The quantity of vault-share claims can still be checked at the token-account level, while the value each claim represents cannot be independently verified from the share supply alone. The gap is therefore specifically one of valuation and backing, not merely of share issuance.

^{1, 2, 149, 150}

Net assessment. Reported NAV is self-produced, off-chain, unattested, and structurally conflicted; there is no third-party verification of the NAV PiggyBank reports; and the protocol’s only external assurance addresses the correctness of code rather than the accuracy of valuation.

5.2.5 The “First on Solana” and Similar Marketing Claims

The claims. PiggyBank advances a first-mover claim in several places, describing itself as bringing “the first DeFi vaults to earn extra yield on xStocks” and, in its footer, as “the first decentralized protocol bringing extra yield to xStocks,” and it describes its xStocks vaults as “Overcollateralized and fully transparent”.²

The broad reading appears overstated by the record. On any reading in which “extra yield on xStocks” encompasses lending yield, the capability predated PiggyBank across multiple venues. xStocks launched on 30 June 2025. Kamino enabled xStocks as lending collateral on 14 July 2025, described contemporaneously as the first major DeFi lender to do so and as letting holders borrow against and earn yield on tokenized equities. The xStocks issuer’s own documentation lists earning yield by lending xStocks “via Kamino, Loopscale, JupLend (Solana) or Euler (Ethereum)”.

PiggyBank itself launched on mainnet only on 30 October 2025. Yield on xStocks therefore existed across several venues roughly three to four months before PiggyBank was live.^{145, 13, 14, 16, 5}

The narrow reading is unverifiable and self-narrowing. Read strictly as “the first to package xStocks into an automated, delta-neutral, funding-rate vault that compounds an additional structured yield,” the claim may be accurate, but it cannot be independently confirmed, because no comprehensive registry of such products exists against which priority could be established. It is, moreover, the kind of narrowly drawn superlative — true only under a precise and self-selected definition — that is endemic to the sector’s marketing. The xStocks promotional statement that, through PiggyBank, tokenized equities “for the first time... have yield” is itself marketing rather than independent verification, and is inaccurate on the broad reading given the lending yield already available.¹³⁸

The “fully transparent” claim is in tension with the findings of this section. The descriptor “Overcollateralized and fully transparent” sits uneasily against the protocol’s actual disclosure posture: undisclosed positions and addresses (4.2.3), self-reported and unreconciled metrics (5.2.2), and self-produced, unattested NAV (5.2.4). Overcollateralization on the lending leg is a verifiable property of the underlying money markets and is fairly claimed at that layer; but “fully transparent,” applied to PiggyBank’s own operation, is not supported by the record.

Net assessment. On the broad reading, the first-mover claim is not independently established and appears overstated in light of prior xStocks lending-yield availability. On the narrow reading, it remains unverifiable from public materials. The “fully transparent” claim is also overstated relative to the protocol’s disclosure posture: selected vault state is visible, but complete strategy, treasury, live-position, and NAV-verification data are not public. These phrases are therefore best characterized as marketing superlatives rather than independently verified factual findings.

6. Conclusion & Assessment

This concluding section draws the analysis into a final verdict. It summarizes the protocol (6.1), presents the risk profile (6.2), states the core tension between PiggyBank’s marketing and observed conduct (6.3), lists the open questions a reviewer should require be answered (6.4), and closes with a measured publication assessment (6.5). Consistent with the scope and method above, the conclusion is evidence-based research analysis, not investment advice.

6.1 Protocol Summary

PiggyBank is a Solana-based decentralized-finance yield protocol that launched on mainnet on 30 October 2025 and operates three deposit vaults — a USDC stablecoin vault, an SPYx vault built on the xStocks tokenized tracker of the SPY S&P 500 ETF, and the JitoSOL vault built on the

underlying JitoSOL liquid-staking token (see Section 1.3; Section 2.1.1). Depositors receive fungible, yield-bearing pb-tokens that represent a pro-rata claim on the managed pool and can be redeemed for the underlying assets plus accrued performance (see Section 2.1.2). Rather than a passive, immutable on-chain primitive, PiggyBank is an actively managed, discretionary, leveraged strategy: an off-chain operator deploys capital into a delta-neutral, funding-rate-arbitrage program across approximately ten perpetual venues at up to approximately five-times leverage, and for the non-stablecoin vaults it posts depositor assets as collateral on Solana money markets such as Kamino and Loopscale to borrow stablecoins that are then routed into the same market-neutral strategy (see Section 2.2; Section 2.3).^{5, 1, 2}

The protocol markets defined target net yields — on the order of 15% on the USDC vault and roughly 7% on the xStocks vaults — and runs on a 48-hour epoch cycle: yield compounds at each epoch, deposits begin accruing from the following epoch, and withdrawals are batched and processed over an epoch as perpetual positions are unwound, so redemptions clear against operator-reported net-asset-value (NAV) prints and typically settle 48 to 96 hours after a request (see Section 5.1.3; Section 4.3.2). The central limitation established in Section 5 is narrower than saying the protocol is invisible: selected vault state is externally readable, but the operator identity, complete address map, live exposures, NAV methodology, and independent attestation layer remain unavailable or insufficiently verifiable.

6.2 Risk Profile Verdict

The five ratings below are relative, evidence-based research judgments on a Low / Medium / High / Critical scale. They are not predictions of imminent failure. In this report, “Critical” means that a risk area is both material to depositor outcomes and not independently measurable from available public information; it does not mean that loss is certain or immediate. The scale weights six factors: likelihood of the risk materializing, potential impact on depositor funds, degree of independent verifiability, reliance on operator discretion, user ability to exit before loss crystallization, and quality of disclosed mitigations. The paragraphs below explain the rating logic directly.

- **Smart-contract risk — Medium.** PiggyBank’s principal external code assurance is a public Formal Land security audit of the on-chain program. That report is useful and prevents a High rating for the narrow smart-contract dimension, because it identifies a real review perimeter, date, repository, and findings. Two facts prevent a Low rating. First, the audit is scoped to the pre-launch program code and does not verify the off-chain NAV process, live strategy positions, venue integrations, deployed-program authority configuration, or economic soundness of the leveraged strategy (see Section 3.1; Section 5.2.4). Second, although selected vault state accounts are visible

through a public adapter, the complete official deployment, authority, treasury, and live-position register is not published in a consolidated verifiable form (see Section 2.1.5; Section 5.2.2). Smart-contract risk is therefore meaningfully mitigated but not low.

- **Strategy risk — High.** The strategy is actively managed, discretionary, and leveraged — a delta-neutral funding-rate-arbitrage book run across approximately ten perpetual venues at up to approximately five-times leverage — and delta-neutrality is a description of intended market exposure, not a guarantee of safety (see Section 5.1.1). Leverage amplifies both losses and liquidation risk; the funding-rate book is exposed to persistent negative funding; the SPYx leg carries off-hours basis risk because the underlying S&P 500 tracker prices outside continuous trading; and illiquid or off-hours legs are valued at model marks rather than realizable prices (see Section 5.1.5; Section 5.2.2). The LAB token episode is direct and recent evidence that these risks are not theoretical: a roughly US\$100,000 position in locked LAB acquired over the counter at a discount and hedged with a perpetual short unwound when LAB liquidity evaporated and funding turned deeply negative, with PiggyBank closing the short and crystallizing a loss that produced reported drawdowns of about 15% on the USDC vault, 12% on the SPYx vault, and 9% on the JitoSOL vault (see Section 4.3.1). That the largest markdown fell on the stablecoin vault — a 15% drawdown far beyond what a stablecoin-yield risk profile implies — is the sharpest illustration of the gap between the strategy’s marketing and its realized behavior. The rating is High rather than Critical because funding-rate arbitrage is a recognized strategy and the realized losses, while material, did not on the record impair the vaults beyond recovery.^{2, 7, 11, 69}

- **Counterparty/composability risk — Medium.** PiggyBank depends on several external systems: the perpetual venues used for hedging, lending markets used for collateralized borrowing, the xStocks issuer Backed and its custody/redemption stack for SPYx, and the Jito liquid-staking system behind JitoSOL (see Section 5.1.4; Section 5.1.5). That dependency is real, but it should not be overstated. Kamino, Jupiter Lend, JitoSOL, Backed/xStocks, and, after its post-launch recovery work, Loopscale are not treated here as weak or inherently suspect counterparties merely because PiggyBank uses them; use of established Solana and tokenized-asset infrastructure is normal for a DeFi yield product. The reason the rating is not Low is narrower: PiggyBank users inherit external smart-contract, oracle, liquidation, market-pause, issuer/custody, redemption, and liquidity dependencies, while PiggyBank does not disclose the live venue-position map, allocation weights, LTVs, collateral buffers, or hedging concentrations needed to size those dependencies independently (see Section 4.2.3; Section 5.2.2). The Drift exploit shows that venue-layer failures can matter to PiggyBank, but the reported loss was approximately US\$106,000 and was absorbed by the operator rather than by depositors (see Section 4.3.1). The LAB episode is treated primarily as strategy, transparency, and team/risk-culture evidence, not as proof that reputable Solana

counterparties are themselves high risk. The rating is therefore Medium: meaningful and unquantified enough to prevent a Low rating, but not High solely because PiggyBank composes with established external protocols.^{146, 11}

- **Transparency risk — Critical.** Transparency is the protocol’s defining deficiency and the only dimension rated Critical, but the criticism must be stated precisely. PiggyBank is not wholly invisible on-chain: DefiLlama’s adapter identifies three vault state accounts and reads their accounting fields. The critical gap is that this partial vault-state visibility does not disclose the team, operating entity, treasury and signer addresses, live venue positions, strategy allocations, leverage, collateral routing, NAV methodology, independent NAV attestation, or proof-of-reserves (see Section 5.2.1; Section 5.2.2; Section 5.2.4). Its interface metrics are partly externally checkable at the vault-account level, but they remain incomplete as evidence of full backing and risk exposure. The transparency rating therefore rests not on the absence of any public address data, but on the absence of the complete information needed to verify the economically material risk profile.

- **Team risk — High.** The operating team is wholly anonymous — no operating entity, operating jurisdiction, named principals, verifiable track record, or identity for the treasury and program-authority signers — yet exercises discretionary, custodial-style control over leveraged depositor capital, the configuration in which the accountability that substitutes for code-is-law is most needed and most absent (see Section 5.2.1; Section 4.2.3). There is a genuine, if partial, offset: PiggyBank absorbed the approximately US\$106,000 Drift loss from its own treasury, disclosed the LAB loss publicly and relatively promptly rather than concealing it, and later placed an official compensation statement on the public record. That offset is weaker than identification combined with ring-fenced funds, because voluntary reimbursement depends on the continued solvency and goodwill of the same undisclosed party, and the public record reviewed here does not provide payment-level verification or per-user compensation terms (see Section 4.3.1; Section 5.2.1). The LAB episode also raises governance and risk-culture questions, with ZachXBT and several commentators asking how a single position could produce an approximately 15% loss in a vault marketed as low-risk, pointing to possible gaps in position limits or risk safeguards. The rating sits at the severe end of High rather than Critical because the demonstrated Drift reimbursement, public disclosure of the LAB loss, and official compensation statement are real accountability signals, even if fragile.^{146, 8, 7, 69, 152}

6.3 The Core Tension

The central tension surfaced by this paper is between PiggyBank’s marketing posture and the risk profile documented in the LAB episode. The marketing narrative is one of neutrality and safety:

the protocol describes strategies that are automated and risk-managed to generate yield “without taking directional market exposure,” the xStocks issuer promoted the vaults as delta-neutral, and the USDC vault is positioned as the conservative, stablecoin entry point (see Section 5.2.3). The LAB record shows a materially different risk profile. The operator deployed depositor capital into a roughly US\$100,000 over-the-counter purchase of locked LAB — a low-float token that ZachXBT had publicly criticized through allegations of insider control, OTC/private-sale irregularities, and market-maker coordination — while attempting to hedge that exposure with a perpetual short. Those allegations against LAB remain disputed unless independently verified, but they were public before the loss crystallized.^{2, 147, 148, 68, 117}

The trade was delta-neutral in form — long spot, short perpetual — but it created concentrated illiquidity, funding, and basis risk. When LAB liquidity evaporated and funding turned deeply negative, maintaining the short became, in the protocol’s own characterization, economically irrational, so PiggyBank closed it, leaving the fund holding an unsellable locked token and economically exposed to the asset it had attempted to hedge. ZachXBT characterized the episode in pejorative terms, while PiggyBank maintained that it had acted within the strategy’s scope of mid-cap basis trades and before the position crossed its risk limits. That dispute — in-scope basis trade or out-of-mandate speculation — is the analytical point: a delta-neutral product whose risk perimeter is defined only by the discretionary operator after the fact offers depositors no public, enforceable boundary.^{7, 11, 68, 8}

Three conclusions about the protocol’s actual risk culture and governance follow. First, “delta-neutral” is not shown in the public materials to be an enforceable constraint embedded in code, governance, or published risk limits; discretion permitted the operator to describe the LAB trade as within the mandate despite its concentrated and illiquid character (see Section 5.1.2). Second, the supposedly conservative framing did not correspond to a ring-fenced book: the loss fell hardest on the USDC vault, the product marketed as safest, indicating that the same discretionary strategy engine exposed all three vaults. Third, the governance response relied on an operator-determined NAV treatment: the LAB leg was excluded from NAV until the August unlock, a choice the protocol described as “the fairest and most transparent way to let users manage their liquidity.” PiggyBank later made an official compensation statement, but the detailed trade-record post-mortem, quantified scheme, and independent payment-level evidence were not present in the public record reviewed here. The episode therefore reveals a protocol whose marketed posture of neutrality, safety, and transparency remains dependent on operator discretion when it is tested.^{69, 8,}

^{7, 152}

6.4 Open Questions

The following questions are the ones the public record cannot answer and that a prospective depositor, analyst, or reviewer should require be answered before engaging with the protocol. Each corresponds to a specific, documented gap identified in Section 5.

Who operates the protocol? The operating entity, its operating jurisdiction, the named principals, and the identity of the treasury and program-upgrade signers, so that accountability attaches to identifiable persons rather than to an anonymous collective (see Section 5.2.1; Section 4.2.3).

Where are the funds, and what do they hold? A complete address register covering treasury, signer, strategy, venue-position, and program-authority accounts, alongside the already visible vault state accounts, would permit independent verification of reserves, live positions, realized leverage, and counterparty concentration (see Section 4.2.3; Section 5.2.2).

Is the reported NAV real? Independent attestation or proof-of-reserves for the off-chain, operator-marked NAV against which deposits and redemptions clear — most acutely for the roughly US\$1.35 million locked LAB position now excluded from NAV until 14 August (see Section 5.2.4).¹¹

How does the public audit map to the live deployment? The Formal Land report establishes a pre-launch review at a specific repository commit, but the public record still needs a clear mapping from that audited commit to the current deployed program ID, upgrade authority, and live bytecode, as well as disclosure of whether any material post-launch changes have occurred (see Section 3.1; Section 5.2.4).

What are the real risk limits? The position-size, leverage, liquidity, and asset-eligibility constraints that define “the strategy’s scope,” whether the LAB position breached them, and what concretely distinguishes a permitted mid-cap basis trade from out-of-mandate speculation (see Section 5.1.2).

8

How will the LAB episode be resolved? The promised detailed post-mortem, the final recovery value of the locked position after the August unlock, the exact amounts and timing attached to PiggyBank’s compensation statement, any payment-level verification of that statement, and the NAV at which redemptions will clear in the interim (see Section 4.3.1).^{7, 152}

What is the protocol’s true scale? Independently verifiable TVL and user figures reconciled against on-chain reserves, in place of the self-reported and internally inconsistent metrics currently presented (see Section 5.2.2).

6.5 Closing Assessment

PiggyBank gets several things right, and a fair assessment records them. It is a real, functioning product built on real infrastructure: it uses established Solana money markets, tokenized-asset infrastructure, and an on-chain share-accounting layer that is at least partly inspectable (see Section 5.2.4). Its sole external assurance comes from a real and reputable formal-verification firm rather than a fabricated or unknown auditor (see Section 3.1). It has shown partial but genuine accountability, disclosing the Drift exposure and stating that it absorbed the related loss from team funds; disclosing the LAB loss publicly rather than concealing it; and later placing an official compensation statement on the public record.^{146, 8, 152}

What remains concerning is more fundamental. PiggyBank is not necessarily unsafe because it is complex. It is risky because the parts that matter most to depositor outcomes are not independently observable at sufficient depth: the operator, the full address and signer set, the live positions, the routing and leverage profile, and the NAV marks. The LAB episode demonstrated that the delta-neutral mandate is not a public, enforceable boundary and that the vault marketed as safest can absorb a discretionary strategy loss. The subsequent response added useful disclosure and a compensation statement, but not the trade-record post-mortem, independently verifiable NAV treatment, or payment-level compensation evidence that would close the core due-diligence gap (see Section 4.3.1; Section 5.2).

For PiggyBank to be assessed with stronger confidence, the required improvements are specific and material. It would need to disclose the operating entity, operating jurisdiction, and principals, together with a complete on-chain address register covering vaults, treasury accounts, strategy accounts, signer authorities, and venue positions, so that reserves and positions can be independently verified beyond the partial vault-state visibility already available (see Section 5.2.1; Section 4.2.3). It would need to replace self-marked NAV with independent attestation or proof-of-reserves and publish a clear mapping from the Formal Land audit to the currently deployed program and authority configuration (see Section 5.2.4). It would need to codify and publish enforceable risk limits — on position size, leverage, liquidity, and eligible assets — and demonstrate adherence to them, turning delta-neutrality from a marketing claim into a governed constraint (see Section 5.1.2). It would need to close the LAB episode credibly by publishing the trade-record-bearing post-mortem, resolving the locked position transparently, and providing independently checkable evidence for any compensation represented as complete (see Section 4.3.1).

On the current record, PiggyBank is best characterized as a functioning but structurally opaque, discretionary, and leveraged yield product. The question is not whether such a product can

generate yield; it can. The question is whether depositors can independently verify the risk being taken to generate that yield. Until the disclosure, valuation, and governance gaps identified in this report are closed, the protocol remains investable only on a substantial trust assumption in an anonymous operator rather than on a fully independently verifiable risk profile. That conclusion is a bounded research judgment, not investment advice.

Source Table

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